Knowledge-Aided Space-Time Adaptive Processing for Functional Ultrasound: Robust Microvascular Detection in High-Clutter Regimes

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Abstract

Functional ultrasound (fUS) and ultrafast Doppler imaging face severe interference from tissue motion, phase aberrations, and aliased high-velocity flow, which can obscure slow microvascular dynamics. Low-rank clutter suppression methods such as motion-compensated singular value decomposition (MC–SVD) are widely used, but can degrade in extreme clutter and in the far right tail of flow evidence.

Motivated by airborne radar and RF array processing, we introduce a space–time adaptive processing (STAP) framework for fUS slow-time flow detection. The method operates on localized spatiotemporal tiles with a fixed-parameter operating profile (tile geometry, slow-time aperture, Doppler band geometry, and covariance conditioning) and produces a matched-subspace STAP score. An optional knowledge-aided (KA) spatial-spectral prior uses local band-energy features to apply a conservative, shrink-only penalty in high-risk regions while protecting a high-confidence set; by construction, the post-KA score is non-increasing on penalized pixels. We also study conditional execution as a decision-rule variant that skips STAP outside a flow-proxy region and quantify its leakage sensitivity.

Quantitative detection performance is evaluated using a strict low-FPR operating discipline (typically 10^{-4}–10^{-3}, subject to empirical support), reflecting the sensitivity of downstream functional mapping to rare flow artifacts and consistent with tail-rate control philosophies in radar detection and tracking. We report results on labeled k-Wave brain simulations and on structurally labeled *in vitro* phantom acquisitions (Gammex; labels derived from B-mode geometry), and we benchmark against MC–SVD, robust PCA, and tensor HOSVD baselines as well as conventional short-ensemble Doppler scores. We further report labeled within-ensemble motion ladders on phantom data via per-shot motion injection, and label-free stability under controlled motion corruption on kilohertz ultrasound localization microscopy (ULM) IQ blocks. Because most open in-vivo fUS releases do not preserve raw high-PRF complex IQ with independent H_1/H_0 labels, we restrict quantitative ROC claims to labeled simulations and structurally labeled phantoms, using in-vivo data only for robustness and telemetry audits. All experiments and figures are accompanied by a command-level reproducibility manifest in the Supplementary Material (exact scripts, arguments, and expected artifact paths).

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1 Introduction

Microvascular flow detection in ultrasound is an extreme-tail problem: flow evidence occupies a small fraction of pixels while tissue motion, phase aberration, and aliased high-velocity clutter induce heavy-tailed background statistics. In this regime, improvements in global summaries such as AUC can be operationally uninformative; the key requirement is stable detection at ultra-low false-positive rates.

We address this setting with a localized, STAP-inspired detector in which adaptation is temporal (slow-time), while spatial localization enters through tile-local covariance estimation. Instead of relying only on global low-rank suppression, we whiten local slow-time snapshots and score energy in a flow subspace, targeting spatially varying interference patterns that dominate extreme-tail false alarms.

We further introduce a knowledge-aided (KA) shrink-only regularization layer driven by local band-energy features. KA can suppress high-risk tail events without score inflation: post-KA scores are pointwise non-increasing, and a protected set is explicitly preserved.

Contributions.

- A tile-based temporal STAP detector with matched-subspace interpretation, evaluated at strict low-FPR operating points.
- A feature-conditioned KA regularizer with shrink-only monotonicity and protected-set invariance.
- A leakage-controlled conditional-execution variant treated explicitly as a changed decision rule (not a cost-free speedup).
- Regime-stratified evidence across labeled simulations, structurally labeled Gammex flow phantom sequences, and label-free instrumentation datasets.

Scope and claims. Claims are stratified by data regime and summarized in Table 2. Quantitative detection efficacy is limited to regimes with independent H_1/H_0 labels; unlabeled in-vivo IQ and PD-only datasets are used for instrumentation and robustness analyses.

Organization. Section 2 defines the detection problem and score taxonomy. Section 3 presents the localized detector and KA regularization. Section 4 specifies fixed-profile experimental design and integrity controls. Section 5 reports results, followed by discussion and conclusion.

2 Background and Problem Formulation

Problem setup. For each spatial location (or tile-aggregated snapshot), let $x \in \mathbb{C}^{L_t}$ denote a slow-time vector. We consider

$$H_0 : \; x = c + n, \qquad H_1 : \; x = s + c + n,$$

where s is flow-consistent signal, c is structured clutter/motion, and n is residual noise. In practice, H_0 is spatially nonstationary and heavy-tailed, motivating local covariance adaptation and tail-focused evaluation.

Core notation and bands. We use band operators (P_o, P_f, P_g, P_a) for clutter/DC, flow, guard, and alias regions, respectively. For short ensembles, these are exact orthogonal DFT-bin projectors; for longer apertures they are leakage-controlled analysis operators and are treated as band-energy statistics with explicit sanity sentinels (Appendix H).

2.1 Related context

STAP and knowledge-aided STAP. Classical MVDR/LCMV/STAP and knowledge-aided variants shape covariances using external structure [42–44, 49, 50, 53]. Our spatial-spectral prior formalizes when such transforms are allowed to act in fUS and keeps them inert when data are out of regime.

Ultrasound clutter suppression. Spatiotemporal SVD clutter filters are a standard tool in ultrafast Doppler and fUS [6,7,11,12,16]. Robust PCA and tensor/HOSVD variants have also been studied for clutter suppression and motion robustness [22, 36]. We benchmark STAP against MC–SVD, RPCA, and HOSVD baselines in brain-like regimes and report conventional short-ensemble Doppler comparators on phantom data.

Relation to eigen-based and local adaptive clutter filtering. Eigen-based clutter filter design for ultrasound color flow imaging and block-wise/local adaptive clutter filters are closely related ultrasound-native approaches to covariance/eigenspace adaptation [13–15]. Our contribution differs in the *detector* formulation and evaluation discipline: we whiten slow-time snapshots using tile-local conditioned covariances and score energy in an explicitly defined flow subspace (matched-subspace statistic), evaluate at empirically supported ultra-low FPR operating points, and treat compute gating (conditional execution) as a distinct decision rule with leakage controls.

Twinkling artifact and Doppler phantoms. The Doppler "twinkling artifact" has been documented as a persistent color-flow phenomenon on calcifications and other strong reflectors, and is widely used as a diagnostic benchmark for Doppler processing chains. Leonov *et al.* provide an open-access database of raw beamformer signals with phantoms including a flow-through Gammex phantom and calculi/metal/plastic objects [35]. We use these data to (i) obtain a structurally labeled phantom ROC via B-mode-only tube segmentation and (ii) construct an artifact-heavy KA hygiene regime where the prior penalty is expected to be conservative and auditable.

fUS detection and hemodynamic analysis. GLM-based detection and atlas-driven ROI analysis are common in hemodynamic fUS [37,38]; our hemodynamic STAP variant reuses the same banded temporal machinery on PD time series and adds an alias-aware shrink-only prior penalty as a safety layer.

2.2 Baseline clutter suppression and score conventions

Beamformed slow-time data. Let $X[t, y, x] \in \mathbb{C}$ denote a beamformed slow-time IQ movie over $t = 0, \ldots, T-1$ and spatial indices (y, x). We first apply a baseline clutter-suppression operator (e.g. motion-compensated SVD), producing a residual cube $X_{\text{base}}[t, y, x]$ used for both baseline scoring and as the input to STAP.

Power Doppler proxy and baseline scores. From the baseline-filtered cube we define a conventional power Doppler (PD) brightness proxy (used only for deterministic preprocessing such as

Symbol / name	What it measures	Used for	Data requirement
S_{base}	baseline right-tail detector score on X_{base}	baseline ROC / low-FPR operating-point reporting	IQ
$S_{\text{stap,pre}}$	pre-KA matched-subspace STAP score	primary STAP ROC / low-FPR operating-point reporting	IQ + local covariance
$S_{\text{stap,post}}$	post-KA shrink-only score $wS_{\text{stap,pre}}$	KA tail-hygiene analyses	IQ + feature map
$\text{PD}^{\uparrow}$, proxy masks	baseline PD-brightness proxy and derived masks	gating, protected sets, visualization	IQ (or PD-only proxy mode)

Table 1: Score taxonomy used throughout. Detector scores are separated from proxy maps used for gating/diagnostics to avoid circular reporting.

proxy masks),

$$\text{PD}^{\uparrow}(y,x) = \frac{1}{T}\sum_{t=0}^{T-1} |X_{\text{base}}(t,y,x)|^2 .$$

For quantitative detection and thresholding we operate strictly on the right tail of score distributions and denote the baseline right-tail score as $S_{\text{base}}(y,x)$ (power Doppler on the baseline-filtered cube). For short-ensemble regimes we additionally report auxiliary, conventional baseline comparators (e.g. log-power Doppler and Kasai lag-1 magnitude) to avoid baseline-score idiosyncrasies dominating conclusions (Notation). Note: $\text{PD}^{\uparrow}$ is the same underlying power statistic as S_{base} (up to fixed scaling); we keep a separate name to enforce the policy that proxy masks are used only for deterministic preprocessing and diagnostics, not as quantitative efficacy claims.

2.3 Score taxonomy

Unless explicitly stated, "STAP" refers to the pre-KA matched-subspace score $S_{\text{stap,pre}}$; KA is analyzed as an optional, shrink-only post-processing regularizer.

3 Method

Algorithm 1 (KA–STAP pipeline, fixed profile).

1. Baseline clutter suppression (e.g. MC–SVD) produces a residual cube X_{base} and baseline score $S_{\text{base}}(y,x)$.
2. For each tile Ω_i: form slow-time snapshots $\{x_{i,k}\}_{k=1}^{K}$ (spatial sampling; optional Hankelization), estimate conditioned covariance R_i, and whiten $y_{i,k} = R_i^{-1/2} x_{i,k}$.
3. Compute the matched-subspace statistic $S_{\text{stap,pre}}(i) = \frac{1}{K}\sum_k y_{i,k}^{\text{H}} P_f\, y_{i,k}$ and aggregate over overlapping tiles to obtain $S_{\text{stap,pre}}(y,x)$.
4. Compute local features $\phi_i = [m_{\text{alias}}(i),\ r_g(i),\ c_f(i)]^{\top}$.
5. If KA enabled: determine regime $\mathcal{R}_0/\mathcal{R}_1/\mathcal{R}_2$ and compute shrink-only weight $w(\phi_i) \in (0,1]$ (protected-set invariance); output $S_{\text{stap,post}}(y,x) = w\, S_{\text{stap,pre}}(y,x)$ and audit logs (region, reason, risk mode).
6. Else: output $S_{\text{stap,post}}(y,x) = S_{\text{stap,pre}}(y,x)$.

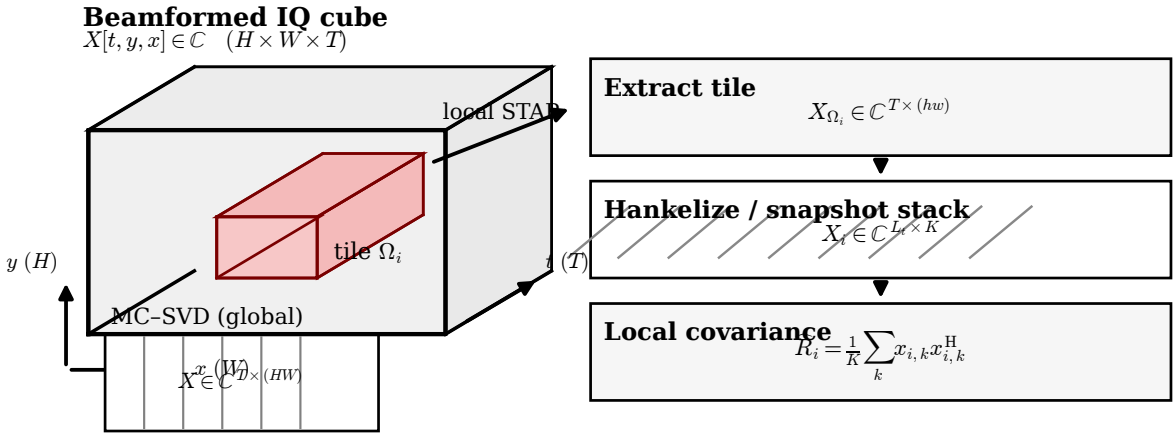

Figure 1: Schematic of spatiotemporal Hankelization and local covariance estimation. A localized spatial tile Ω_i is extracted from the beamformed slow-time data cube $X[t, y, x] \in \mathbb{C}$ $(H \times W \times T)$. Overlapping slow-time snapshots of length L_t are stacked (optionally Hankelized) to estimate the localized covariance matrix $\widehat{R}_i$. Unlike global low-rank filters (e.g. MC–SVD) that operate on a global Casorati matrix, this localized formulation preserves spatial-spectral heterogeneity of microvascular flow and highly structured tissue clutter.

In practice STAP is intended as a second-stage residual covariance whitener: a coarse baseline clutter filter removes gross stationary tissue clutter, and STAP targets the remaining structured, spatially varying interference that dominates extreme-tail false alarms. A STAP-only ablation on phantom data is provided in Appendix B.

3.1 Temporal STAP core and Doppler band geometry

Tile-based STAP. To mitigate the degraded tail sensitivity of global low-rank filters in high-clutter regimes, we apply a localized temporal STAP core on overlapping spatial tiles. We use "STAP" in a STAP-inspired sense: the adaptive filter operates in slow time, while spatial localization enters through tile-local covariance adaptation. Let Ω_i denote tile i and L_t the slow-time aperture. From X_{base} we construct a set of slow-time training snapshots $\{x_{i,k} \in \mathbb{C}^{L_t}\}_{k=1}^{K}$ for each tile (via spatial sampling and, when applicable, Hankelization in slow time). Here K denotes the number of training snapshots per tile; in the fully Hankelized case $K = T - L_t + 1$ per spatial sample (or a subsampled variant when training support is capped). We estimate a local covariance

$$\widehat{R}_i := \frac{1}{K} \sum_{k=1}^{K} x_{i,k} x_{i,k}^{\mathrm{H}},$$

apply conditioning/diagonal loading to obtain R_i, and whiten snapshots as $y_{i,k} := R_i^{-1/2} x_{i,k}$. Figure 1 illustrates the local tile extraction and (optional) Hankelization used to form $\widehat{R}_i$ and contrasts it with the global Casorati stacking used by MC–SVD.

Band operators and matched-subspace STAP score. We partition the Doppler spectrum using band projectors (P_o, P_f, P_g, P_a) (Notation). For short ensembles, these are exact, orthogonal DFT-bin projectors with $P_o + P_f + P_g + P_a = I$; for longer apertures they are leakage-controlled operators that are approximately orthogonal and are treated as band-energy statistics. We define a matched-subspace STAP test statistic by flow-subspace energy on whitened snapshots, e.g.,

$$S_{\text{stap,pre}}(i) := \frac{1}{K} \sum_{k=1}^{K} y_{i,k}^{\text{H}} P_f \, y_{i,k},$$

and form a pixel-level score map $S_{\text{stap,pre}}(y, x)$ by overlap-add aggregation of the contributing tile statistics. This pre-KA score is the primary STAP detector output used for quantitative reporting unless explicitly stated otherwise.

Regularization and self-normalization avoidance. If we whitened using the exact sample covariance formed from the same snapshots being scored, the statistic collapses: the whitened sample covariance becomes (approximately) identity and averaged subspace energies concentrate near constants (e.g. $\text{tr}(P_f)$), destroying discrimination. We avoid this by whitening with a *regularized* covariance R_i (robust estimation plus shrinkage/diagonal loading), so that typically $R_i \neq \widehat{R}_i$ and the matched-subspace energy remains informative. We do not rely on an analytic null distribution for CFAR operation; operating points are empirically calibrated on labeled H_0 sets under the fixed-profile protocol of Section 4.1.

Spatial-spectral feature vector. In addition to the detector score, we compute per-tile band energies

$$E_f(i) := \frac{1}{K} \sum_{k=1}^{K} y_{i,k}^{\text{H}} P_f \, y_{i,k},$$

$$E_g(i) := \frac{1}{K} \sum_{k=1}^{K} y_{i,k}^{\text{H}} P_g \, y_{i,k},$$

$$E_a(i) := \frac{1}{K} \sum_{k=1}^{K} y_{i,k}^{\text{H}} P_a \, y_{i,k},$$

$$E_o(i) := \frac{1}{K} \sum_{k=1}^{K} y_{i,k}^{\text{H}} P_o \, y_{i,k},$$

and a total $E_{\text{tot}}(i) := E_f(i) + E_g(i) + E_a(i) + E_o(i) + \epsilon$ with $\epsilon > 0$. We use ratios of these energies as label-free statistics for interpretation and for the KA prior, e.g.,

$$m_{\text{alias}}(i) := \log \frac{E_a(i) + \epsilon}{E_f(i) + \epsilon}, \qquad m_{\text{guard}}(i) := \log \frac{E_g(i) + \epsilon}{E_f(i) + \epsilon}, \qquad r_g(i) := \frac{E_g(i)}{E_{\text{tot}}(i)}.$$

For short ensembles, we additionally compute a non-DC peak-frequency indicator from a pooled whitened tile vector $z_i := \frac{1}{K} \sum_k y_{i,k}$ and its DFT; defining $f_{\text{peak}}(i)$ as the maximizing (non-DC) frequency, we record $\pi_f(i) := \mathbf{1}\{f_{\text{peak}}(i) \in \mathcal{B}_f\}$ and summarize peak occupancy on proxy sets by averages of π_f. Together with structural proxy coverage (below), these local descriptors form a per-tile *spatial-spectral feature vector* used to construct the KA regularizer.

3.2 Feature-conditioned score regularization

While the matched-subspace statistic S_{pre} effectively isolates flow in Gaussian clutter, highly structured artifacts can still elevate false-alarm rates in the extreme right tail. To systematically suppress these artifacts, we introduce a feature-conditioned regularization layer. While we conceptually view this penalty as a prior-odds modifier in log-score space, the construction is deterministic, using local spectral features as auxiliary statistics rather than producing a calibrated probability output.

Matched-subspace statistic as a whitened subspace detector. Fix tile i and consider a single slow-time snapshot $x \in \mathbb{C}^{L_t}$. Under a Gaussian clutter-plus-noise model,

$$H_0 : \; x \sim \mathcal{CN}(0, R_i), \qquad H_1 : \; x = A_f s + n, \quad s \sim \mathcal{CN}(0, \sigma_s^2 I), \quad n \sim \mathcal{CN}(0, R_i),$$

where the columns of A_f span the flow subspace (equivalently the range of P_f). After whitening $y = R_i^{-1/2} x$, the GLRT for $\sigma_s^2 > 0$ versus $\sigma_s^2 = 0$ is monotone with the whitened subspace energy

$$T(x) = x^{\mathrm{H}} R_i^{-1/2} P_f R_i^{-1/2} x = y^{\mathrm{H}} P_f \, y,$$

which is a standard matched-subspace / adaptive energy detector. [51,52] Averaging $T(x)$ over the K local snapshots yields our pre-KA tile statistic $S_{\text{stap,pre}}(i)$.

Local spatial-spectral feature vector. For each tile i, we define a feature vector $\phi_i \in \mathbb{R}^3$ that summarizes local spectral kinematics and structural proxy support:

$$\phi_i := \begin{bmatrix} m_{\text{alias}}(i) \\ r_g(i) \\ c_f(i) \end{bmatrix},$$

where $m_{\text{alias}}(i) = \log \frac{E_a(i)+\epsilon}{E_f(i)+\epsilon}$ is the alias-to-flow energy ratio, $r_g(i) = E_g(i)/E_{\text{tot}}(i)$ is the guard fraction, and $c_f(i)$ is a deterministic flow-proxy coverage fraction defined below.

Feature-conditioned shrinkage (log-score view). We implement knowledge aid via a bounded weight $w(\phi_i) \in (0, 1]$ applied in score space:

$$S_{\text{stap,post}}(i) = w(\phi_i) \, S_{\text{stap,pre}}(i), \qquad w(\phi_i) \in (0, 1].$$

Equivalently, define a scale factor $\gamma(\phi_i) := 1/w(\phi_i) \geq 1$ and write

$$S_{\text{stap,post}}(i) = \frac{S_{\text{stap,pre}}(i)}{\gamma(\phi_i)}.$$

For brevity, we refer to $S_{\text{stap,pre}}$ and $S_{\text{stap,post}}$ as S_{pre} and S_{post} when discussing the feature-conditioned penalty and in figure captions. Equivalently, for $S_{\text{pre}}(i) > 0$ define $T_{\text{pre}}(i) := \log S_{\text{pre}}(i)$ and $T_{\text{post}}(i) := \log S_{\text{post}}(i)$, giving

$$T_{\text{post}}(i) = T_{\text{pre}}(i) + \log w(\phi_i).$$

We use this additive log-score form as an *interpretive lens*: it mirrors adding a feature-dependent penalty term to a log-likelihood score, but the construction is deterministic and uses ϕ_i as auxiliary statistics rather than producing calibrated probabilities. We constrain it to be shrink-only by enforcing $w(\phi_i) \leq 1$. For numerical stability in plots we display $\log_{10}(S + \epsilon)$ with $\epsilon > 0$.

Feature-space regimes. For auditability and conservative behavior, we partition the feature space into three regimes that control the strength of the prior penalty:

- **Uninformative regime $\mathcal{R}_0$.** If sample support is insufficient, band discretization is invalid, or risk features are effectively flat, the prior defaults to uninformative: $w(\phi_i) \equiv 1$, so $S_{\text{stap,post}} = S_{\text{stap,pre}}$.
- **Guarded regularization regime $\mathcal{R}_1$.** If features indicate uncertainty or leakage (e.g. elevated guard fractions or moderate alias ratios) but do not satisfy stronger actionability evidence, we apply a bounded penalty $w(\phi_i) \in [w_{\min}, 1]$ while preserving a protected high-confidence set.
- **Actionable alias-dominance regime $\mathcal{R}_2$ (research-only).** If features indicate alias-dominant structure and are *actionable* in the sense that they explain high-score background-tail events under proxy sets, we permit a stronger but still bounded penalty, while maintaining shrink-only and protected-set invariance.

Proxy sets (label-free) and protected-set invariance. When ground-truth H_1/H_0 labels are unavailable, we define proxy sets deterministically from a structural proxy (e.g. $\text{PD}^{\uparrow}$ or simulator geometry). When using $\text{PD}^{\uparrow}$, we define a high-quantile flow proxy mask

$$M_{\text{flow}}(y, x) := \mathbf{1}\{\text{PD}^{\uparrow}(y, x) \geq Q_{q_{\text{flow}}}(\text{PD}^{\uparrow})\},$$

a per-tile flow coverage

$$c_f(i) := \frac{1}{|\Omega_i|} \sum_{(y,x)\in\Omega_i} M_{\text{flow}}(y, x),$$

and background/flow proxy sets

$$\mathcal{T}_{\text{bg}} := \{i : \ c_f(i) \leq c_{\text{bg}}\}, \qquad \mathcal{T}_{\text{flow}} := \{i : \ c_f(i) \geq c_{\text{flow}}\}.$$

To enforce protected-set invariance, we define a protected set $\mathcal{T}_{\text{prot}}$ that includes high-confidence flow-proxy tiles (and optionally extreme-score tiles) and restrict penalization to a fixed candidate set of non-protected tiles in an uncertain score window; outside this candidate set we set $w(\phi_i) \equiv 1$. Region membership ($\phi_i \in \mathcal{R}_0/\mathcal{R}_1/\mathcal{R}_2$) is determined by deterministic integrity checks and by label-free actionability evidence that risk features are associated with high-score background tails (Appendix A).

Penalty mapping (summary). Let $S_{\text{pre}}(i)$ denote the (right-tail) pre-KA tile score and let $m_{\text{risk}}(i)$ denote the chosen risk feature (typically $m_{\text{alias}}(i)$; a guard-risk lane may substitute m_{guard} when guard dominates). Using frozen design quantiles $q_{\text{lo}}, q_{\text{hi}}, q_{\text{risk}}$, we define a candidate set

$$\mathcal{T}_{\text{cand}} := \Big\{i \notin \mathcal{T}_{\text{prot}} : \ Q_{q_{\text{lo}}}(S_{\text{pre}}) \leq S_{\text{pre}}(i) \leq Q_{q_{\text{hi}}}(S_{\text{pre}})\Big\},$$

set a background-proxy risk threshold $\tau_{\text{risk}} := Q_{q_{\text{risk}}}(m_{\text{risk}} \mid \mathcal{T}_{\text{bg}})$, and gate tiles by

$$g(i) = \mathbf{1}\{i \in \mathcal{T}_{\text{cand}}, \ m_{\text{risk}}(i) \geq \tau_{\text{risk}}\}.$$

On gated tiles we apply a bounded shrink factor

$$w(i) = \max(w_{\min}, \exp(-k\, u(i))), \quad u(i) = \max\left(0, \frac{m_{\text{risk}}(i) - \tau_{\text{risk}}}{\text{MAD}(m_{\text{risk}} \mid \mathcal{T}_{\text{bg}}) + \epsilon}\right),$$

and set $w(i) = 1$ for $g(i) = 0$ and for all $i \in \mathcal{T}_{\text{prot}}$. This yields $\gamma(i) = 1/w(i) \geq 1$ and $S_{\text{post}}(i) = S_{\text{pre}}(i)/\gamma(i)$.

Lemma 1 (Shrink-only monotonicity)**.** *Let $w(i) \in (0, 1]$ and define $S_{\text{post}}(i) = w(i)\, S_{\text{pre}}(i)$. For any threshold τ,*

$$\{S_{\text{post}} \geq \tau\} \subseteq \{S_{\text{pre}} \geq \tau\}.$$

Consequently, on any subset A (e.g. H_0), the tail probability cannot increase at fixed τ:

$$\mathbb{P}\{S_{\text{post}} \geq \tau \mid A\} \leq \mathbb{P}\{S_{\text{pre}} \geq \tau \mid A\}.$$

Proof. If $S_{\text{post}}(i) \geq \tau$, then $S_{\text{pre}}(i) = S_{\text{post}}(i)/w(i) \geq \tau$ since $w(i) \leq 1$. □

When reporting strict low-FPR operating points where τ is reselected post-KA to satisfy $\mathbb{P}\{S_{\text{post}} \geq \tau \mid H_0\} = \alpha$, changes in TPR reflect background-tail compression despite scores never increasing.

3.3 Algorithmic integrity and deterministic safeguards

To make the architecture auditable and to avoid leakage or discretization artifacts masquerading as performance gains, we enforce deterministic, label-free safeguards:

- **Conditional execution as a decision-rule variant.** To reduce latency, STAP can be skipped outside a flow-proxy region and replaced by the baseline score there. Because this changes the overall decision rule, any conditional-execution reporting is treated as a distinct algorithmic variant and is accompanied by leakage controls (same-window vs. disjoint-window vs. randomized proxy masks).
- **Short-ensemble band discretization sanity.** For extremely short ensembles (e.g. $N \in \{5, 9, 17\}$), the coarse frequency resolution $\Delta f = \text{PRF}/N$ can cause requested bands to collapse or overlap after discretization. If the discretized flow and alias bands are empty, DC-only, or overlapping (violating orthogonality in the DFT-bin construction), we force $\phi_i \in \mathcal{R}_0$ (uninformative prior; $w \equiv 1$) with an explicit reason.
- **Sample-support and tail-sizing constraints.** Tail-association metrics are defined on a background-tail subset and become unstable when sample support is insufficient. We use a size-aware background-tail fraction $q^{\star}_{\text{tail}} = \max(q_{\text{tail}}, n_{\text{tail,min}}/n_{\text{bg}})$ and refuse to enter $\mathcal{R}_2$ (actionable prior) when tail sizes or proxy supports fall below deterministic floors.
- **Proxy-mask circularity control.** Proxy masks are used only for label-free band-energy aggregation, candidate/protected-set definitions, and conditional execution. Quantitative ROC claims are restricted to regimes with independent H_1/H_0 labels (simulator truth or B-mode structural geometry).

For auditability, we log the selected feature-space region and explicit reason labels so that "KA off" ($w \equiv 1$) is an expected success mode under flat or non-actionable evidence. Appendix A provides modality-agnostic sufficient conditions and additional sentinels used to interpret when the KA prior should be active or inert.

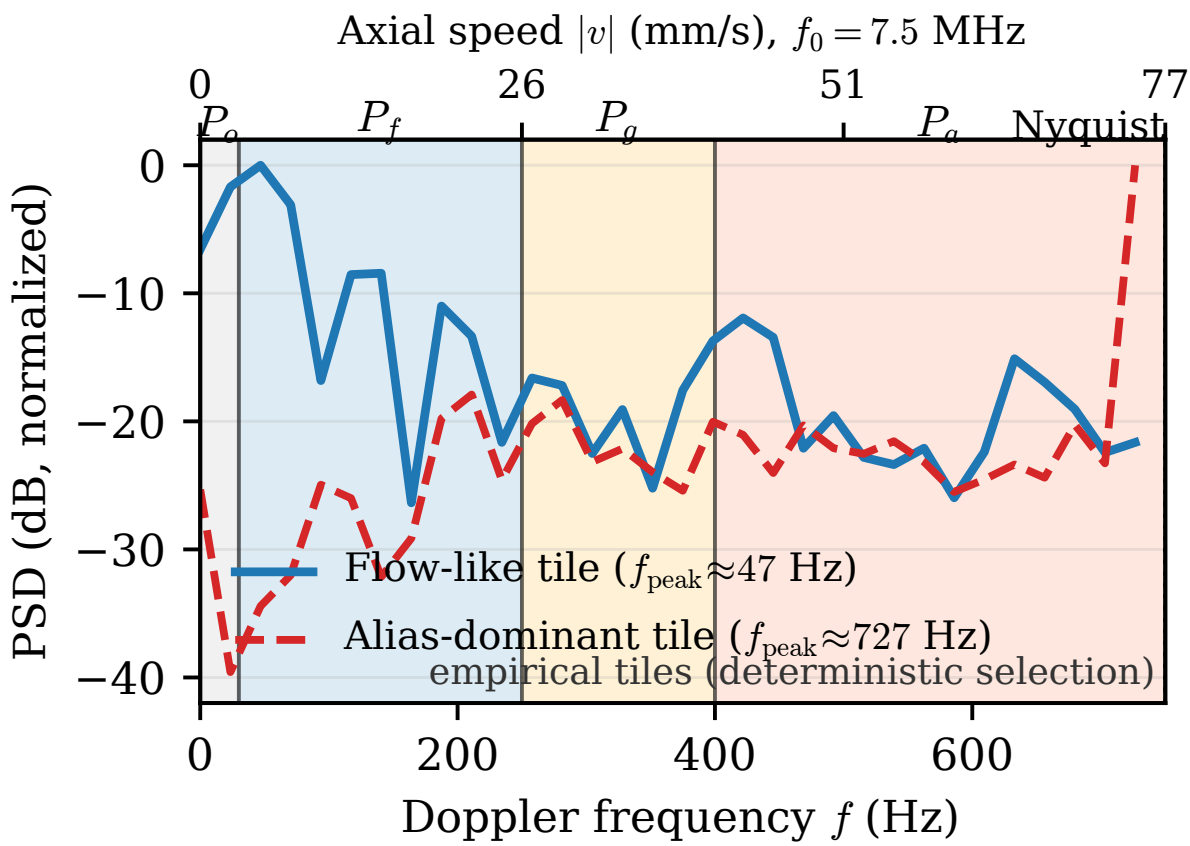

Figure 2: Doppler power spectral density (PSD) partitioning and kinematic band geometry (empirical example). Curves show one-sided PSDs of the tile-mean slow-time signal on a Brain-AliasContract k-Wave pilot window (PRF = 1500 Hz): a flow-like tile (peak in P_f) and an alias-dominant tile (peak in P_a). The Nyquist interval is deterministically partitioned into DC/high-amplitude clutter (P_o), microvascular flow (P_f), a transition guard band (P_g), and aliased high-velocity artifacts or vibration (P_a); the feature vector ϕ_i aggregates band energies in these regimes to drive the spatial-spectral prior.

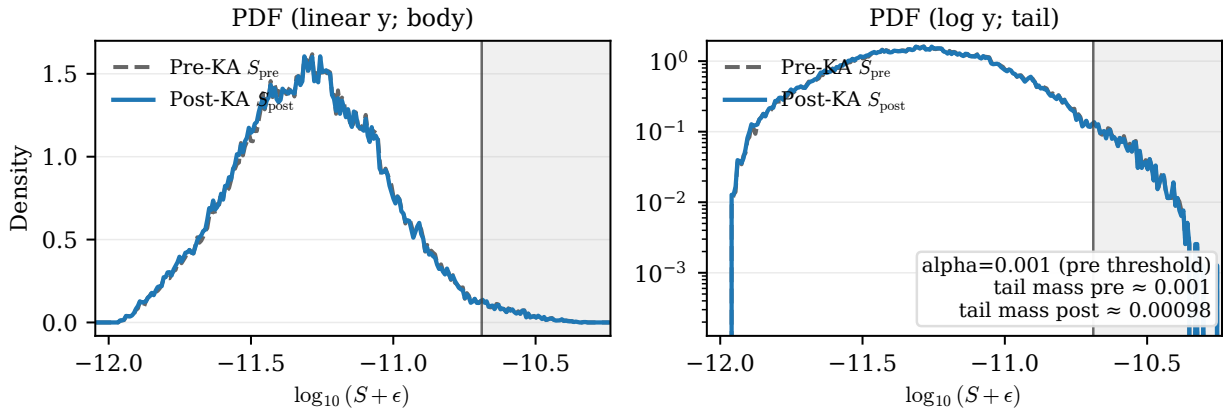

Figure 3: Empirical statistical effect of the strictly bounded, shrink-only penalty on a heavy-tailed background-proxy score distribution (Twinkling calculi artifact sequence). The uninformed matched-subspace statistic S_{pre} exhibits a heavy right tail characteristic of structured artifacts. Applying the non-increasing penalty $\gamma(\phi_i) \geq 1$ yields the post-regularization score $S_{\text{post}} = S_{\text{pre}}/\gamma(\phi_i)$, compressing the background tail while preserving a protected high-confidence set (shrink-only and protected-set invariance). The vertical line indicates the pre-KA threshold τ_α chosen so that $\mathbb{P}\{S_{\text{pre}} \geq \tau_\alpha \mid \text{bg proxy}\} = \alpha$ (here $\alpha = 10^{-3}$); the shaded region highlights the corresponding right tail, and the post-KA tail mass is evaluated at the same τ_α.

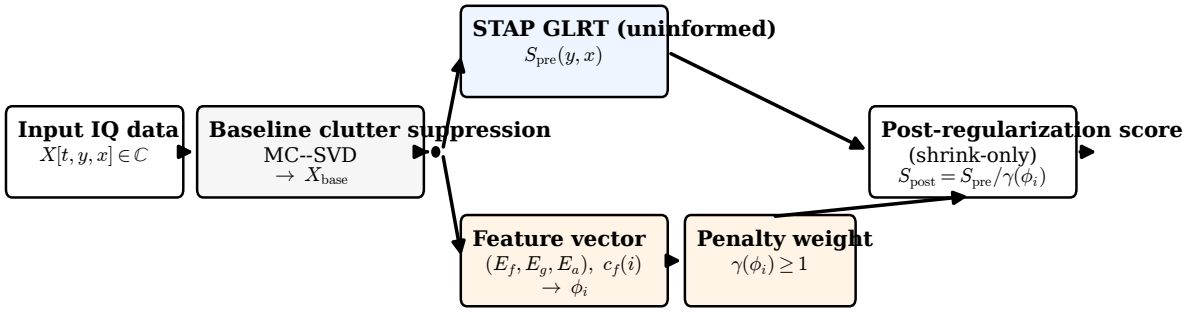

Figure 4: System block diagram for the proposed Knowledge-Aided Space-Time Adaptive Processing (KA–STAP) architecture. Beamformed kilohertz data initially undergoes baseline motion-compensated clutter filtering (MC–SVD), yielding a residual cube X_{base}. The residual is processed via two parallel branches: (top) the uninformed STAP core computes the baseline matched-subspace statistic $S_{\text{pre}}(y,x)$; (bottom) the spatial-spectral feature extractor evaluates localized band energies to construct the feature vector ϕ_i, which governs a bounded penalty $\gamma(\phi_i) \geq 1$. The branches recombine to form the strictly bounded post-regularization score $S_{\text{post}}(y,x) = S_{\text{pre}}(y,x)/\gamma(\phi_i)$ (shrink-only).

4 Experimental Design

4.1 Fixed-profile protocol and operating-point reporting

All hyperparameters (tile geometry, aperture, band definitions, conditioning targets, and prior thresholds) are fixed per regime before reporting results. Detection is evaluated in the right tail at fixed FPR targets:

$$\mathbb{P}\{S \geq \tau_\alpha \mid H_0\} = \alpha, \qquad \mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}.$$

For each experiment, we report the supported floor $fpr_{\min} = 1/n_{\mathrm{neg}}$ and bootstrap uncertainty at the natural unit of variation (windows or frames, depending on regime). These choices enforce a strict low-FPR operating discipline and avoid over-interpreting unsupported tail points.

4.2 Integrity checks and deterministic sentinels

This work is a reproducible processing pipeline (decode → baseline → STAP score → spatial-spectral features → optional KA regularization). Several practical pathologies can invalidate tail-rate reporting or spuriously activate the KA layer (e.g. missing acquisition metadata, coarse short-ensemble band discretization, proxy-mask circularity, or compute-gating leakage). We therefore enforce deterministic, label-free integrity checks that either (i) record provenance and proceed conservatively or (ii) force $\phi_i \in \mathcal{R}_0$ (uninformative prior; $w \equiv 1$) with an explicit reason label. A concise checklist (and where each sentinel is logged) is provided in Appendix C.

4.3 Regimes and claim scope

Claims are stratified by regime to separate quantitative detection efficacy from label-free instrumentation.

- **Brain-k-Wave simulations (labeled).** Across Brain-OpenSkull / Brain-AliasContract / Brain-SkullOR regimes, the fixed profile yields a nontrivial right-tail shoulder in the primary detector scores (baseline power Doppler and matched-subspace STAP; Notation), with the strongest low-FPR gains in the alias-augmented Brain-AliasContract and skull/OR-inspired Brain-SkullOR regimes (Table 4).
- **Phantom acquisitions (structurally labeled Gammex flow phantom).** We decode raw beamformer outputs, derive B-mode-only tube masks (flow/background), benchmark conventional short-ensemble Doppler baselines, and report structural-label low-FPR operating points and within-ensemble motion ladders.
- **ULM Zenodo 7883227 (real kHz IQ).** We report label-free motion-robustness curves on beamformed IQ blocks using controlled rigid/nonrigid motion injection, with fairness alignment and degeneracy checks (no task ROC).
- **Supplementary analyses.** Additional label-free audits (e.g. Shin Fig3 real IQ; PD-only whole-brain fUS) and diagnostic panels are provided in Appendix B.

5 Results

Results roadmap. We first report labeled Brain-* simulation results at ultra-low FPR, then structurally labeled Gammex flow phantom results (including motion ladders), and finally label-

Regime	Data type	Claim scope
Brain-* (k-Wave)	simulated IQ	low-FPR ROC and latency (simulation)
Gammex flow phantom	real Doppler IQ (phantoms)	structural-label ROC + motion ladders (phantom; not clinical)
ULM Zenodo 7883227	real kHz IQ	motion robustness proxies (no task ROC)
Supplementary datasets	real IQ and PD-only	descriptive/audit analyses (no task ROC)

Table 2: Claims discipline by regime. Quantitative detection performance is reported on labeled Brain-* simulations and on structurally labeled phantom data (Gammex flow phantom). ULM 7883227 is used for label-free motion-robustness evidence on real kHz IQ. Additional datasets are used only for descriptive/audit analyses and are reported in Appendix B.

Regime (sampling)	Baseline	STAP core (fixed profile)	Doppler bands (Hz)
Brain-* (k-Wave; PRF 1500; 64-frame windows)	MC–SVD (energy fraction $e = 0.90$); RPCA; HOSVD (baselines scored by PD on residual)	tiles 8×8, stride 3; $L_t = 8$; robust covariance (Tyler-type) with diagonal loading $\lambda = 0.07$; matched-subspace flow-band score	$P_f = [30, 250]$, guard $[250, 400]$, $P_a = [400, 750]$
Gammex flow phantom (linear17; PRF 2500; $N = 17$ shots; pooled over cine frames)	baseline SVD band-pass keep $[2, 17]$; conventional short-ensemble comparators (PD/log-PD/Kasai)	tiles 8×8; $L_t = 16$; Tyler-type covariance with $\lambda = 0.07$; stride 6 (along) / 4 (across); matched-subspace flow-band score	$P_f = [150, 450]$, guard $[450, 700]$, $P_a = [700, 1200]$
ULM Zenodo 7883227 (PRF 1000; 128-frame blocks)	MC–SVD baseline with energy fraction $e = 0.975$ frozen by a one-time label-free sweep	tiles 8×8, stride 3; $L_t = 64$; SCM covariance with $\lambda = 0.07$; matched-subspace flow-band score	$P_f = [10, 150]$, guard $[150, 200]$, $P_a = [200, 500]$

Table 3: Frozen operating profiles for the main regimes. All settings are fixed per regime prior to reporting results; exact commands and exported metadata are provided in Appendix L.

free motion-robustness evidence on real kHz IQ (ULM 7883227). Additional dataset audits and diagnostic panels are provided in Appendix B.

Scope and low-FPR operating protocol. Quantitative detection performance is reported on labeled Brain-* simulations and on structurally labeled phantom regimes (Gammex flow phantom sequences; labels derived from B-mode geometry) using the strict low-FPR operating protocol of Section 4.1 at ultra-low FPR. Real kHz IQ datasets are used for instrumentation (band-energy integrity checks and label-free motion-robustness proxies), not task ROC. Conditional execution is a decision-rule variant (compute gating) that uses a flow-proxy mask; leakage ablations and additional diagnostics are reported in Appendix B.

5.1 Brain-k-Wave regimes (labeled): low-FPR ROC and latency

We report labeled low-FPR operating points on Brain-* k-Wave simulations using right-tail detector scores (Notation): a baseline power Doppler score and the proposed matched-subspace STAP score. Baseline methods differ only in clutter suppression (MC–SVD, RPCA, HOSVD) and are scored using the same power Doppler statistic; STAP is evaluated on top of the MC–SVD residual. Across Brain-* regimes, these global baselines degrade sharply in the ultra-low-FPR tail, while the localized STAP score retains a distinct, non-saturated shoulder, including in the SkullOR regime where alias

Method / score	Brain-OpenSkull			Brain-AliasContract			Brain-SkullOR		
	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}
MC–SVD (power Doppler)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
RPCA (power Doppler)	0.0015 (0.0000,0.0046)	0.0031 (0.0000,0.0062)	0.0062 (0.0000,0.0231)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
HOSVD (power Doppler)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0015)	0.0062 (0.0031,0.0073)	0.0015 (0.0000,0.0015)	0.0015 (0.0015,0.0046)	0.0077 (0.0069,0.0092)	0.0015 (0.0000,0.0015)	0.0015 (0.0000,0.0015)	0.0031 (0.0015,0.0031)
STAP (pre-KA matched-subspace) on MC–SVD residual	0.0223 (0.0031,0.3922)	0.2107 (0.0077,0.4610)	0.2834 (0.1496,0.5302)	0.3499 (0.2649,0.3756)	0.4033 (0.3283,0.4560)	0.5063 (0.4383,0.5113)	0.3791 (0.2622,0.3987)	0.4460 (0.3230,0.4648)	0.5102 (0.4364,0.5456)

Table 4: Brain-* k-Wave low-FPR operating points using right-tail detector scores: baseline power Doppler and matched-subspace STAP. Each regime uses five non-overlapping 64-frame windows (offsets 0/64/128/192/256); each window has $n_{\text{neg}} = 50199$ ($fpr_{\min} \approx 1.99 \times 10^{-5}$). Entries report median (IQR) TPR across windows at each target FPR. Baseline methods differ only in clutter suppression (MC–SVD, RPCA, HOSVD); STAP is applied on top of MC–SVD. Full per-window outputs are provided in Appendix L.

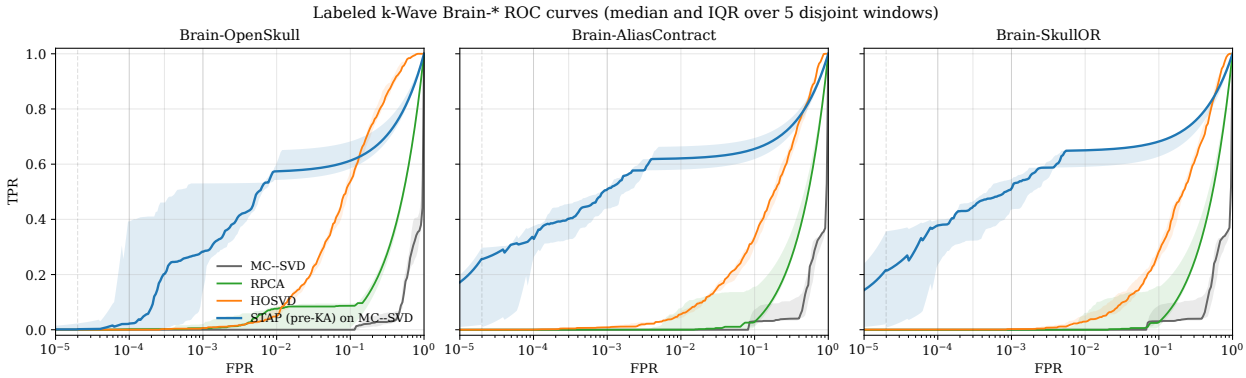

Figure 5: Brain-* k-Wave labeled regimes: ROC-style curves (TPR vs FPR on a log-x axis) under the same 64-frame window protocol used for the Brain-* low-FPR operating-point reporting (Table 4). Curves show the median and interquartile range over five disjoint windows. The vertical dashed line indicates the supported floor $fpr_{\min} = 1/n_{\text{neg}}$.

deliberately leaks into the positive class. Table 4 summarizes low-FPR operating points for representative pilot realizations over five non-overlapping 64-frame windows (offsets 0/64/128/192/256); each window has $n_{\text{neg}} = 50199$ background pixels ($fpr_{\min} = 1/n_{\text{neg}} \approx 1.99 \times 10^{-5}$), so FPR targets down to 10^{-4} are empirically supported. Numbers are medians (IQR) over windows and are computed by choosing τ per method per window on that window's negative set to match the target FPR, then evaluating TPR on the corresponding positive set. The wider IQR at $\alpha = 10^{-4}$ in Brain-OpenSkull reflects window-to-window nonstationarity and finite tail support under a fixed, local-covariance profile (shoulder windows versus near-floor windows). Per-window thresholds and realized FPRs are provided in Appendix L. A cross-window threshold-transfer audit (calibrate τ_α on one window and evaluate on a disjoint window) is reported in Appendix B.2.

Baseline fairness and frozen settings. All baseline hyperparameters are fixed prior to reporting results. For Brain-* pilots, the MC–SVD energy-fraction cutoff is selected by a one-time sweep and then frozen at $e = 0.90$ for all reported windows (sweep report in `reports/brain_mcsvd_energy_sweep_seed1.csv` / `reports/brain_mcsvd_energy_sweep_seed1.json`). RPCA and HOSVD baselines use standard, reproducible settings documented in Appendix L. All baselines are scored using the same right-tail power Doppler statistic so that differences isolate the clutter-suppression stage rather than the scoring rule.

Latency and conditional execution. We also report representative runtimes for baseline clutter suppression and STAP under a conservative fixed operating profile (tile geometry, slow-time aperture L_t, and covariance-conditioning targets fixed a priori). Conditional execution reduces wall-clock time by skipping tiles with zero overlap with a proxy flow mask and falling back to the baseline there, but it is a *different* decision rule and can change operating characteristics; we therefore treat it as a separate algorithm variant and require explicit leakage controls when reporting conditional-execution performance.

Regime	MC–SVD (s)	STAP full (s)	STAP conditional (s)
Brain-OpenSkull	≈ 1.2	≈ 61	≈ 4.7
Brain-AliasContract	≈ 1.0	≈ 45	≈ 3.9
Brain-SkullOR	≈ 1.2	≈ 56	≈ 3.2

Table 5: Representative latency summary for the conservative reference profile on Brain-* k-Wave regimes (per 64-frame window). Conditional execution is a compute-gating variant that uses a proxy mask and changes the decision rule; it is reported as a deployment optimization requiring leakage ablations rather than as a cost-free speedup. All runs are CPU-only (`stap_device=cpu`); times are approximate wall-clock durations from runtime telemetry.

A leakage ablation confirms that conditional execution can inflate apparent performance under same-window proxy gating, while leakage-controlled gating yields a small TPR decrease (order 10^{-3}) at the target FPRs; we therefore treat conditional execution as a distinct decision rule rather than a cost-free speedup. Table 6 summarizes the effect on Brain-OpenSkull pilots; full bootstrap CIs and details are provided in Appendix B.

5.2 Phantom acquisitions (Gammex flow phantom): structural ROC and within-ensemble motion ladders

Dataset and acquisition parameters. We use the open-access Doppler Twinkling Artifact database [35], which provides complex beamformer output from phantom acquisitions on a clinical scanner. Each cine frame contains a contemporaneous B-mode image and a short-ensemble color-flow (CFM) slow-time block with $N \in \{5, 9, 17\}$ pulses. Because Doppler PRF is not reliably available in the provided metadata for these acquisitions, all Twinkling experiments treat PRF as an explicit experimental assumption frozen per sequence family (listed in Appendix E) and enforce band-discretization sanity checks that force $\phi_i \in \mathcal{R}_0$ when discretized Pf/Pa bands are empty or overlapping (Section 3). Decoding/QC details are provided in Appendix E (PRF overrides and decoder sanity); reproduction commands and manifests are provided in Appendix L.

B-mode-only structural tube masks (independent H_1/H_0). For the Gammex flow phantom sequences we derive evaluation masks from B-mode structure only (no Doppler/STAP outputs). Using acquisition geometry metadata that specifies the CFM region-of-interest (ROI) within the B-mode grid (beam and sample offsets and any beam subsampling), we map the B-mode ROI onto the CFM grid and segment the tube lumen on a median log-envelope reference image. This yields (i) a conservative lumen (flow) mask, eroded to avoid wall pixels and (ii) a background mask that excludes a guard band around the lumen and bright reflectors. Mask provenance and QC metrics

Conditional gating mask	ΔTPR@10^{-4} ($\times 10^{-3}$)	ΔTPR@3×10^{-4} ($\times 10^{-3}$)	ΔTPR@10^{-3} ($\times 10^{-3}$)
Same-window proxy mask	+0.31	+0.81	+3.05
Disjoint-window proxy mask	−3.05	−2.90	−1.74
Random mask (matched coverage)	−3.05	−2.97	−2.36

Table 6: Conditional-execution leakage ablation on Brain-OpenSkull pilots (conditional minus full STAP; five independent realizations). Entries report mean ΔTPR at fixed FPR targets; full bootstrap CIs and details are provided in Appendix B.

are recorded with each experiment; Figure 6 shows representative overlays. These structural labels support a defensible phantom ROC.

Frozen short-ensemble Doppler profile and baseline fairness scores. For the linear-probe Gammex sequences (PRF = 2500 Hz, $N = 17$ shots), we freeze a short-ensemble operating profile: baseline SVD band-pass keep range $[2, 17]$, STAP aperture $L_t = 16$, tiles 8×8, a Tyler-type robust covariance estimator, and diagonal loading $\lambda = 0.07$. To keep the profile auditable while avoiding sample-support limitations on small ROIs and excessive overlap/mixing on large ROIs, we use a deterministic geometry-only stride policy: choose the largest tile stride ≤ 6 such that the implied tile count is ≥ 500. On the two Gammex views this yields stride 6 (along; large ROI) and stride 4 (across; small ROI). Band definitions are frozen to $P_f = [150, 450]$ Hz and $P_a = [700, 1200]$ Hz. For baseline context on short ensembles, we also report conventional comparators derived from the baseline-filtered cube (power Doppler, log power Doppler, and Kasai lag-1 power). Reproducibility details are provided in Appendix L.

Structural-label low-FPR ROC (Gammex). We pool pixel scores across cine frames and report low-FPR operating points using right-tail thresholding on pooled background pixels, then evaluate TPR on pooled lumen pixels. In these short-ensemble Doppler acquisitions, the background set includes strong twinkling/artifact pixels whose right-tail scores dominate conventional baselines; consequently, low-FPR thresholds at $\alpha \leq 10^{-3}$ can lie above essentially all lumen pixels, yielding numerically zero baseline TPR at the reported operating points. Figure 9 visualizes this effect for a representative baseline comparator. To avoid treating pooled pixels as independent samples, we report 95% confidence intervals via a frame-level bootstrap: we resample cine frames as the unit of variation while holding the pooled threshold fixed. Table 7 summarizes the resulting structural-label operating points for the two Gammex views under their frozen profiles; full pooled summaries (including baseline-score variants and realized thresholds) are provided in Appendix L. Because STAP is evaluated on the same baseline-filtered residual cube, the strong STAP TPR at strict FPR rules out the interpretation that the baseline filter simply removed lumen flow; rather, the failure mode is heavy-tailed background scores dominating the right-tail threshold (see Figure 9).

Within-ensemble motion ladders (labeled, real acquisition). To obtain a labeled motion-robustness curve on a real acquisition without external motion ground truth, we inject synthetic *within-ensemble* motion directly into the CFM cube: for each cine frame we apply a per-shot rigid translation $(\Delta y_t, \Delta x_t)$ to the N CFM shots prior to baseline and STAP, with motion kinds `rw` (random walk) and `step`. Because the structural tube masks are derived from B-mode and fixed per frame, this produces a labeled "motion ladder" where the only change is within-ensemble

	Along (linear17 @ PRF 2500)			Across (linear17 @ PRF 2500)		
Method / score	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}
Baseline (power Doppler)	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]
Baseline (log power Doppler)	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]
Baseline (Kasai lag-1 power)	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]
STAP (pre-KA matched-subspace)	0.634 [0.628,0.640]	0.803 [0.798,0.808]	0.946 [0.943,0.948]	0.846 [0.842,0.850]	0.891 [0.888,0.893]	0.938 [0.936,0.940]

Table 7: Gammex flow phantom structural-label operating points using B-mode-derived structural masks. All scores are right-tail (higher indicates more flow evidence). Thresholds are chosen per method on pooled background pixels to match each target FPR and evaluated on pooled lumen pixels. Brackets show 95% frame-bootstrap CIs ($n = 2000$ resamples over cine frames) at the fixed pooled threshold. Pooled background counts are $n_{\mathrm{bg}} = 3.85 \times 10^6$ for the along view ($fpr_{\min} \approx 2.6\times10^{-7}$) and $n_{\mathrm{bg}} = 1.97\times10^6$ for the across view ($fpr_{\min} \approx 5.1\times10^{-7}$). Reproducibility details are provided in Appendix L.

corruption. We generate motion-corrupted ensembles (85 cine frames; amplitudes 0–2 px) and aggregate using frame-level bootstraps for CIs together with an explicit amp=0 equivalence check against the non-motion reference. Figure 10 shows the resulting curves; reproducibility details are provided in Appendix L.

Supplementary KA hygiene (artifact-heavy calculi). Additional KA hygiene analyses on artifact-heavy calculi sequences (region/reason summaries and representative pre/post tail examples) are provided in Appendix B.

5.3 ULM Zenodo 7883227 (real kHz IQ): baseline calibration and nonrigid motion robustness

Dataset and loader. We use a real kHz IQ dataset released for ultrasound localization microscopy benchmarking [33, 34] and distributed as HDF5 blocks with accompanying acquisition metadata. Each block contains a dataset `iq` of shape $(2, H, W, T) = (2, 256, 180, 800)$ float32 representing I/Q over $T = 800$ slow-time frames. We load blocks on demand and return complex IQ as a cube of shape (T, H, W) (reproduction details in Appendix L).

One-time label-free baseline selection. For ULM we select a single MC–SVD energy-fraction cutoff by a one-time label-free sweep on blocks 1–3 over frames 0:128, and then freeze it for all subsequent ULM motion experiments. The sweep uses baseline-only runs and relies on label-free integrity and stability criteria (no task labels); it yields a frozen energy fraction $e = 0.975$ (full sweep details in Appendix L).

Nonrigid motion injection and fairness alignment ablation. We evaluate motion robustness by injecting controlled motion directly into the beamformed IQ cube and re-running baseline+STAP under a frozen ULM profile (PRF = 1000 Hz; $L_t = 64$; tiles 8×8; stride 3; SCM covariance; diagonal loading $\lambda = 0.07$; baseline MC–SVD with frozen $e = 0.975$). Because motion is injected at the beamformed IQ stage, this test stresses post-beamforming robustness to residual motion mismatch; it does not model channel-domain decorrelation or skull-induced phase-aberration physics. Motion ladders span two nonrigid families: `brainlike` (rigid random-walk translation plus a smooth, depth-decaying elastic warp with temporally correlated coefficients and micro-jitter) and `elastic` (nonrigid-only; rigid and micro-jitter disabled). For each amplitude

we compute label-free robustness endpoints on the baseline score S_{base} and the matched-subspace STAP score $S_{\text{stap,pre}}$ relative to the no-motion reference under three *common* alignment modes (none / common B-mode / injected GT proxy) plus a diagnostic per-method "self" alignment. We additionally report degeneracy checks (std ratios, top-k overlap, and tail quantile ratios) to rule out "stability by collapse." Figure 11 shows the main `brainlike` sweep; additional motion variants are provided in Appendix B.

Endpoint definition. For a score map $S_{\text{meth}}(a)$ at motion amplitude a and the corresponding no-motion reference $S_{\text{meth}}(0)$ (same block), we compute $\rho_{\text{meth}}(a)$ as the Pearson correlation between the aligned maps on a central crop Ω_a that excludes a margin of $\lceil a \rceil + 2$ pixels to avoid boundary artifacts from warping. Correlations are computed directly on the exported right-tail scores (no log transform) and use the same crop/alignment for baseline and STAP in the "common" alignment modes.

Representative quantitative endpoints (ULM). At a 3 px motion amplitude under the injected GT alignment proxy, the median correlation on the baseline score S_{base} drops to ≈ 0.20 (`brainlike`), while the STAP score $S_{\text{stap,pre}}$ remains ≈ 0.75 (medians over blocks 1–3; full tabular summaries and additional motion variants are provided in Appendix B). Degeneracy checks confirm that this stability is not explained by map collapse: the median std ratio and 0.999-quantile ratio of $S_{\text{stap,pre}}$ remain nontrivial across amplitudes, and top-k overlap decays gradually rather than saturating to trivial values.

Supplementary datasets and diagnostics. Additional label-free audits (e.g. Shin Fig3 real IQ; PD-only whole-brain fUS) and diagnostic panels (e.g. conditional-execution leakage ablations, KA hygiene panels) are provided in Appendix B.

6 Discussion

A) Robustness in the ultra-low-FPR tail

This work adapts space–time adaptive processing (STAP), a foundational approach in radar target detection under heavy clutter, to functional ultrasound (fUS) slow-time flow detection. Under the strict low-FPR operating protocol (Section 4.1), we find that common global or semi-global clutter suppressors (MC–SVD, RPCA, HOSVD) can lose sensitivity in the extreme right tail, while the proposed tile-based matched-subspace STAP score preserves a distinct, non-saturated shoulder at ultra-low FPR. This behavior is demonstrated on labeled k-Wave simulations and structurally labeled *in vitro* phantoms, where flow/background masks are defined independently from Doppler/STAP outputs via simulator truth or B-mode geometry. The within-ensemble motion ladders on phantom acquisitions further indicate that STAP retains detection stability under controlled per-shot motion corruption, consistent with its reliance on localized covariance rather than global low-rank structure.

B) Knowledge-aided prior as a fail-safe

The optional knowledge-aided (KA) layer is designed as a conservative safety regularizer rather than an unconstrained signal amplifier. It acts as a shrink-only prior penalty with an auditable (region, reason) output: when spatial-spectral evidence is flat or non-actionable, the prior defaults

to $\mathcal{R}_0$ (uninformative; $w \equiv 1$) and leaves the pre-KA detector score unchanged. When the prior lies in $\mathcal{R}_1/\mathcal{R}_2$, it enforces protected-set invariance and applies only non-increasing shrink on high-risk regions, so that "KA does nothing" is an intended success mode out of regime. This posture prioritizes transparent regime diagnostics and avoids score inflation in regimes where spectral evidence cannot justify aggressive intervention.

C) Data limitations and the phase-history barrier

A key limitation in evaluating advanced Doppler-side covariance methods is field-wide data availability: many open-access *in vivo* fUS datasets are temporally downsampled, post-filtered, or released only as power-Doppler equivalents, which can irreversibly destroy the high-rate complex phase history needed for spatiotemporal covariance estimation and band-energy integrity checks. Accordingly, while we report real-data instrumentation (band-energy integrity, prior-regime behavior, and label-free stability under controlled motion corruption), comprehensive *in vivo* ROC evaluation of Doppler-side STAP/KA requires high-PRF raw IQ acquisitions prior to proprietary clutter suppression, with reliable metadata and task/landmark structure to define independent evaluation labels.

7 Conclusion and future work

This study establishes a deterministic, auditable KA-STAP framework for fUS slow-time flow detection under a tune-once-then-freeze discipline. Using a strict low-FPR operating protocol and independent structural labels where available, we show that localized matched-subspace STAP can preserve separability in regimes where conventional low-rank baselines degrade in the extreme tail, while the KA spatial-spectral prior provides shrink-only, protected-set-preserving risk control with explicit out-of-regime behavior.

Future work is to (i) evaluate Doppler-side STAP/KA on pre-registered, labeled raw-IQ *in vivo* datasets that preserve phase history and support independent task/landmark labels, (ii) reduce latency via hardware-aware implementations of the batched covariance and band-energy kernels toward real-time deployment, and (iii) explore compact, frozen learned regularizers only when they can be constrained to preserve the shrink-only auditability and low-FPR reporting guarantees established here.

Data and code availability

All datasets used in this work are either publicly available (as cited) or derived deterministically from those sources. A complete reproduction manifest (commands, configurations, and artifact naming) is provided in Appendix L.

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A KA Prior (Modality–Agnostic)

In this part we describe regime conditions and operator-level sufficient structure for a knowledge-aided (KA) temporal covariance transform layered on top of a baseline slow-time detector. The analysis is formulated in terms of abstract band projectors (P_o, P_f, P_g, P_a), alias features, and score distributions, and does not depend on any particular imaging modality. Only in later sections do we instantiate these conditions for functional ultrasound (fUS) brain imaging with concrete band choices, acquisition parameters, and baselines.

Label group	Plain-English meaning (what it checks)
R–*	Regime preconditions: do the bands exist, do risk features vary, and is there tail headroom worth regularizing?
BG–*	Band geometry: after discretization, are Pf/guard/Pa non-empty and non-overlapping for the chosen (PRF, T)?
OC–*	Operator/covariance constraints: KA can only apply bounded, bandwise structure without uncontrolled mixing or conditioning pathologies.
G–*	Gating/localization: KA is allowed to act only where artifacts are likely and positives are unlikely (protected-set invariance).
Safety–*	Safety guards: if any check indicates score inflation or loss of invariance, KA is forced off.

Table 8: Appendix label groups in plain English. Failing any condition triggers a conservative fallback ($w = 1$), which is an intended success mode out of regime.

How to read this appendix (plain-English map)

This appendix is a *contract*, not a hyperparameter sweep: each condition is a deterministic integrity check that either (i) permits the KA layer to act conservatively or (ii) forces the safe default $w \equiv 1$ (KA off). The labels below are grouped by purpose.

A.1 Regime Preconditions (Data–Level, Non–Negotiable)

The most important empirical lesson is: even if the KA operator Q and band design (P_f, P_a) are perfect in an abstract sense, *uplift requires the data themselves to be in a compatible regime*. This section codifies that.

We write $S(f)$ for the (multi–taper) PSD of a baseline–whitened tile, and define

$$E_f := \sum_{f \in \mathcal{B}_f} S(f), \quad E_a := \sum_{f \in \mathcal{B}_a} S(f), \quad E_g := \sum_{f \in \mathcal{B}_g} S(f),$$

with $\mathcal{B}_f, \mathcal{B}_a, \mathcal{B}_g$ the flow, alias, and guard bands in frequency.

A.2 R–1: Band Occupancy Separation

We require that the fixed bands $(\mathcal{B}_f, \mathcal{B}_a)$ are physically meaningful for the current PRF, slow–time length T, and DPSS parameters, in the sense that label statistics show actual occupancy.

On a held–out labeled sample:

$$\mathbb{P}\{f_{\text{peak}} \in \mathcal{B}_f \mid H_1\} \geq p_{f,\min}, \qquad p_{f,\min} \in [0.5, 0.7], \tag{1}$$

$$\mathbb{P}\{f_{\text{peak}} \in \mathcal{B}_a \mid H_0\} \geq p_{a,\min}, \qquad p_{a,\min} \in [0.2, 0.3], \tag{2}$$

where f_{peak} is the frequency at which $S(f)$ attains its maximum and H_1/H_0 denote flow/background tiles.

Interpretation: a majority of flow tiles have their dominant PSD peak *inside* the flow band, and a nontrivial fraction of background tiles have their dominant peak in the alias band. This ensures that the band projectors (P_f, P_a) actually correspond to distinct physical processes.

Implementation in Brain–* fixed profiles. For the main Brain-* simulation regimes, the raw k-Wave RF tensor $x(t, a, e)$ (angle index a, element index e) lives on an angle and element grid that does not align pixelwise with the STAP PD masks ($N_y \times N_x$). We therefore implement R–1 using the whitened multi–taper PSD and band-ratio feature statistics computed on the *same* spatial grid as the exported PD scoring maps (the right-tail score S_{PD} stored as `score_pd_*.npy`). We take f_{peak} from the whitened PSD peaks used in the band-ratio recorder and report the fraction of H_1 tiles whose peak lies in $\mathcal{B}_f$ and the fraction of H_0 tiles whose peak lies in $\mathcal{B}_a$ from those tilewise peak statistics. When the RF tensor is spatially aligned with the masks (e.g. 1D pilots), we instead use a direct FFT on $x(t, \cdot, \cdot)$ as in the definition above.

A.3 R–2: Alias Score Class Separation

We generalize the alias metric used for localization.

Alias score. Let $m_{\mathrm{alias}}(z)$ be any scalar score derived from the baseline PSD and/or covariance satisfying:

- m_{alias} increases when E_a grows relative to E_f,
- m_{alias} is larger when Pa energy is more coherent or concentrated (e.g. a strong tone),
- m_{alias} is penalized when guard/edge energy E_g is large.

Examples (not exhaustive):

- Log band ratio: $m_{\mathrm{alias}}(z) = \log \frac{E_a+\epsilon}{E_f+\epsilon}$,
- Combination of band fractions: $m_{\mathrm{alias}}(z) = \frac{E_a}{E_f+E_a+E_g} \cdot c_a$ where c_a is an alias–band coherence metric (e.g. leading eigenvalue fraction of a band–restricted covariance),
- Peak–location based scores (peak in Pa, high tone prominence).

For KA to be meaningfully guided by alias evidence, we require *class–differential separation* of m_{alias}:

$$\mathrm{median}(m_{\mathrm{alias}} \mid H_0) - \mathrm{median}(m_{\mathrm{alias}} \mid H_1) \;\geq\; \Delta_{\min}, \tag{3}$$

with $\Delta_{\min}$ set so that alias dominance differs by at least a factor $\approx e^{\Delta_{\min}}$ in expectation (for log ratio, $\Delta_{\min} \approx 0.3$ corresponds to a $\approx 35\%$ relative difference in E_a/E_f medians).

If (3) fails (alias score almost identical on H_0 and H_1), the alias-localization rule (Appendix J.8) has no reliable signal and KA uplift cannot be expected.

A.4 R–3: Baseline Headroom

Let S^- be the baseline score distribution on negatives, and $\mathrm{pAUC}_{\mathrm{base}}(\alpha)$ be the partial AUC up to FPR α.

We require:

- Sufficient *quantile density* at the target FPR α (as already assumed implicitly in SL–2): the density $f_-(t_\alpha)$ at the α–quantile t_α is bounded away from 0.
- Baseline not already numerically saturated: for a chosen $\alpha_{\max}$ (e.g. 10^{-3}),

$$\mathrm{pAUC}_{\mathrm{base}}(\alpha \leq \alpha_{\max}) \leq \mathrm{pAUC}_{\max} - \varepsilon_{\mathrm{AUC}}, \qquad \varepsilon_{\mathrm{AUC}} > 0,$$

  where $\mathrm{pAUC}_{\max}$ is the ideal partial AUC under perfect separation.

If the baseline already achieves near–maximal pAUC at the FPR of interest, even a perfect KA transform cannot provide significant gains; in this regime KA can serve as a regularizer or safety mechanism, but not as a source of measurable uplift.

A.5 Operator–Level Sufficient Structure (Non–Negotiable Core)

This section is unchanged at the core: it describes conditions on Q alone under which KA behaves like a band–wise scalar rescaling of scores, up to small perturbations.

A.6 OC–1: Approximate Band Commutation

We require Q to be nearly block–diagonal with respect to (P_o, P_f, P_g, P_a):

$$\|[Q, P_f]\|_2 \le \varepsilon_{\text{mix}}, \qquad \|[Q, P_a]\|_2 \le \varepsilon_{\text{mix}},$$

for some small $\varepsilon_{\text{mix}} \in [0, 1)$ (typical target ≤ 0.05).

Interpretation: Q preserves band subspaces up to small mixing; it does not rotate energy from Pf into Pa or vice versa in any substantial way. This rules out the failure mode where KA accidentally feeds alias into the flow numerator.

Runtime label–free check. Let $\widehat{Q} := R_\beta^{-1/2}\widehat{R}^{1/2}$ be the realized KA transform. Estimate

$$\widehat{\varepsilon}_{\text{mix}} := \max\left\{ \frac{\left\|P_f\widehat{Q}P_a\right\|_F}{\left\|P_f\widehat{Q}P_f\right\|_F}, \frac{\left\|P_a\widehat{Q}P_f\right\|_F}{\left\|P_a\widehat{Q}P_a\right\|_F} \right\},$$

and require $\widehat{\varepsilon}_{\text{mix}} \le \varepsilon_{\text{mix}}$.

A.7 OC–2: Differential Band Gains with Spectral Bounds

We require that $Q^{\text{H}}Q$ is well approximated by bandwise scalars on the ranges of P_f, P_a, P_o, i.e., there exist $s_f, s_a, s_o > 0$ and tolerances $\delta_f, \delta_a, \delta_o \in [0, 1)$ such that

$$\begin{aligned}
s_f^2(1-\delta_f)\,P_f &\preceq P_f Q^{\text{H}} Q P_f \preceq s_f^2(1+\delta_f)\,P_f,\\
s_a^2(1-\delta_a)\,P_a &\preceq P_a Q^{\text{H}} Q P_a \preceq s_a^2(1+\delta_a)\,P_a,\\
s_o^2(1-\delta_o)\,P_o &\preceq P_o Q^{\text{H}} Q P_o \preceq s_o^2(1+\delta_o)\,P_o,
\end{aligned}$$

with *strict separation*

$$\frac{s_a}{s_f} \ge \Lambda > 1 + c\,\varepsilon_{\text{mix}}, \qquad c \in [2, 4].$$

Interpretation: within each band, $Q^{\text{H}}Q$ behaves like a scalar (s_f^2, s_a^2, s_o^2) up to modest anisotropy $\delta_\bullet$, and the alias band gains are strictly larger than flow band gains.

Runtime label–free check. Compute eigenvalues of the band–projected operators:

$$\lambda_{\min}^f, \lambda_{\max}^f := \lambda_{\min\,/\,\max}(Q^{\text{H}}Q|_{P_f}),$$

and similarly for a, o. Set

$$\widehat{s}_f^2 := \tfrac{1}{2}(\lambda_{\min}^f + \lambda_{\max}^f), \quad \widehat{\delta}_f := \frac{\lambda_{\max}^f - \lambda_{\min}^f}{\lambda_{\max}^f + \lambda_{\min}^f},$$

and analogously for $\widehat{s}_a, \widehat{s}_o$ and $\widehat{\delta}_a, \widehat{\delta}_o$. Require $\widehat{\delta}_\bullet \le \delta_\bullet$ and $\widehat{s}_a/\widehat{s}_f \ge \Lambda$.

A.8 Consequence: Approximate Multiplicative Effect on Scores

Under OC–1–OC–2, for any z one has

$$\mathrm{BR}(Qz) = \frac{z^{\mathrm{H}} Q^{\mathrm{H}} P_f Q z}{z^{\mathrm{H}} Q^{\mathrm{H}} P_a Q z} \approx \frac{s_f^2}{s_a^2} \cdot \mathrm{BR}(z) \cdot (1 + \Delta_{\mathrm{BR}}(z))$$

with $|\Delta_{\mathrm{BR}}(z)| \le C_1(\delta_f + \delta_a) + C_2 \varepsilon_{\mathrm{mix}}$, for constants C_1, C_2 depending only on band dimensions. A similar approximation holds for $\mathrm{GLRT}_\rho(z)$ with denominators involving s_a^2, s_o^2.

Thus, if $\Lambda = s_a/s_f > 1$ and tolerances are small, BR/GLRT on a tile are rescaled approximately by a *bandwise constant* factor, with predictable sign (e.g. downwards when $s_a > s_f$).

A.9 Band Geometry & Leakage Budget

This section remains as in the original doc: its role is to ensure that the projectors P_f, P_a correspond to well–separated spectral bands given the PRF, T, and DPSS parameters, using DPSS tapers and multi-taper spectral estimates in the sense of Slepian and Thomson [39–41].

A.10 BG–1: Leakage Separation Margin

Let $G(f, g)$ be the multi–taper spectral leakage kernel associated with your DPSS design (known from design). Require

$$\max_{f \in \mathcal{B}_f} \sum_{g \in \mathcal{B}_a} G(f, g) \le \ell_{\max}, \qquad \ell_{\max} \le 0.15.$$

Interpretation: at most 15% of energy from an on–grid flow bin leaks into the alias band, and vice versa. This supports keeping the band mixing in OC–1 small after whitening.

A.11 BG–2: Band–Edge Guard

Define a guard band $\mathcal{B}_g$ separating $\mathcal{B}_f$ and $\mathcal{B}_a$ such that

$$\mathrm{dist}(\mathcal{B}_f, \mathcal{B}_a) \ge b_g,$$

for some guard width b_g in frequency (or bins). Require that any energy within b_g of $\mathcal{B}_f$ leaks only modestly into $\mathcal{B}_a$:

$$\sum_{g:\, \mathrm{dist}(g, \mathcal{B}_f) \le b_g} G(f, g) \le \ell_{\mathrm{edge}}, \qquad \ell_{\mathrm{edge}} \le 0.1.$$

Interpretation: avoid designing $\mathcal{B}_f, \mathcal{B}_a$ so tightly that off–grid peaks straddle the boundary; the guard ensures band–edge leakage is limited.

A.12 BG–3: Within–Band Condition Numbers

From the baseline whitened covariance in each band, require

$$\kappa_f := \frac{\lambda_{\max}(\widehat{R}|_{P_f})}{\lambda_{\min}(\widehat{R}|_{P_f})} \le \kappa_f^{\max}, \qquad \kappa_a := \frac{\lambda_{\max}(\widehat{R}|_{P_a})}{\lambda_{\min}(\widehat{R}|_{P_a})} \le \kappa_a^{\max},$$

with $\kappa^{\max} \in [10, 40]$ depending on T and sample support.

Interpretation: this makes it feasible to construct R_β that achieves OC–2 with moderate β, δ without overly crushing the flow band.

A.13 Covariance blending and localization (summary)

KA reshapes covariances only within bands. We use either a commuting band-scalar transform with target gains (s_f, s_a, s_o) or a trace-normalized blend R_β with small diagonal lift; both enforce band-wise structure (no off-block priors). Localization restricts KA to alias-dominated, low-coverage tiles using alias scores, guard ratios, optional flow coherence/coverage checks, baseline score windows, depth/PD/registration cues, and is validated via coverage rates p_-, p_+. Full localization definitions are provided in Appendix J.8 (PD-only instantiation in Appendix G).

A.14 Score–Level Guarantees

This section is unchanged in structure; we only make the dependence on p_-, p_+ explicit.

A.15 SL–1: Per–Tile Multiplicative Effect

Under OC–1–OC–2, for any z,

$$\mathrm{BR}_{\mathrm{KA}}(z) = \mathrm{BR}(Qz) \approx \frac{s_f^2}{s_a^2} \cdot \mathrm{BR}_{\mathrm{base}}(z) \cdot (1 + O(\delta_\bullet + \varepsilon_{\mathrm{mix}})).$$

Similarly,

$$\mathrm{GLRT}_{\rho,\mathrm{KA}}(z) \approx \frac{s_f^2}{\alpha_a(z)\, s_a^2 + \alpha_o(z)\, s_o^2} \cdot \mathrm{GLRT}_{\rho,\mathrm{base}}(z),$$

where

$$\alpha_a(z) := \frac{z^{\mathrm{H}} P_a z}{z^{\mathrm{H}}(P_a + \rho I)z} \in [0,1], \quad \alpha_o(z) := \frac{\rho\, \|z\|^2}{z^{\mathrm{H}}(P_a + \rho I)z} \in [0,1].$$

On alias–dominant tiles (large $\alpha_a(z)$), both BR and GLRT are approximately multiplied by $\mu := s_f^2/s_a^2 < 1$.

A.16 SL–2: Tail Threshold Improvement at Fixed FPR

Let S^-_{base} be the baseline negative score distribution, with CDF F_-. Let t_α be the α–quantile: $F_-(t_\alpha) = 1 - \alpha$. After KA with gating, the negative CDF becomes a mixture:

$$F'_-(t) = (1 - p_-)F_-(t) + p_- F_-(t/\mu),$$

ignoring higher–order mixing terms and assuming $\mu < 1$ is the multiplicative shrinkage factor on gated negatives. Under mild regularity and for α in the low–FPR regime,

$$t'_\alpha \approx t_\alpha - \frac{p_-(1-\mu)}{f_-(t_\alpha)},$$

so that $t'_\alpha < t_\alpha$ whenever $p_- > 0$ and $\mu < 1$. In the PD-mode experiments in this paper we report operating points using the exported right-tail PD score map S_{PD} (stored as `score_pd_*.npy`): choose τ so that $\mathbb{P}\{S_{\mathrm{PD}} \geq \tau \mid \mathrm{bg}\} = \alpha$ and evaluate $\mathbb{P}\{S_{\mathrm{PD}} \geq \tau \mid \mathrm{flow}\}$. For compatibility, acceptance bundles may also include `pd_*.npy`, which we denote $\mathrm{PD}^{\downarrow}$ and relate deterministically by $S_{\mathrm{PD}} = -\mathrm{PD}^{\downarrow}$.

Interpretation (plain language). KA is designed to shrink scores only on a subset of artifact-dominated background tiles. This selectively compresses the extreme right tail of H_0 scores, which lowers the background threshold needed to hit a target FPR. Lowering that threshold can reveal additional true flow pixels without increasing the false-positive rate, because scores never increase on penalized pixels. Equivalently, scaling down a fraction p_- of top–tail negatives removes mass from the negative tail and shifts t_α left; uplift is only plausible when p_+ (positive gating rate) is small.

We emphasize that the SL–2 guarantee is conditioned on:

- Regime preconditions R–1/R–2/R–3,
- Gating coverage in the range $0.2 \le p_- \le 0.5$, $p_+ \le 0.1 p_-$.

A.17 SL–3: Positive Collateral Bound

Let p_+ be the positive gating rate and assume gated positives suffer at most a multiplicative drop $\nu := s_f^2/s_a^2 \cdot (1 + O(\delta_\bullet + \varepsilon_{\text{mix}}))$. For any threshold t'_α satisfying $F'_-(t'_\alpha) = 1 - \alpha$, the TPR under KA satisfies

$$\text{TPR}_{\text{KA}}(t'_\alpha) \ge \text{TPR}_{\text{base}}(t_\alpha) - \eta\, \Phi(\nu),$$

where $0 \le \eta \lesssim \frac{p_+}{p_-}$ (up to geometry) and $\Phi(\nu)$ is the fraction of positives near t'_α that are moved below the threshold. For $p_+ \ll p_-$ and ν close to 1, the loss term is dominated by the gain from SL–2.

A.18 Evaluation Discipline

This section remains as in the original document: it covers GLRT parameter sharing, block/cluster–aware tail CIs, and effective negatives. These aspects ensure fair comparison but do not themselves cause uplift.

A.19 Feature Statistics and Pass/Fail Checklist

At runtime, label–free feature statistics must support the following checks:

(i) Band design: BG–1/BG–2/BG–3 satisfied for current DPSS/PRF/T.
(ii) KA operator: $\widehat{\varepsilon}_{\text{mix}} \le \varepsilon_{\text{mix}}$ (OC–1); $\widehat{s}_a/\widehat{s}_f \ge \Lambda$ and $\widehat{\delta}_\bullet \le 0.1$ (OC–2).
(iii) Localization primitives: risk scores m_{alias} and m_{guard}, guard ratios, flow coverage, FPC, depth, PD, registration; localization indicator g constructed via G–1/G–2/G–2'/G–3/G–4.
(iv) Invariance sentinel: if $\widehat{s}_a/\widehat{s}_f \in [1-\epsilon, 1+\epsilon]$ with $\epsilon \le 0.02$, KA is effectively a scalar; disable it (see Section A.21).

A.20 Knowledge-Aided Prior: shrink-only score-space penalty

To keep KA conservative under heterogeneity, we implement a deterministic, feature-conditioned *prior* that acts in score space as a shrink-only penalty. Concretely, the post-KA score is defined as

$$S_{\text{post}}(i) = w(i)\, S_{\text{pre}}(i), \qquad w(i) \in (0, 1],$$

equivalently $T_{\text{post}} = T_{\text{pre}} + \log w$ when working with log-likelihood scores (MAP log-odds decomposition). By construction, $w \equiv 1$ (no KA effect) is an explicit, expected outcome whenever the feature statistics are uninformative or fail integrity checks.

A.20.1 Feature-space regimes (R0/R1/R2)

We partition the feature space into three regimes per volume (or per slice/plane in PD-only mode):

- $\mathcal{R}_0$ **(uninformative).** The prior is uninformative and $w \equiv 1$; output equals the baseline detector (STAP on IQ; PD-only detector on PD-only). Feature statistics are still recorded.
- $\mathcal{R}_1$ **(guarded shrink).** A bounded, shrink-only penalty may be applied on a localized set of high-risk tiles while strictly protecting a high-confidence set. No uplift claim is implied.
- $\mathcal{R}_2$ **(actionable; research-only).** The same shrink-only mechanism may be applied more strongly, but only when additional *actionability* evidence suggests that shrink preferentially targets false activations. This regime is used to gate reporting of KA deltas; it is not a default deployment mode.

For auditability, the implementation also logs a reason label (why $w \equiv 1$) and a *risk mode* (alias vs guard) so that $\mathcal{R}_0$ behavior is interpretable. (For backward compatibility, exported metadata may also include legacy string labels such as `C0_OFF`/`C1_SAFETY`/`C2_UPLIFT` mapped to $\mathcal{R}_0/\mathcal{R}_1/\mathcal{R}_2$.)

A.20.2 Feature primitives (band-energy statistics)

On each tile i we record band energies using fixed, disjoint projectors (P_o, P_f, P_g, P_a) with $P_o + P_f + P_g + P_a = I$:

$$E_o(i) = z_i^{\mathrm{H}} P_o z_i, \quad E_f(i) = z_i^{\mathrm{H}} P_f z_i, \quad E_g(i) = z_i^{\mathrm{H}} P_g z_i, \quad E_a(i) = z_i^{\mathrm{H}} P_a z_i,$$

where $z_i \in \mathbb{C}^T$ (conjugate-transpose $(\cdot)^{\mathrm{H}}$; in real PD-only instantiations $(\cdot)^{\mathrm{H}}$ reduces to $(\cdot)^\top$). We optionally track a DC subcomponent $E_{\mathrm{dc}}(i)$ (e.g. energy in a narrow DC tone within P_o) as an auxiliary feature statistic. We define

$$E_{\mathrm{tot}}(i) = E_o + E_f + E_g + E_a + \epsilon,$$

and form robust, label-free ratios:

$$m_{\mathrm{alias}}(i) = \log \frac{E_a(i) + \epsilon}{E_f(i) + \epsilon}, \quad r_g(i) = \frac{E_g(i)}{E_{\mathrm{tot}}(i)}, \quad r_f(i) = \frac{E_f(i)}{E_{\mathrm{tot}}(i)}.$$

We also use a flow-coverage proxy $c_f(i) \in [0, 1]$ (fraction of tile pixels in the flow mask), a depth proxy $d(i) \in [0, 1]$ when available, and a baseline detector score $S_{\mathrm{base}}(i)$ on which KA operates (defined as a right-tail score so that higher means "more flow-like"; in PD-mode this is $S_{\mathrm{base}} = S_{\mathrm{PD}} = \texttt{score_pd}$).

A.20.3 Proxy sets and actionability tests

Because class labels are unavailable at runtime, the regime selector uses two proxy sets:

$$\mathcal{T}_{\mathrm{bg}} := \{i : c_f(i) \le c_{\mathrm{bg}}\}, \qquad \mathcal{T}_{\mathrm{flow}} := \{i : c_f(i) \ge c_{\mathrm{flow}}\},$$

with frozen design values (typical $c_{\mathrm{bg}} \approx 0.05$, $c_{\mathrm{flow}} \approx 0.20$). These thresholds are intentionally conservative, order-of-magnitude safety bounds chosen a priori to avoid penalizing dense-flow tiles; they are not optimized to maximize ROC on the evaluated datasets. The uplift-eligibility checks are reframed as *label-free actionability*:

- **Background-vs-flow separation:** $\Delta_{bf} = \mathrm{median}(m_{\mathrm{alias}} \mid \mathcal{T}_{\mathrm{bg}}) - \mathrm{median}(m_{\mathrm{alias}} \mid \mathcal{T}_{\mathrm{flow}})$.

- **Tail association in background:** within $\mathcal{T}_{\text{bg}}$, define a tail subset $\mathcal{T}_{\text{tail}}$ as the top 1% of S_{base} and a mid subset $\mathcal{T}_{\text{mid}}$ as a central quantile band; require that m_{alias} is higher in $\mathcal{T}_{\text{tail}}$ than in $\mathcal{T}_{\text{mid}}$ (risk-feature association explains "false-alarm-looking" background tiles).

When guard fractions dominate, the actionable regime $\mathcal{R}_2$ is disallowed but guarded shrink $\mathcal{R}_1$ may still be permitted. In this case we switch to a guard-risk mode using $m_{\text{guard}}(i) = \log \frac{E_g(i)+\epsilon}{E_f(i)+\epsilon}$ (or equivalently $r_g(i)$) and apply the same tail association check using m_{guard}.

A.20.4 Protected set, candidate window, and shrink mapping

KA must not disturb high-confidence flow tiles. We define a protected set

$$\mathcal{T}_{\text{prot}} := \{i : c_f(i) \geq c_{\text{flow}}\} \cup \{i : S_{\text{base}}(i) \geq Q_{q_{\text{hi}}}(S_{\text{base}})\} \cup \{i : r_f(i) \geq r_{f,\text{prot}}\},$$

and a candidate window $\mathcal{T}_{\text{cand}}$ consisting of non-protected tiles whose baseline scores lie in an "uncertain" quantile band $[Q_{q_{\text{lo}}}, Q_{q_{\text{hi}}}]$. On $\mathcal{T}_{\text{bg}}$ we set an adaptive risk threshold $\tau_{\text{risk}} = Q_{q_{\text{risk}}}(m_{\text{risk}} \mid \mathcal{T}_{\text{bg}})$ where m_{risk} is m_{alias} or m_{guard} depending on risk mode. The localization indicator is

$$g(i) = \mathbf{1}\{i \in \mathcal{T}_{\text{cand}},\ m_{\text{risk}}(i) \geq \tau_{\text{risk}}\}.$$

For penalized tiles (those with $g(i) = 1$) we compute a bounded shrink factor

$$w(i) = \max\left(w_{\min}, \exp(-k\, u(i))\right), \qquad u(i) = \max\left(0, \frac{m_{\text{risk}}(i) - \tau_{\text{risk}}}{\text{MAD}(m_{\text{risk}} \mid \mathcal{T}_{\text{bg}}) + \epsilon}\right),$$

with $w_{\min} \in (0, 1]$ so the corresponding scale $\gamma(i) = 1/w(i)$ is finite, and set $w(i) = 1$ on non-penalized and protected tiles. In Doppler/IQ PD-mode bundles we export the per-pixel scale $\gamma(i) = 1/w(i) \geq 1$ (averaged over overlapping tiles) and apply it on the union of penalized tiles as

$$\text{PD}^{\downarrow}_{\text{post}}(i) = \gamma(i)\, \text{PD}^{\downarrow}_{\text{pre}}(i),$$

which inflates the legacy PD-mode map $\text{PD}^{\downarrow}$ on penalized regions. Since $S_{\text{PD}} = -\text{PD}^{\downarrow}$, this is equivalent to $S_{\text{post}} = w\, S_{\text{pre}}$ and therefore is non-increasing in score space. For PD-only detectors such as PD-z that take both signs, we use hard-suppression/soft-shrink variants (Appendix G) so that shrink-only is literal for signed scores.

A.20.5 Sentinels, reason codes, and auditability

The regime selector defaults to $\mathcal{R}_0$ ($w \equiv 1$) when any of the following hold: insufficient background proxy support, flat risk statistics (e.g. $\text{IQR}(m_{\text{risk}} \mid \mathcal{T}_{\text{bg}})$ too small), guard domination that is not actionable, coverage outside bounds (localization inert or global), invariance (penalized w nearly constant), or a monotonicity guard failure on near-protected tiles. When $w \not\equiv 1$, we record coverage and strength summaries (e.g. $p_{\text{shrink}} = \mathbb{P}\{g = 1 \mid \mathcal{T}_{\text{cand}}\}$ and quantiles of w) along with a short reason label (e.g. `alias_metric_flat`, `guard_dominates_not_actionable`, `coverage_out_of_bounds`, `monotonicity_guard_failed`). This makes KA behavior reviewable and supports the intended posture: $\mathcal{R}_0$ is an expected success regime, $\mathcal{R}_1$ is a conservative safety regime, and $\mathcal{R}_2$ is a research-only regime used to gate uplift reporting.

A.20.6 PD-only instantiation

On PD-only datasets (Macé/Urban, fUS-BMI, cranial-window PD), the same regime selection is applied after replacing Doppler bands with hemodynamic bands and treating tile-averaged PD time series as the slow-time signal. The feature primitives (E_f, E_a, E_g) and the shrink-only map $w(i)$ are defined identically, and the regime label is produced per plane. This yields a uniform export schema across Doppler IQ and hemodynamic PD regimes.

Figure 12 shows a compact prior/regime summary panel over a representative sweep across regimes: most runs remain in $\mathcal{R}_0$ with an explicit disable reason, while a minority enter $\mathcal{R}_1$ and apply a mild, localized shrink in an alias- or guard-risk lane (never boosting scores).

A.21 Regime Sentinel and Safety Guard

Experience shows that even when OC/BG/CB/G conditions are roughly satisfied, there can exist regimes where KA is either inert or harmful (e.g. risk feature flat, localization never triggers or triggers everywhere, baseline ROC saturated). This section formalizes when KA should be *explicitly disabled*.

A.22 RS–1: Regime Sentinel (Design–Time)

On a held–out labeled configuration, check:

- **RS–1a: Band occupancy.** If R–1 fails (e.g. most flow peaks near DC instead of Pf, or alias peaks rarely in Pa), the band design is not physically realized; do not deploy KA.
- **RS–1b: Alias separation.** If R–2 fails (alias scores m_{alias} show negligible difference between classes), alias localization has no reliable signal; do not deploy KA.
- **RS–1c: Localization coverage.** If, after choosing $q_{\text{alias}}, c_{f,\max}, c_{\text{pc}}$, the empirical localization rates satisfy $p_- \approx 0$ or $p_- \approx 1$, or $p_+ \approx p_-$, then localization is either inert, global, or non–differential; do not expect uplift and default to $\mathcal{R}_0$.
- **RS–1d: Baseline saturation.** If $\text{pAUC}_{\text{base}}$ in the target FPR range is already at or near maximum and the score ratio distribution under KA has median close to 1 with tiny spread, you are in a ROC–invariant regime; KA can be left off.

A.23 Safety–1: Score Monotonicity Guard (Runtime)

Even when regime preconditions and localization coverage hold, we require that KA does not produce grossly pathological score transformations on *penalized flow tiles*. Let $S_{\text{base}}, S_{\text{KA}}$ denote baseline and KA scores on the chosen metric (BR or GLRT).

For the set of penalized flow tiles, define

$$r_i := \frac{S_{\text{KA}}(i)}{\max(S_{\text{base}}(i), \varepsilon)}, \quad u_i := \log r_i,$$

with a small $\varepsilon > 0$.

Let

$$\text{median}_f := \text{median}(u_i), \quad \text{IQR}_f := Q_{0.75}(u_i) - Q_{0.25}(u_i),$$

and τ_f the Kendall τ rank correlation between S_{KA} and S_{base} on these pixels.

We require:

- **Sample size.** Apply the guard only if the number of gated flow pixels exceeds a threshold $N_{\min}$ (e.g. $N_{\min} = 200$). If not, mark KA as "too sparse to evaluate" but do not disable it solely on this basis.
- **Flow monotonicity.** KA is marked unsafe if either

$$\begin{aligned} &\mathrm{IQR}_f > \log 10 \quad (\text{heterogeneity } > \times 10), \quad \text{or} \\ &\mathrm{median}_f > \log 10 \quad \text{and} \quad |\tau_f| < 0.1, \end{aligned}$$

  i.e. if KA systematically boosts flow scores by more than an order of magnitude while destroying rank consistency, or applies extremely heterogeneous multiplicative effects.

If Safety–1 fails, KA is disabled at runtime for that configuration and the system falls back to pure STAP.

A.24 What the Regime Does Not Promise (Updated)

KA–STAP may *fail to provide uplift* or may be disabled by the regime sentinel / safety guard in the following cases:

- If $\Lambda \approx 1$ (band–constant scaling, invariance sentinel) or $\varepsilon_{\mathrm{mix}}$ is large, global band–wise transforms are nearly ROC–invariant or harmful; expect parity or degradation.
- If R–2 alias separation fails, i.e. m_{alias} shows no class–differential behavior, alias localization cannot preferentially capture alias–dominated negatives; KA is out of regime.
- If localization is not class–differential (offline $p_+ \approx p_-$) or p_- is either tiny or nearly one, KA is either inert or global; SL–2 no longer applies.
- If baseline ROC is already saturated in the target FPR range, even an ideal KA transform cannot move the operating point significantly.
- If Safety–1 indicates that KA acts in a grossly non–monotone way on penalized flow tiles (e.g. large heterogeneous boosts), KA is disabled for that configuration.

In all these regimes, MC–SVD or STAP–only should be preferred; KA can be viewed as an optional regularizer but not a source of guaranteed uplift.

A.25 Safety and Regime Sentinel Summary

For deployed use it is helpful to summarize, in plain language, when KA is allowed to act and when it is automatically turned off.

- *When KA is allowed to act.* KA is enabled only in regimes that satisfy the structural preconditions (R–1/R–2/R–3), have useful but not global localization coverage (roughly $0.2 \le p_- \le 0.5$ with $p_+ \ll p_-$), and pass the invariance and Safety–1 checks. In this case KA behaves as a controlled, band–aware regularizer on top of STAP: it is permitted to shrink scores on alias–dominated negatives but must remain monotone and mild on flow tiles.
- *When KA is turned off.* KA defaults to $\mathcal{R}_0$ by design when band occupancy or alias separation fail (R–1/R–2), when localization becomes inert or non–differential, when the baseline ROC is already saturated in the target low–FPR band (R–3), or when Safety–1 detects non–monotone, highly heterogeneous score changes on penalized flow tiles. The invariance sentinel also forces $\mathcal{R}_0$ when the realized gains satisfy $\widehat{s}_a/\widehat{s}_f \approx 1$ so that KA is effectively a scalar rescaling.

- *Behavior in skull/OR–like regimes.* In skull/OR realism regimes such as Brain–SkullOR, alias power deliberately leaks into H_1 and R–2 fails by construction. In these regimes KA is expected to be off or ROC–neutral; the primary detector is the fixed STAP PD profile, with MC–SVD/RPCA/HOSVD as reference baselines. This is a safety feature, not a failure of the method: whenever the regime preconditions are not met, the system falls back to STAP alone rather than forcing KA to act.

B Supplementary Results

This appendix collects additional dataset audits and diagnostic panels that support the main paper's methodology, but are not required for the primary labeled low-FPR detection claims.

Evaluation discipline (summary). Quantitative ROC claims are restricted to regimes with independent labels (Brain-* simulator truth; Gammex B-mode structural masks). Operating points are reported under a fixed-profile protocol by selecting thresholds on labeled (or structurally labeled) H_0 sets at target FPR and evaluating TPR on the corresponding H_1 set; the empirically supported floor $fpr_{\min} = 1/n_{\text{neg}}$ is reported. Uncertainty is quantified at the natural unit of variation (windows for Brain-*; frames for cine acquisitions) rather than treating pooled pixels as independent samples. Conditional execution is treated as a changed decision rule and is reported only with explicit leakage controls. For Twinkling sequences, PRF is treated as an explicit experimental assumption (frozen overrides) and band-discretization sentinels force conservative behavior when discretized bands are invalid. When KA is enabled, it is shrink-only with protected-set invariance; labeled ROC reporting uses the pre-KA STAP score.

Where to find what.

- Full ROC curve shapes for labeled regimes: Figure 5 (Brain-*) and Figure 7 (Gammex).
- STAP-only ablation and PRF override sensitivity (Gammex): Figure 8.
- Conditional-execution leakage ablation (Brain-OpenSkull): Table 10.
- Shin proxy-metric comparisons, runtime, and measured rigid-motion telemetry: Figure 18 and Figure 19.

B.1 Full ROC curves for labeled regimes

The main text reports strict low-FPR operating points. Here we provide representative ROC-style curves to show the overall curve shape under the same masks and scoring conventions.

ROC curves are reported with the same unit-of-variation discipline as the operating-point tables. For Brain-* simulations, we compute ROC curves per 64-frame window and summarize them as median and IQR across **windows**; the supported floor $fpr_{\min} = 1/n_{\text{neg}}$ is fixed by the window's negative-set size. For Gammex, thresholds are calibrated on pooled pixels for stability, while uncertainty is quantified via a frame-level bootstrap (resampling **cine frames** as the unit of variation at fixed pooled thresholds).

In the full build (main text + appendices), these curves are included in the main Results section (Figure 5 and Figure 7); this appendix focuses on complementary audits and diagnostics.

Regime	α	within-window TPR med (IQR)	cross-window TPR med (IQR)	cross-window realized FPR med (IQR)
Brain-OpenSkull	10^{-4}	0.0223 (0.0031,0.3922)	0.0254 (0.0018,0.2226)	1.59e-04 (0,1.25e-03)
Brain-OpenSkull	3×10^{-4}	0.2107 (0.0077,0.4610)	0.2253 (0.0045,0.3409)	3.19e-04 (0,1.82e-03)
Brain-OpenSkull	10^{-3}	0.2834 (0.1496,0.5590)	0.3026 (0.1158,0.4226)	1.15e-03 (7.47e-05,3.38e-03)
Brain-AliasContract	10^{-4}	0.3499 (0.2649,0.3756)	0.3303 (0.2534,0.3998)	1.10e-04 (0,5.28e-04)
Brain-AliasContract	3×10^{-4}	0.4033 (0.3283,0.4560)	0.4016 (0.3188,0.4569)	2.99e-04 (5.48e-05,9.21e-04)
Brain-AliasContract	10^{-3}	0.5063 (0.4383,0.5113)	0.4927 (0.4096,0.5416)	9.76e-04 (3.54e-04,2.18e-03)
Brain-SkullOR	10^{-4}	0.3791 (0.2622,0.3987)	0.3491 (0.2624,0.4213)	1.20e-04 (0,7.37e-04)
Brain-SkullOR	3×10^{-4}	0.4460 (0.3230,0.4648)	0.4010 (0.3242,0.4741)	3.19e-04 (2.99e-05,1.39e-03)
Brain-SkullOR	10^{-3}	0.5102 (0.4364,0.5456)	0.5136 (0.4131,0.5377)	9.86e-04 (1.74e-04,2.72e-03)

Table 9: Brain-* cross-window threshold calibration audit for the pre-KA STAP score. "Within-window" matches the main reporting protocol (calibrate τ_α on a window's negatives and evaluate on that same window). "Cross-window" calibrates τ_α on one window and evaluates on a disjoint window ($n = 20$ ordered window pairs). Reported values are median (IQR) over windows/pairs. Full outputs are written to `reports/brain_crosswindow_calibration.csv` / `reports/brain_crosswindow_calibration_summary.json`.

B.2 Cross-window threshold calibration audit (Brain-*)

The main Brain-* tables calibrate the operating threshold τ_α on the negative set of the *same* 64-frame window being evaluated. To quantify how sensitive these strict low-FPR operating points are to window-to-window distribution shift (and to avoid over-interpreting quantized tail points), we perform a simple cross-window audit: for each regime and FPR target α, we calibrate τ_α on one window's negatives and apply it to the other windows. Table 9 reports within-window TPR (as in the main text) alongside cross-window TPR and the realized cross-window FPR.

B.3 STAP-only ablation and PRF override sensitivity (Gammex)

To isolate the role of the STAP core from baseline prefiltering, we run a small STAP-only ablation on the Gammex flow phantom by applying STAP directly to raw (registered) CFM IQ without the SVD band-pass baseline. We also provide a small PRF override sensitivity check on the along view to assess brittleness to the frozen PRF assumption required by Twinkling metadata limitations. In the full build (main text + appendices), this audit is shown in the main Results section (Figure 8).

B.4 Conditional execution leakage ablation (Brain-OpenSkull pilots)

Conditional execution (compute gating) skips STAP outside a flow-proxy region and falls back to the baseline score. Because this changes the overall decision rule, we evaluate it explicitly under leakage controls by varying only the *gating* mask: (i) a proxy mask computed from the same scored window, (ii) a proxy mask carried from a disjoint window, and (iii) a randomized mask with matched coverage. Table 10 reports the corresponding change in TPR relative to full STAP at fixed low-FPR targets.

Conditional-execution mask	ΔTPR@10^{-4} ($\times 10^{-3}$)	ΔTPR@3×10^{-4} ($\times 10^{-3}$)	ΔTPR@10^{-3} ($\times 10^{-3}$)
Same-window proxy mask	$+0.31\,[0.23, 0.39]$	$+0.81\,[0.54, 1.04]$	$+3.05\,[2.67, 3.43]$
Disjoint-window proxy mask	$-3.05\,[-5.69, -0.92]$	$-2.90\,[-5.34, -0.88]$	$-1.74\,[-4.64, 0.28]$
Random mask (matched coverage)	$-3.05\,[-5.65, -0.92]$	$-2.97\,[-5.68, -0.71]$	$-2.36\,[-4.96, -0.06]$

Table 10: Conditional-execution leakage ablation on Brain-OpenSkull pilots (five independent realizations; 64-frame window; matched-subspace STAP score). Entries report mean ΔTPR (conditional minus full STAP) at fixed FPR targets, with 95% bootstrap CIs over realizations ($n = 5$). Evaluation masks are identical across conditions; only the conditional-execution gating mask varies.

This behavior is expected because gating changes the decision rule: leakage-controlled gating produces a small TPR decrease in exchange for a large runtime reduction, making it a clear engineering tradeoff rather than a free performance gain.

B.5 KA hygiene on artifact-heavy calculi sequences (Twinkling)

To illustrate the shrink-only behavior of the KA layer in a strongly artifact-dominated regime, we evaluate Twinkling calculi sequences under a frozen short-ensemble profile. Figure 13 summarizes the KA prior outputs (region and reason) across Gammex and calculi runs, and Figure 14 shows a representative calculi example where the post-KA score compresses the background tail without increasing any scores on penalized pixels.

B.6 ULM 7883227: additional motion variants

In addition to the main `brainlike` motion sweep reported in the main paper, we report an `elastic` (nonrigid-only) motion ladder under the same frozen baseline and STAP configuration.

B.7 Macé whole-brain fUS: hemodynamic band-energy statistics and hold-out check

We treat the Macé/Urban whole-brain dataset as a PD-only, open-skull analogue of Brain-OpenSkull and compute tile-level hemodynamic band-energy statistics with fixed bands $P_f = [0, 0.5]$ Hz, guard $[0.5, 1.0]$ Hz, and $P_a = [1.0, 3.0]$ Hz on 8×8 tiles (stride 3). Offline we also compute atlas-derived *region strata* (e.g. VIS/SC/LGd versus CP/CA/DG) for descriptive summaries; these strata are not treated as ground-truth flow labels and are not used as KA inputs. Figure 16 summarizes a systematic sweep in "Pf occupancy vs alias separation" space. We provide an explicit atlas-alignment sanity check in Appendix F. Here PD-z denotes a per-tile change-from-baseline z-score on the PD time series (distinct from the Doppler/IQ PD-mode score maps S_{PD} exported by the STAP pipeline).

Hold-out discipline (avoid plane selection bias and label-tuned thresholds). To address the selection-bias critique and to avoid per-plane PD-z thresholds chosen using atlas strata (e.g. setting thresholds to force a fixed activation fraction in a preferred stratum), we add a scan-level hold-out evaluation (reproducibility details in Appendix L). We (i) define label-free proxy sets from

Held-out subset (scan4–scan5)	n	offline VIS/SC/ LGd / CP/CA/DG hits (pre)	offline VIS/SC/ LGd / CP/CA/DG hits (post)	VIS/SC/ LGd retention	CP/CA/DG reduction
All planes	40	23/142	22/95	0.957	47
Selected ($\text{PfPeak}_{\text{flow}} \geq \tau_{\text{sel}}$)	5	19/22	19/19	1.000	3

Table 11: Macé hold-out alias-gate evaluation with label-free thresholding. For each plane we set the PD-z threshold τ by a fixed background-proxy tail rule $\mathbb{P}\{\text{PD-}z \geq \tau \mid \mathcal{T}_{\text{bg}}\} = \alpha_{\text{pd}}$ with $\alpha_{\text{pd}} = 0.01$, and set the alias suppression threshold as the 0.8 quantile of $m_{\text{alias}} = \log((E_{\text{Pa}} + \epsilon)/(E_{\text{Pf}} + \epsilon))$ on the same background proxy set. "Selected" planes are those whose Pf-peak fraction on flow-proxy tiles exceeds a training-derived threshold τ_{sel} (the 0.95 quantile of $\text{PfPeak}_{\text{flow}}$ over the training scans `scan1/2` and `scan3/6`; $\tau_{\text{sel}} \approx 0.145$). Hit counts use atlas strata (VIS/SC/LGd and CP/CA/DG) only for offline descriptive evaluation (not KA inputs and not a Doppler efficacy claim). Reproducibility details are provided in Appendix L.

a mean-PD vascularity mask and set a per-plane PD-z operating point by a fixed background-proxy tail rule $\mathbb{P}\{\text{PD-}z \geq \tau \mid \mathcal{T}_{\text{bg}}\} = \alpha_{\text{pd}}$ (no atlas labels), (ii) define the alias suppression threshold as a fixed background-proxy quantile of $m_{\text{alias}} = \log((E_{\text{Pa}} + \epsilon)/(E_{\text{Pf}} + \epsilon))$, and (iii) select Pf-rich planes in held-out scans using a label-free band-energy statistic (Pf-peak fraction on flow-proxy tiles), with the selection threshold fixed from the training scans. Table 11 summarizes the resulting held-out counts on `scan4`–`scan5` (training: `scan1/2` and `scan3/6`; note `scan2` duplicates `scan1` and `scan6` duplicates `scan3` in this loader).

On held-out scans (`scan4`–`scan5`), this fixed, shrink-only alias suppression reduces proxy activations in the CP/CA/DG stratum while modestly affecting activations in the VIS/SC/LGd stratum at the chosen label-free operating point. At $\alpha_{\text{pd}} = 0.01$ on the background proxy, aggregate (VIS/SC/LGd / CP/CA/DG hit counts) change from 23/142 to 22/95 (VIS/SC/LGd retention ≈ 0.96). On the Pf-peak-selected subset of held-out planes ($\text{PfPeak}_{\text{flow}} \geq \tau_{\text{sel}}$), hit counts change from 19/22 to 19/19 (VIS/SC/LGd retention = 1.0). We interpret this as a conservative safety-regularization behavior rather than an efficacy claim: Macé is PD-only and provides no Doppler-side flow labels, so its primary role here is to validate the band-energy/proxy machinery and the no-peeking threshold discipline.

Pixel-level separability vs. the vascular atlas (independent structural labels). To add one real-data quantitative check with an *independent* label source (not derived from PD), we map the dataset-provided Allen atlas vascular volume `atlas.Vascular` into each Macé scan plane using the affine in `Transformation.mat`. We threshold `atlas.Vascular` at 0.5 to define vascular vs. non-vascular pixels and restrict evaluation to pixels that map inside the atlas and to non-background atlas regions. This yields $n_{\text{neg}} \approx 1.7 \times 10^4$ non-vascular pixels per plane (median $fpr_{\min} \approx 5.9 \times 10^{-5}$), so FPR targets down to 10^{-4} are empirically meaningful at pixel level. As expected for PD-only hemodynamics, extreme-tail separability is weak; the purpose of this check is independent-label auditability and an empirical FPR scale rather than a performance headline.

We then evaluate three families of scalar maps: (i) a structural baseline based on the mean PD image (reported as a right-tail score using a left-tail convention, $S_{\text{inv_meanPD}} = -\log(\overline{P}+\epsilon)$), (ii) PD-z as a per-pixel change-from-baseline z-score on log-PD (the maximum z-score after an early baseline segment), and (iii) a hemodynamic alias band-energy statistic $m_{\text{alias}} = \log((E_a + \epsilon)/(E_f + \epsilon))$ computed from simple FFT band energies on log-PD with fixed hemodynamic bands $P_f = [0, 0.5]$ Hz

Score (vascular vs. non-vascular)	TPR@10^{-4} ($\times 10^{-3}$)	TPR@3×10^{-4} ($\times 10^{-3}$)	TPR@10^{-3} ($\times 10^{-3}$)
$S_{\text{inv_meanPD}} = -\log(\overline{P} + \epsilon)$ (structural baseline)	0.00 (0.00, 0.00)	0.00 (0.00, 1.04)	2.76 (1.84, 3.20)
PD-z (log-PD; max z after baseline)	0.00 (0.00, 0.00)	0.00 (0.00, 0.81)	0.94 (0.00, 1.78)
$m_{\text{alias}} = \log((E_a + \epsilon)/(E_f + \epsilon))$ (hemodynamic alias statistic)	0.00 (0.00, 0.00)	0.00 (0.00, 0.75)	1.05 (0.00, 2.04)
PD-z + label-free alias penalty (shrink-only)	0.00 (0.00, 0.00)	0.00 (0.00, 0.81)	0.92 (0.00, 1.74)

Table 12: Macé pixel-level quantitative evaluation using an independent vascular atlas label source (`atlas.Vascular`; $n = 80$ planes from four deduplicated scans). Entries report median (IQR) TPR at fixed FPR targets for vascular vs. non-vascular pixels. The median negative-pixel count per plane is $\approx 1.69 \times 10^4$ (median $fpr_{\min} \approx 5.9 \times 10^{-5}$). Reproducibility details are provided in Appendix L.

and $P_a = [1.0, 3.0]$ Hz. Finally, to mirror the conservative posture on real data, we apply a label-free alias penalty as a shrink-only score-space regularizer on PD-z: protect a flow-proxy set defined by a high mean-PD quantile, then down-weight non-proxy pixels whose m_{alias} lies above the 0.90 quantile of m_{alias} on the background proxy. Table 12 summarizes the resulting low-FPR operating points.

PD-only prior summary panel. To keep PD-only Macé reporting aligned with the same audit posture used for Doppler IQ, we summarize the PD-only KA prior outcomes over all planes using a compact summary panel that mirrors the Doppler-IQ summary but operates per plane. The panel reports (region, reason) histograms and shrink-only coverage/strength band-energy summaries under label-free proxy sets (Appendix G), and presents atlas-stratified summaries only as a clearly separated offline/descriptive panel.

B.8 Shin RatBrain Fig3 (LOCA-ULM): real kHz IQ viability and KA-prior diagnostics

To demonstrate that the frozen STAP + band-energy + KA-prior stack is operational on real Doppler-like IQ, we use the LOCA-ULM RatBrain Fig3 dataset released with Shin et al. as beamformed, post-compounded complex IQ sequences [31, 32]. This is a microbubble ULM IQ dataset; we use it solely as a kHz-IQ ingestion and band-energy/prior audit anchor, not as a physiological fUS benchmark. The Fig3 archive contains 80 clips (`IQData001.dat`–`IQData080.dat`) and a `SizeInfo.dat` file describing the tensor shape. Each clip stores $T = 250$ slow-time frames on a (238×130) spatial grid as little-endian float64 arrays containing I then Q in column-major order (matching the authors' reference MATLAB loader); the effective slow-time sampling is 1 kHz (post-compounding), with center frequency $f_0 \approx 15.6$ MHz [32]. We treat this dataset as a *real-IQ viability and safety anchor* rather than a labeled detection benchmark: there are no H_1/H_0 labels, so we do not report ROC uplift; instead we report prior-regime stability (tilewise band-energy summaries and coverage), shrink-only invariants, and label-free endpoints defined on pixel-level proxy masks (flow/background).

Processing workflow. We mirror the authors' loader and convert IQ clips into a common analysis format used throughout this work (reproduction details in Appendix L). For a chosen clip and time window we:

- Apply an SVD clutter filter on the Casorati matrix $X \in \mathbb{C}^{T\times HW}$ to form a baseline PD-derived map pd_{base} and a residual IQ cube (the input to STAP).
- Construct pixel-level proxy masks from pd_{base} (used here as a conventional PD-brightness proxy $\text{PD}^{\uparrow}$ distinct from the ROC score maps): a flow/vessel proxy (top quantile; minimum pixel count) and a background proxy (complement of the flow mask in a central crop).
- Run the temporal STAP core on the residual IQ cube with fixed tile geometry (8×8 tiles, stride 3) and a longer slow-time aperture $L_t = 64$ suited to 1 kHz slow time (bin spacing 15.6 Hz), producing a pre-KA PD-mode score map.
- Apply the KA spatial-spectral prior (Section 3) to produce a region index, reason label, and (when active) a non-increasing post-KA score map together with gate/scale maps.
- Record tilewise band-energy statistics (peak frequency, guard fraction, alias/guard risk metrics) and the penalization maps in the same analysis format used throughout this work.

The proxy masks are used only for label-free proxying/protection; prior-regime decisions and hygiene/robustness proxies follow the same right-tail score convention used elsewhere in this manuscript: operating points are expressed as a threshold on a right-tail PD score map, i.e. $S_{\text{PD}} \geq \tau$ at fixed background-proxy tail rate.

Scale-map convention. When the KA prior is active ($w < 1$ on a nontrivial set), we record an equivalent per-pixel scale factor $\gamma(i) \geq 1$ and apply it to the PD score map as

$$\text{PD}_{\text{post}}(i) = \text{PD}_{\text{pre}}(i)/\gamma(i), \qquad \gamma(i) \geq 1,$$

which makes the PD score map non-increasing on penalized regions. For any fixed threshold τ this is a pure shrink-only penalty; when thresholds are recalibrated post-KA to match a target background tail rate, any TPR change reflects background-tail suppression rather than score inflation.

Frozen Shin operating profiles and baseline calibration. To avoid retuning the conservative brain Doppler profile to Shin until outcomes appear favorable, we treat Shin as its own frozen operating regime and explicitly calibrate only the baseline clutter filter once. We sweep the MC–SVD energy-fraction cutoff (smallest number of singular modes removed to exceed the target energy fraction) over a calibration subset (10 representative clips; window 0:128) and select a single frozen baseline setting based on band-energy integrity and stability, not on prior-regime counts. Table 13 reports $\mathcal{R}_1/\mathcal{R}_2$ counts for transparency, but selection is based on the label-free integrity/actionability metrics (Δ_{bf}, Δ_{tail}) and stability/hygiene proxies. Overly aggressive baseline removal increases Pf peak occupancy but can invert actionability evidence (background-vs-flow separation becomes negative) and suppress $\mathcal{R}_2$ eligibility. We therefore freeze MC–SVD with energy fraction ≈ 0.97 as a balanced setting for the Shin-U profile (reproducibility details in Appendix L).

Baseline comparisons (proxy separability on real IQ). Although Shin provides no independent H_1/H_0 labels, it is useful to test whether the STAP core improves *proxy* separation on real complex IQ. We run a fixed scenario matrix on 10 clips and three overlapping windows (0:128, 64:192, 122:250), exporting right-tail score maps for MC–SVD, RPCA, HOSVD, STAP (on the MC–SVD residual), and an *STAP-only* ablation that runs STAP directly on raw/registered IQ. Proxy masks are derived deterministically from the MC–SVD baseline PD map via a `pd_auto` quantile

Energy frac	$\mathcal{R}_1/\mathcal{R}_2$	Δ_{bf} med	Δ_{tail} med	Pf-peak(flow) med	BG clusters Δ	BG area Δ
0.90	8/2	+1.52	+0.65	0.011	−17	−32
0.95	8/2	+1.50	+0.65	0.019	−17	−32
0.97	8/2	+1.49	+0.73	0.050	−19	−45
0.975	9/1	−0.38	+0.77	0.237	−30	−89
0.98	10/0	−0.38	+0.78	0.253	−30	−89
0.99	10/0	−0.47	+0.58	0.510	−6	−59

Table 13: Baseline clutter-filter calibration for the frozen Shin-U profile (10 clips; window 0:128). Δ_{bf} is the median background-vs-flow proxy separation of the risk metric; Δ_{tail} is the background tail association (actionability) metric; Pf-peak(flow) is the fraction of flow-proxy tiles (tiles with high flow-mask coverage) whose PSD peak lies in Pf. BG cluster/area deltas are computed from a fixed background-proxy tail operating point on the PD score map S_{PD} (clusters/area after KA minus before KA). We freeze energy fraction ≈ 0.97 because it yields nontrivial Pf realization while preserving positive actionability.

rule (0.995 quantile within depth fraction $[0.25, 0.85]$ with dilation), with the generic default-mask union disabled; masks are then held fixed across methods. We report (i) AUC between flow-proxy and background-proxy score distributions (threshold-free), and (ii) flow-proxy hit rate at matched background tail rates $\alpha \in \{10^{-1}, 10^{-2}, 3\times10^{-3}, 10^{-3}, 3\times10^{-4}, 10^{-4}\}$ (Figure 18; Table 14). In this setup $n_{\text{bg}} \approx 2.6 \times 10^4$ per scenario ($fpr_{\min} \approx 3.8 \times 10^{-5}$), so $\alpha = 10^{-4}$ corresponds to roughly 2–3 background pixels.

Across scenarios, STAP improves proxy separation (median AUC 0.951 vs 0.922 for MC–SVD) and yields nonzero flow-proxy hit rates at strict tails where MC–SVD is typically zero (e.g. median hit rate 0.070 at $\alpha = 10^{-3}$). The STAP-only ablation is comparable to STAP-on-residual, indicating that the localized STAP core is the primary driver of the improvement on this dataset. At moderate strictness ($\alpha \approx 10^{-2}$), RPCA/HOSVD can exhibit higher proxy hit rates than STAP despite lower AUC, consistent with a broader, lower-contrast score elevation rather than strict-tail selectivity. Finally, a conditioning-target sweep ($\kappa_{\text{target}} = 80$ vs 200) produced identical STAP score maps under this frozen profile, indicating that the conditioning target is non-binding in this particular configuration. Figure 20 shows a representative qualitative comparison.

Measured rigid-motion telemetry. Because Shin clips are short (≈ 0.25 s) and there are no external motion sensors, we treat rigid registration telemetry as a descriptive motion proxy rather than a full motion benchmark. Figure 19 summarizes the phase-correlation shifts estimated during baseline preprocessing (RMS shift magnitude per scenario) and relates them to the proxy AUC uplift (STAP minus MC–SVD). In these windows the estimated rigid motion is typically subpixel, so we do not treat Shin as a stratified motion-stress regime; the proxy-separation gains are not explained by measured rigid shifts.

Proxy circularity audit. To reduce dependence on any single baseline-derived proxy mask, we report metrics under (i) a flow-proxy mask derived from MC–SVD (pd_{base} via `pd_auto`) and (ii) a consensus proxy mask formed by majority vote over MC–SVD/RPCA/HOSVD `pd_auto` masks; rankings are consistent under this cross-mask audit.

Cross-mask audit (proxy-label sensitivity). Table 14 includes both the baseline-derived mask and the consensus mask results; full cross-mask outputs (including per-scenario thresholds

	MC–SVD proxy mask			Consensus proxy mask		
Method	AUC (med [IQR])	hit@$\alpha = 10^{-3}$ (med [IQR])	hit@$\alpha = 10^{-4}$ (med [IQR])	AUC (med [IQR])	hit@$\alpha = 10^{-3}$ (med [IQR])	hit@$\alpha = 10^{-4}$ (med [IQR])
MC–SVD	0.922 [0.916, 0.934]	0.000 [0.000, 0.023]	0.000 [0.000, 0.000]	0.913 [0.908, 0.924]	0.000 [0.000, 0.029]	0.000 [0.000, 0.000]
RPCA	0.902 [0.893, 0.907]	0.024 [0.000, 0.031]	0.008 [0.000, 0.016]	0.909 [0.898, 0.915]	0.028 [0.000, 0.036]	0.010 [0.000, 0.020]
HOSVD	0.881 [0.873, 0.906]	0.040 [0.030, 0.056]	0.008 [0.000, 0.015]	0.883 [0.874, 0.909]	0.055 [0.036, 0.069]	0.010 [0.000, 0.019]
STAP	0.951 [0.948, 0.956]	0.070 [0.034, 0.081]	0.014 [0.006, 0.025]	0.944 [0.932, 0.947]	0.051 [0.018, 0.082]	0.009 [0.003, 0.032]
STAP-only	0.948 [0.944, 0.951]	0.065 [0.024, 0.078]	0.020 [0.003, 0.032]	0.939 [0.933, 0.944]	0.065 [0.012, 0.083]	0.021 [0.000, 0.033]

Table 14: Shin RatBrain Fig3 proxy-metric summary over 30 scenarios (10 clips $\times$ 3 windows) under the frozen Shin-U profile (PRF = 1 kHz, $L_t = 64$, MC–SVD energy fraction ≈ 0.97). AUC is computed between flow-proxy and background-proxy score distributions; hit@α is the flow-proxy hit rate when the threshold is chosen to achieve background tail rate α. Metrics are shown under two proxy-mask definitions: an MC–SVD-derived mask and a consensus mask (majority vote over MC–SVD/RPCA/HOSVD masks). Full per-scenario outputs are in `reports/shin_ratbrain_baseline_matrix_shinU_e970_Lt64_nomaskunion_k80.csv` and `reports/shin_ratbrain_crossmask_audit_shinU_consensus3.csv`.

and realized FPRs) are in `reports/shin_ratbrain_crossmask_audit_shinU_consensus3.csv`.

Profile variants (Shin-U vs Shin-S/L). In addition to Shin-U, we evaluated two alternative frozen band profiles (Shin-S and Shin-L) designed to push the "flow" band Pf toward lower frequencies and increase Pf peak occupancy (an interpretability-forward choice for microvascular-like content under 1 kHz slow time). Across the same 10-clip calibration subset, these variants typically increase Pf-peak(flow) but render the contract actionability check Δ_{tail} negative under our flow/background proxy sets, so the prior lies predominantly in $\mathcal{R}_1$ (safety-only) rather than $\mathcal{R}_2$ (actionable). We therefore treat Shin-S/L as conservative "deployment-mode" profiles ($\mathcal{R}_1$-only by design) and use Shin-U as the primary "real IQ viability + safety" anchor reported here.

KA behavior, stability, and hygiene endpoints. Under the frozen Shin-U profile (MC–SVD energy fraction ≈ 0.97, STAP $L_t = 64$, fixed Pf/Pa band definitions, and KA enabled in `auto` mode), we run the pipeline on 10 pre-selected clips (`IQData001`–`IQData005`, `IQData010`, `IQData020`, `IQData040`, `IQData060`, `IQData080`) and evaluate stability across three overlapping time windows (0:128, 64:192, 122:250). Under Shin-U, Pf realization is weak (Pf-peak(flow)$\ll p_{f,\min}$), so Shin is not treated as a KA-uplift feasibility case under R–1; we use it to validate end-to-end determinism, invariants, and robustness proxies. Because there are no labels, we report only label-free endpoints (reproducibility details in Appendix L):

- **KA regime stability.** $\mathcal{R}_1$ dominates in all windows (8–10 of 10 clips), while $\mathcal{R}_2$ is rare (0–2 clips) due to the Pf-realization eligibility condition for $\mathcal{R}_2$ based on flow-proxy tiles (tiles with high flow-mask coverage); all decisions emit auditable reason labels.
- **Shrink-only and protected invariance.** When KA is active it applies a mild, localized shrink (median $p_{\text{shrink}} \approx 0.09$; median scaled pixel fraction ≈ 0.20) and preserves the protected set exactly (maximum absolute PD change on protected pixels is 0 in all evaluated clips).
- **Background-tail hygiene.** A label-free background-tail metric shows consistent tail compression on early windows: for each clip/window we choose a threshold τ so that the background proxy satisfies $\mathbb{P}\{S_{\text{PD}} \geq \tau \mid \text{bg proxy}\} = 10^{-3}$ on the pre-KA map, then count clusters/area of pixels satisfying $S_{\text{PD}} \geq \tau$ before vs. after KA. Total background-tail clusters decrease from 125 $\rightarrow$ 106 (window 0:128) and 121 $\rightarrow$ 104 (window 64:192), while the late window is closer to neutral (129 $\rightarrow$ 124). This is interpreted as "map hygiene" rather than

Window	$\mathcal{R}_0/\mathcal{R}_1/\mathcal{R}_2$	Pf-peak(flow) med	guard q90 med	p_{shrink} med	scaled-px frac med	scale p90 (γ) med
0:128	0/8/2	0.050	0.104	0.091	0.201	1.167
64:192	0/10/0	0.032	0.110	0.097	0.198	1.168
122:250	0/9/1	0.039	0.108	0.090	0.205	1.167

Table 15: Shin RatBrain Fig3 prior-regime summary band-energy statistics for the frozen Shin-U profile ($e \approx 0.97$, $L_t = 64$) across three overlapping windows (10 clips each).

Window	background clusters (pre→post)	background area (pre→post)
0:128	125 → 106	263 → 218
64:192	121 → 104	264 → 212
122:250	129 → 124	264 → 257

Table 16: Shin RatBrain Fig3 label-free hygiene endpoints under Shin-U ($e \approx 0.97$; $L_t = 64$; 10 clips). "Pre" and "post" denote STAP PD-mode outputs before and after the non-increasing KA prior penalty. The background-tail metric targets flow-like tail activations under the right-tail score convention: for each clip/window instance we choose τ so that the background proxy satisfies $\mathbb{P}\{S_{\text{PD}} \geq \tau \mid \text{bg proxy}\} = 10^{-3}$ on the corresponding pre-KA map, then count connected components (clusters) and pixels (area) with $S_{\text{PD}} \geq \tau$. Reproducibility details are provided in Appendix L.

ROC uplift.

Table 15 summarizes prior-regime counts and key band-energy summaries across windows.

All-clips KA regime summary (n=80). To reduce dependence on the 10-clip subset, we also run a diagnostics-only sweep over all 80 clips (window 0:128), recording the prior regime outputs. In this sweep we skip the STAP pass and do not apply the score-space KA mapping; we treat the result as an activation-rate audit (region/reason frequencies and band-energy summary statistics), not a performance result. On all clips the prior lies in $\mathcal{R}_1$ for 73/80 and in $\mathcal{R}_2$ for 7/80 (all with `reason=ok`, risk mode `alias`); median Pf-peak(flow)$\approx$ 0.091, guard $q_{0.90} \approx 0.105$, and median predicted $p_{\text{shrink}} \approx 0.095$. Reproducibility details are provided in Appendix L.

Synthetic motion robustness sweeps (label-free). To probe motion robustness on real IQ without ground-truth labels, we implement controlled motion-injection sweeps on Shin IQ cubes (reproduction details in Appendix L). In addition to simple per-frame subpixel translations, global phase modulation, and spatially smooth low-frequency clutter tones, we define a *brain-like* motion model designed to be mechanistically comparable to Brain-SkullOR: a bulk rigid translation (correctable by phase-correlation registration) plus a smooth, depth-decaying elastic warp with temporally correlated coefficients (not removable by translation-only registration). Motion is applied directly to the beamformed IQ cube (a stress test rather than a full scatterer-level simulator), after which we rerun the same baseline and STAP pipeline (KA disabled to isolate STAP).

We report two label-free robustness endpoints as functions of motion amplitude: (i) PD-mode score-map correlation versus the no-motion reference (after rigid alignment and a boundary crop), and (ii) a flow-proxy hit rate at a matched background-proxy tail operating point (background-proxy tail rate = 10^{-3}), expressed in score space as $\mathbb{P}\{S_{\text{PD}} \geq \tau \mid \text{bg proxy}\} = 10^{-3}$. Concretely, for each clip/window we define a fixed pixel-level flow-proxy mask from the no-motion baseline PD-derived map pd_{base} (top quantile; minimum pixel count) and a background-proxy mask from

Window	$\mathrm{corr}_{\mathrm{self}}$ base	$\mathrm{corr}_{\mathrm{self}}$ STAP (pre-KA)	frac(STAP>base) $\mathrm{corr}_{\mathrm{self}}$	hit $\mathrm{rate}_{\mathrm{flow}}$ base @ bg tail = 10^{-3}	hit $\mathrm{rate}_{\mathrm{flow}}$ STAP @ bg tail = 10^{-3}	frac(STAP>base) hit $\mathrm{rate}_{\mathrm{flow}}$
0:128	0.188	0.454	1.0	0.000	0.041	1.0
64:192	0.186	0.399	0.9	0.000	0.040	1.0
122:250	0.182	0.364	0.8	0.000	0.041	1.0

Table 17: Shin RatBrain Fig3 motion-sweep endpoints at 2 px amplitude (median over 10 clips) under the frozen Shin-U profile, with rigid registration enabled and KA disabled (STAP reported pre-KA). $\mathrm{corr}_{\mathrm{self}}$ is Pearson correlation between each method's PD-mode score map at the given motion amplitude and its own no-motion reference, after phase-correlation alignment and a boundary crop. hit $\mathrm{rate}_{\mathrm{flow}}$ is computed on a fixed, method-independent flow-proxy mask derived from the no-motion baseline PD-derived map pd_{base}: for each run/method we choose τ so that the background proxy achieves a tail rate 10^{-3} under the right-tail score rule $\mathbb{P}\{S_{\mathrm{PD}} \geq \tau \mid \text{bg proxy}\} = 10^{-3}$, then evaluate the fraction of flow-proxy pixels satisfying $S_{\mathrm{PD}} \geq \tau$. frac(STAP>base) reports the paired fraction of clips where STAP exceeds MC–SVD at 2 px on the corresponding endpoint.

its complement; for each run/method we choose τ on that run's background proxy and then report the fraction of flow-proxy pixels satisfying $S_{\mathrm{PD}} \geq \tau$. As a degeneracy check (to rule out "stability by collapse"), the sweep also records non-collapse diagnostics (map std/entropy and background variance ratios). A post-hoc sensitivity sweep recomputes correlation under different crop margins and with alignment disabled. Over 10 pre-selected clips and three overlapping time windows (0:128, 64:192, 122:250) under the frozen Shin-U profile (PRF = 1 kHz, $L_t = 64$, baseline MC–SVD with rigid registration enabled), STAP PD-mode score maps are consistently more stable than MC–SVD as motion increases: at a 2 px motion amplitude, median correlation improves from ≈ 0.18–0.19 (MC–SVD) to ≈ 0.36–0.45 (STAP) across windows, and the flow-proxy hit rate improves from 0 to ≈ 0.04 at bg tail rate $= 10^{-3}$. Table 17 summarizes these endpoints; Figures 21–22 show motion curves.

Figure 21 shows the canonical window (0:128) motion curves, while Figure 22 shows the same endpoints across all three overlapping windows; Figure 23 provides a rigid-registration ablation (`reg0`).

Figure 24 shows the corresponding KA prior summary panel for the frozen Shin-U profile. Figure 25 shows a representative per-clip multipanel view separating the baseline PD-derived map, the STAP PD-mode score map pre-KA, the KA gate/scale, and the final PD-mode score map post-KA, along with key band-energy maps and the flow-proxy mask that defines the protected set.

Limitations and claim boundaries (Shin). The Shin RatBrain Fig3 analysis is intended as an end-to-end *real-IQ viability and generalization* anchor, not a clinical detection benchmark. Because there are no task labels and no ground-truth vessel segmentation, quantitative endpoints reported here are label-free proxies: baseline comparisons use deterministic flow/background proxy masks and report both threshold-free proxy separation (AUC) and tail-calibrated flow-proxy hit rates, while motion-sweep endpoints measure within-method consistency relative to each method's own no-motion reference and persistence of a fixed flow proxy under a matched background-proxy tail operating point. In addition, the "brain-like" motion stress test is an image-domain warp applied to beamformed IQ (rigid + depth-decaying elastic components) rather than a scatterer-level forward model; it is designed to mimic the specific high-rank, spatially varying residual-motion mismatch represented in Brain-SkullOR, not to reproduce full ultrasound physics. Accordingly, we interpret Shin results as evidence of (i) deterministic real-IQ ingestion and band-energy summaries, (ii) KA prior auditability and invariants (shrink-only; exact protected-set invariance), and (iii) self-

Pitfall (what can go wrong)	Why it matters	Mitigation sentinel (what we enforce and where it is logged)
Vendor raw-container decode ambiguity (header sizing, line ordering, beam/shot layout)	A mis-decode can silently invalidate structural masks and invalidate Doppler scores and band-energy statistics.	Decoder sanity checks reconstruct B-mode and compare to dataset screenshots; ordering/coherence checks validate CFM layout. Summaries are provided in Appendix E.
Missing or incorrect Doppler PRF (Twinkling metadata does not reliably store PRF)	PRF determines discrete band binning and peak-occupancy statistics; wrong PRF can make Pf/Pa empty or overlapping under short ensembles, creating misleading "actionability".	PRF is treated as an explicit, frozen experimental assumption per sequence family (Appendix E) and recorded in bundle metadata. If binned Pf/Pa are empty/overlap, we force $\phi_i \in \mathcal{R}_0$ with an explicit reason.
Short-ensemble band discretization (coarse $\Delta f = \mathrm{PRF}/N$)	Hz-narrow bands can collapse to DC-only or become indistinguishable with $N \in \{5, 9, 17\}$.	Band operators for short ensembles are constructed as exact orthogonal DFT-bin projectors with non-overlapping bin partitions; bin membership and overlap counts are exported per run and audited by the band-sanity sentinel (Section 3).
Tail sample insufficiency ($n_{\mathrm{tail}} < n_{\mathrm{tail,min}}$)	Tail association metrics (e.g. Δ_{tail}) become undefined or noisy if the background tail is too small, which can spuriously toggle $\mathcal{R}_2$ eligibility.	Tail sizing is made explicit: we record tail sizes and use a size-aware tail fraction so that Δ_{tail} is only computed when supported; otherwise we remain conservative ($\mathcal{R}_0/\mathcal{R}_1$ with an explicit reason).
Proxy masks misread as ground truth (real data)	Proxy-derived labels can create circular evaluation or leakage.	We restrict quantitative ROC to regimes with explicit labels: simulator-truth masks (Brain-*) and B-mode structural masks (Gammex flow phantom sequences). Proxy masks are used only for band-energy statistics/protected sets and for conditional execution, never for detection efficacy claims.
Conditional execution (compute gating) changes the decision rule	Same-window proxy gating can measurably change operating characteristics (including improving them), so it cannot be presented as a cost-free speedup.	Conditional execution is treated as a separate algorithm variant; any performance reporting is accompanied by leakage controls (same-window vs disjoint-window vs random-mask). Reproducibility details are provided in Appendix L.

Table 18: Integrity checks and regime sentinels. Each sentinel is deterministic and label-free; it either records provenance/QC metrics or forces conservative KA behavior (e.g. $\phi_i \in \mathcal{R}_0$, $w \equiv 1$) with an explicit reason label.

consistency under controlled residual-motion corruption, while leaving ROC uplift claims to labeled regimes (Brain-* simulations) and to future raw-IQ datasets with task/landmark structure.

C Integrity checks and regime sentinels

We implement a reproducible processing pipeline (decode → baseline → STAP score → spatial-spectral features → optional KA regularization) and treat tail-regime validity as part of the experimental design. This appendix enumerates common pitfalls (metadata ambiguity, discretization artifacts, proxy-mask circularity, and compute-gating leakage) and the deterministic, label-free sentinels used to detect them. When a sentinel triggers, we either record provenance and proceed conservatively or force $\phi_i \in \mathcal{R}_0$ (uninformative prior; $w \equiv 1$) with an explicit reason label so that "KA off" is an auditable success mode under flat or non-actionable evidence.

D Practical Band Design for Brain fUS (PRF 1.5–5 kHz, T 8–16)

We provide practical band guidelines for selecting Pf/guard/Pa with explicit physical anchors.

For brain fUS at center frequencies $f_0 \approx 6$–15 MHz and PRF ≈ 1.5–5 kHz [1, 3, 6, 7, 25]:

- Microvascular brain flows (arterioles/venules) in humans and rodents are typically in the range $v \approx 3$–20 mm/s for the vessels that dominate functional responses [1, 3]. At $f_0 = 7.5$ MHz and $c \approx 1540$ m/s, this corresponds to Doppler shifts $f_D \approx 30$–200 Hz.
- Low–frequency tissue motion (respiration, cardiac pulsation) produces strong spectral content below ≈ 20 Hz; this should be contained in Po and largely removed by high–pass filtering and/or SVD.
- Large–vessel flows (e.g. MCA–like arteries) reach $v \approx 0.3$–0.8 m/s, corresponding to f_D in the kHz range [4, 5]. Depending on PRF and angle, a portion of this energy appears in high Doppler bands and can alias into the >400 Hz region corresponding to Pa [16, 18].

Thus a reasonable fixed band design for brain fUS with $T \approx 8$–16 and PRF ≈ 1.5–3 kHz is:

- Flow band $\mathcal{B}_f$: $[30, 250]$ Hz — microvascular flows ≈ 3–20 mm/s.
- Guard band $\mathcal{B}_g$: $[250, 400]$ Hz — protects against band–edge leakage.
- Alias band $\mathcal{B}_a$: $[400, \text{Nyquist}]$ Hz — high Doppler where large–vessel/aliased flows and certain motion artifacts live.

In discrete terms, for $T = 8$ at PRF=1.5 kHz (bin spacing 187.5 Hz), this maps to:

- P_f on first off–DC bin (with DPSS leakage covering ≈ 30–250 Hz),
- guard around the second bin,
- P_a on higher bins with bin–level guard as per BG–1/BG–2.

E Twinkling PRF overrides and decoder sanity

The Twinkling RawBCF sidecars do not reliably store Doppler PRF, so all Twinkling experiments in this manuscript use an explicit PRF override (`prf_hz`) frozen per sequence family and recorded in each bundle's `meta.json`. Table 19 lists the PRF overrides used for the Gammex structural-label experiments and the calculi hygiene experiments; a full extracted table from bundle metadata is written to `reports/twinkling_prf_override_table.csv` / `reports/twinkling_prf_override_table.json`.

F Macé atlas alignment sanity check

To reduce the risk that offline atlas-derived summaries are dismissed as misregistered, Figure 26 overlays the Allen atlas alignment inputs used throughout the Macé section: atlas region masks (`atlas.Regions`) and the vascular atlas volume (`atlas.Vascular`) mapped into scan planes using the provided affine transform (`Transformation.mat`).

G PD-only KA-prior summary panel (hemodynamic): band-energy statistics and aggregation

This appendix specifies a PD-only instantiation of the KA spatial-spectral prior and the associated summary-panel aggregation used for systematic plane sweeps (e.g. Macé/Urban whole-brain mouse

Sequence family (RawBCF)	N shots	PRF override (Hz)
Flow in Gammex phantom (along – linear probe; `RawBCFCine.par`)	17	2500
Flow in Gammex phantom (across – linear probe; `RawBCFCine_*_17.par`)	17	2500
Twinkling/Flash artifacts on artificial calculi (`calcifications.par`)	9	500
Twinkling artifact on artificial calculi (`RawBCFCine_29062016_174207.par`)	9	500

Table 19: Frozen PRF overrides for Twinkling RawBCF sequences used in this manuscript. For the Gammex flow phantom we use PRF= 2500 Hz as reported for the Gammex example acquisition in [35]. For the calculi sequences the dataset provides screenshots that encode PRF (e.g. `picture (prf=500HZ).png`); we treat PRF as an explicit experimental assumption and enforce band-sanity sentinels to avoid empty/overlapping bands under short ensembles. Values are consistent with `reports/twinkling_prf_override_table.csv`.

Sequence family	B-mode vs PNG corr (best)	coherence ratio (beam/shot)
Gammex along – linear17 (`RawBCFCine.par`)	0.737	2.31
Gammex across – linear17 (`RawBCFCine_*_17.par`)	0.610	2.46
Calculi calcifications (`calcifications.par`)	0.382	2.92
Calculi artificial (`RawBCFCine_29062016_174207.par`)	0.415	1.08

Table 20: Twinkling RawBCF decoder sanity metrics. "B-mode vs PNG corr" is the correlation between a decoded B-mode envelope image (log-compressed) and a dataset-provided screenshot (after bilinear resize; max over left/right flips). "Coherence ratio" is the ratio of mean temporal coherence under beam-major ordering versus shot-major ordering; values > 1 support the beam-major assumption used throughout. Per-sequence decode reports are written to `reports/twinkling_decode_sanity/*/decode_report.json`; the aggregate summary is written to `reports/twinkling_decode_sanity_summary.csv` / `reports/twinkling_decode_sanity_summary.json`.

fUS). The key discipline is unchanged: *runtime* prior selection is label-free (proxy sets), while atlas-defined H_1/H_0 labels are used only *offline* for descriptive reporting and are never used as prior inputs.

A) PD-only band-energy feature definitions (math)

PD movie and tiling. Fix one coronal plane (one "configuration") with a PD movie

$$P[t, y, x] \in \mathbb{R}_{\geq 0}, \qquad t = 0, \ldots, T-1,$$

sampled at rate $f_s = 1/dt$ (Macé: $f_s \approx 10$ Hz). Tile each frame into overlapping spatial tiles Ω_i of size $h \times w$ (Macé: 8×8) with stride s (Macé: $s = 3$).

Tile PD time series. Define the tile-averaged PD time series

$$p_i[t] := \frac{1}{|\Omega_i|} \sum_{(y,x)\in\Omega_i} P[t, y, x].$$

For robustness across tiles, we optionally use a log-stabilized, mean-removed series

$$u_i[t] := \log(p_i[t] + \epsilon_{\mathrm{pd}}), \qquad \tilde{u}_i[t] := u_i[t] - \frac{1}{T}\sum_{t'=0}^{T-1} u_i[t'].$$

Hankel snapshots and whitening. Fix a slow-time aperture L_t (tens of frames in PD-only). Define $K = T - L_t + 1$ and Hankel snapshots

$$x_{i,k} := \begin{bmatrix} \tilde{u}_i[k] \\ \tilde{u}_i[k+1] \\ \vdots \\ \tilde{u}_i[k+L_t-1] \end{bmatrix} \in \mathbb{R}^{L_t}, \qquad k = 0, \ldots, K-1.$$

Estimate a per-tile covariance (SCM or robust variant)

$$\widehat{R}_i := \frac{1}{K}\sum_{k=0}^{K-1} x_{i,k} x_{i,k}^{\mathrm{H}},$$

apply conditioning/diagonal loading to obtain R_i, and whiten

$$y_{i,k} := R_i^{-1/2} x_{i,k}.$$

We pool whitened snapshots to a single per-tile slow-time vector

$$z_i := \frac{1}{K}\sum_{k=0}^{K-1} y_{i,k} \in \mathbb{R}^{L_t}.$$

Hemodynamic band projectors and energies. Let $\mathcal{B}_f, \mathcal{B}_g, \mathcal{B}_a$ denote Pf/guard/Pa hemodynamic bands (Macé: $\mathcal{B}_f = [0, 0.5]$ Hz, $\mathcal{B}_g = (0.5, 1.0]$ Hz, $\mathcal{B}_a = (1.0, 3.0]$ Hz). Build fixed temporal projectors (P_f, P_g, P_a) onto the span of discrete Fourier tones whose frequencies lie in these bands, and define $P_o := I - P_f - P_g - P_a$ ("other"/out-of-band). Define band energies

$$E_f(i) := z_i^{\mathrm{H}} P_f z_i, \quad E_g(i) := z_i^{\mathrm{H}} P_g z_i, \quad E_a(i) := z_i^{\mathrm{H}} P_a z_i, \quad E_o(i) := z_i^{\mathrm{H}} P_o z_i,$$

and $E_{\mathrm{tot}}(i) := E_f + E_g + E_a + E_o + \epsilon$.

Risk metrics and Pf-peak statistics. We use the same KA feature primitives as in the Doppler regime:

$$m_{\mathrm{alias}}(i) := \log\frac{E_a(i)+\epsilon}{E_f(i)+\epsilon}, \qquad m_{\mathrm{guard}}(i) := \log\frac{E_g(i)+\epsilon}{E_f(i)+\epsilon}, \qquad r_g(i) := \frac{E_g(i)}{E_{\mathrm{tot}}(i)}.$$

Define a whitened periodogram of the pooled vector z_i,

$$\widehat{z}_i[n] := \sum_{\ell=0}^{L_t-1} z_i[\ell]\, e^{-2\pi i n\ell/L_t}, \qquad S_i[n] := |\widehat{z}_i[n]|^2, \qquad n = 0, \ldots, \lfloor L_t/2 \rfloor,$$

and the (non-DC) peak frequency $f_{\text{peak}}(i) = \frac{n_{\text{peak}}}{L_t dt}$ where $n_{\text{peak}}(i) := \arg\max_{n\geq 1} S_i[n]$. The Pf-peak indicator and occupancy on a set $\mathcal{T}$ are

$$\pi_f(i) := \mathbf{1}\{f_{\text{peak}}(i) \in \mathcal{B}_f\}, \qquad \text{PfPeakFrac}(\mathcal{T}) := \frac{1}{|\mathcal{T}|}\sum_{i\in\mathcal{T}} \pi_f(i).$$

B) PD-only proxy sets and prior inputs (math)

The PD-only prior uses label-free proxy sets based on a PD-derived flow-coverage proxy, analogous to the IQ flow-mask coverage c_f. Let $\bar{P}[y,x] := \frac{1}{T}\sum_t P[t,y,x]$ be the mean PD image (optionally restricted to a brain mask). Define a high-quantile flow proxy mask

$$M_{\text{flow}}(y,x) := \mathbf{1}\{\bar{P}[y,x] \geq Q_{q_{\text{flow}}}(\bar{P})\},$$

and per-tile flow coverage

$$c_f(i) := \frac{1}{|\Omega_i|}\sum_{(y,x)\in\Omega_i} M_{\text{flow}}(y,x) \in [0,1].$$

With frozen design thresholds c_{bg} and c_{flow} (typical $c_{\text{bg}} \approx 0.05$, $c_{\text{flow}} \approx 0.20$), define

$$\mathcal{T}_{\text{bg}} := \{i:\ c_f(i) \leq c_{\text{bg}}\}, \quad \mathcal{T}_{\text{flow}} := \{i:\ c_f(i) \geq c_{\text{flow}}\}, \quad \mathcal{T}_{\text{nonbg}} := \{i:\ c_f(i) > c_{\text{bg}}\}.$$

If $\mathcal{T}_{\text{flow}}$ is too small (insufficient proxy support), a conservative fallback is to define proxy sets via tilewise mean-$\bar{P}$ quantiles and record an explicit "proxy_fallback" reason code.

C) Prior-regime selection for PD-only (math)

Baseline score for PD-only prior. For Macé, the baseline scalar detector is PD-z on tile time series: choose a baseline segment B (e.g. first third of frames) and define

$$s_{\text{base}}(i) := \max_{t\notin B} \frac{u_i[t] - \mu_i^{(B)}}{\sigma_i^{(B)} + \epsilon_z},$$

with $\mu_i^{(B)}, \sigma_i^{(B)}$ the mean and standard deviation of $u_i[t]$ over $t \in B$. Higher s_{base} means more activation.

Risk-mode selection and integrity checks. The PD-only prior follows the same posture as the Doppler-IQ prior: if risk statistics are flat (e.g. $\text{IQR}(m_{\text{alias}} \mid \mathcal{T}_{\text{bg}})$ too small) or proxy support is insufficient, we default to the uninformative regime ($\mathcal{R}_0$; $w \equiv 1$) with an explicit reason label (e.g. `alias_metric_flat`, `insufficient_bg_samples`). If guard dominates (e.g. $Q_{0.90}(r_g \mid \mathcal{T}_{\text{bg}})$ above a frozen threshold), we disallow the actionable regime $\mathcal{R}_2$ and either remain in a guard-risk safety lane ($\mathcal{R}_1$; requiring tail association of r_g) or default to $\mathcal{R}_0$ as out of regime (e.g. `guard_extreme_out_of_regime`).

Actionability checks (actionable regime $\mathcal{R}_2$, label-free). The actionable regime $\mathcal{R}_2$ (research-only eligibility; reporting-only) requires additional evidence that risk statistics explain high-score background tails. In the selected risk mode ($m_{\text{risk}} = m_{\text{alias}}$ or m_{guard}), define

$$\Delta_{bf} := \text{median}(m_{\text{risk}} \mid \mathcal{T}_{\text{bg}}) - \text{median}(m_{\text{risk}} \mid \mathcal{T}_{\text{flow}}),$$

and within $\mathcal{T}_{\text{bg}}$ define a tail subset $\mathcal{T}_{\text{tail}}$ as the top 1% of s_{base} and a mid subset $\mathcal{T}_{\text{mid}}$ as a central quantile band; require

$$\Delta_{\text{tail}} := \text{median}(m_{\text{risk}} \mid \mathcal{T}_{\text{tail}}) - \text{median}(m_{\text{risk}} \mid \mathcal{T}_{\text{mid}}) \geq \Delta_{\text{tail,min}}.$$

Finally, we enforce a Pf-realization eligibility condition for $\mathcal{R}_2$:

$$\text{PfPeakFrac}(\mathcal{T}_{\text{flow}}) \geq p_{f,\text{min}}.$$

If these checks fail, the prior remains in $\mathcal{R}_0$ or $\mathcal{R}_1$ (guarded only) and does not enter $\mathcal{R}_2$.

Coverage sentinels. As in the Doppler-IQ prior, penalization must be neither inert nor global: after defining protected/candidate sets (below), the penalized fraction

$$p_{\text{shrink}} := \mathbb{P}\{g = 1 \mid \mathcal{T}_{\text{cand}}\}$$

must lie in frozen bounds (e.g. $p_{\text{min}} \leq p_{\text{shrink}} \leq p_{\text{max}}$), otherwise we default to $\mathcal{R}_0$ ($w \equiv 1$) with reason `coverage_out_of_bounds`.

D) PD-only shrink-only prior mapping (optional; summary-panel parity)

The PD-only instantiation supports both a hard suppression used in the Macé systematic-sweep reporting and a soft shrink mapping for summary-panel parity. Define a protected set $\mathcal{T}_{\text{prot}}$ (tiles with high flow coverage and/or very high baseline scores) and a candidate set $\mathcal{T}_{\text{cand}}$ (non-protected tiles in an uncertain score window). Using a background-calibrated risk threshold $\tau_{\text{risk}} = Q_{q_{\text{risk}}}(m_{\text{risk}} \mid \mathcal{T}_{\text{bg}})$, define the penalization indicator

$$g(i) = \mathbf{1}\{i \in \mathcal{T}_{\text{cand}},\ m_{\text{risk}}(i) \geq \tau_{\text{risk}}\}.$$

Hard suppression (Macé sweep): apply the binary indicator g as a hard suppression on positive activations,

$$s_{\text{KA}}(i) = s_{\text{base}}(i) - g(i)\max(s_{\text{base}}(i), 0),$$

which is equivalent to setting positive penalized tiles to zero while leaving negative scores unchanged. *Soft shrink (summary-panel parity):* set $w(i) = \max(w_{\text{min}}, \exp(-k\,u(i)))$ with a robust normalized exceedance $u(i) \geq 0$ and apply $s_{\text{KA}}(i) = s_{\text{base}}(i) - (1 - w(i))\max(s_{\text{base}}(i), 0)$. In both cases, $g(i) = 0$ (and $w(i) = 1$) on $\mathcal{T}_{\text{prot}}$ enforces the protected-set invariance and $s_{\text{KA}}(i) \leq s_{\text{base}}(i)$ enforces shrink-only behavior.

E) Summary-panel schema and aggregation (PD-only)

For PD-only datasets the summary panel is computed per plane (configuration) and written as a CSV with one row per (scan, plane). Each row records: (i) identifiers and frozen settings (scan/plane, T, f_s, L_t, tile geometry, band edges), (ii) proxy support counts ($|\mathcal{T}_{\text{bg}}|$, $|\mathcal{T}_{\text{flow}}|$), (iii) prior outputs (region $\in \{\mathcal{R}_0, \mathcal{R}_1, \mathcal{R}_2\}$, reason label, risk_mode; exported in code as `state/reason`), (iv) key band-energy statistics (PfPeakFrac($\mathcal{T}_{\text{flow}}$), $Q_{0.90}(r_g \mid \mathcal{T}_{\text{bg}})$, IQR($m_{\text{risk}} \mid \mathcal{T}_{\text{bg}}$), Δ_{bf}, Δ_{tail}), and (v) KA coverage/strength when active (p_{shrink} and scale/shrink quantiles). Atlas-labeled descriptive fields are prefixed `offline_` (e.g. `offline_pf_peak_frac_H1`, `offline_pf_peak_frac_H0`, `offline_hits_pre/post`) to make clear that they are not prior inputs.

Summary-panel aggregation mirrors the Doppler-IQ diagnostic panel: histograms of prior regions and (region, reason) pairs, a risk-mode histogram, distributions of key band-energy statistics (Pf peak fraction, guard q90, risk IQR), and (for active planes) shrink coverage/strength summaries. Any atlas-stratified Macé plots (e.g. PfPeak(VIS/SC/LGd) vs PfPeak(CP/CA/DG), alias separation) are presented as a separate, explicitly offline/descriptive panel.

F) "Do not overclaim" messaging (PD-only summary panel)

When reporting a PD-only prior summary panel, we use the following posture in captions and limitations:

- PD-only band-energy statistics operate on low-rate PD time series and are an auditability check; they do not validate Doppler-side uplift on raw IQ.
- The uninformative regime ($\mathcal{R}_0$; $w \equiv 1$) is an intended success outcome when feature evidence is flat or non-actionable; the summary panel is designed to make these cases explicit via reason labels.
- Atlas-stratified results on Macé are descriptive; prior-positive slices identified by scanning the same dataset are not treated as held-out confirmation.
- The actionable regime ($\mathcal{R}_2$) is a reporting discipline requiring additional label-free actionability evidence; $\mathcal{R}_1$ is guarded only and should not be presented as a clinical performance guarantee.

H Algorithmic details (supplementary)

H.1 Knowledge-Aided Prior Regime Selection (code defaults)

The KA prior regime selection is implemented in `pipeline/stap/ka_contract_v2.py`. The entry point is `evaluate_ka_contract_v2`. It consumes per-tile arrays S_{base}, m_{alias}, r_g, c_f, and PfPeak and returns a JSON-serializable report with fields `state` (region), `reason`, and a `metrics` dictionary. The frozen defaults used throughout this repository are the `KaContractV2Config` dataclass defaults:

- **Proxy thresholds:** $c_{\text{bg}} = 0.05$, $c_{\text{flow}} = 0.20$.
- **Candidate/protected score quantiles:** candidate window $q_{\text{lo}} = 0.30$, $q_{\text{hi,cand}} = 1.0$; optional score-protect quantile $q_{\text{hi,prot}} = 0.99999$ (`protect_hi_by_score=true`).
- **Risk threshold:** $q_{\text{risk}} = 0.90$ applied to m_{alias} on $\mathcal{T}_{\text{bg}}$.
- **Feature integrity (flatness + guard):** IQR($m_{\text{alias}} \mid \mathcal{T}_{\text{bg}}$) ≥ 0.25; actionable eligibility disabled if $Q_{0.90}(r_g) > 0.25$; "guard safety" lane allowed up to $Q_{0.90}(r_g) \leq 0.70$ if guard is

actionable, requiring $\mathrm{IQR}(r_g \mid \mathcal{T}_{\mathrm{bg}}) \geq 0.03$ and $\Delta_{\mathrm{tail,guard}} \geq 0.01$.

- **C2 actionability thresholds:** $\Delta_{bf} \geq 0.30$ and $\Delta_{\mathrm{tail}} \geq 0.20$ (computed on $\mathcal{T}_{\mathrm{bg}}/\mathcal{T}_{\mathrm{flow}}$ proxies as in Section 3).
- **C2 Pf-realization eligibility:** $\mathrm{PfPeakFrac}(\mathcal{T}_{\mathrm{flow}}) \geq 0.05$ with $|\mathcal{T}_{\mathrm{flow}}| \geq 50$.
- **Coverage bounds:** safety lane $p_{\mathrm{shrink}} \in [0.02, 0.25]$ (guard lane $[0.02, 0.15]$), actionable lane $p_{\mathrm{shrink}} \in [0.05, 0.40]$.
- **Invariance sentinel:** if $\mathrm{IQR}(\log w \mid \text{penalized}) < 0.02$ then return C0 with `invariance_no_effect`.
- **Sample support:** $n_{\min} = 500$ valid tiles, $n_{\mathrm{bg,min}} = 200$ background-proxy tiles, $n_{\mathrm{cand,min}} = 200$ candidate tiles, $n_{\mathrm{tail,min}} = 20$, $n_{\mathrm{mid,min}} = 20$.

The implementation exports `C0_OFF`, `C1_SAFETY`, or `C2_ACTIONABLE` (exported as `C2_UPLIFT` in current metadata for backward compatibility). These correspond to feature-space regimes $\mathcal{R}_0/\mathcal{R}_1/\mathcal{R}_2$ in the main text. Disable reasons are explicit and frozen in code; common ones include: `missing_inputs`, `shape_mismatch`, `insufficient_samples`, `insufficient_bg_samples`, `alias_metric_flat`, `guard_metric_missing`, `guard_extreme_out_of_regime`, `guard_dominates_not_actionable`, `pf_peak_flow_below_min`, `too_few_candidates`, `coverage_out_of_bounds`, and `invariance_no_effect`. Successful enablement returns `reason=ok`. For short-ensemble Doppler runs, bundle writers may additionally force C0 with `reason=band_sanity_invalid` when discrete band binning yields empty/overlapping Pf/Pa after discretization (see `sim/kwave/icube_bundle.py`).

H.2 Score-space shrink mapping and exported maps

When score-space KA is enabled, per-tile shrink factors $w(i) \in (0, 1]$ are derived from the regime report by `derive_score_shrink_v2_tile_scales` in `pipeline/stap/ka_contract_v2.py` and applied in score space as

$$S_{\mathrm{post}}(i) = w(i)\, S_{\mathrm{pre}}(i), \qquad w(i) \in (0, 1].$$

For convenience (and compatibility with legacy `pd` arrays), bundles store an equivalent per-pixel *scale* map $\gamma(i) = 1/w(i) \geq 1$ as `ka_scale_map.npy`, averaged over overlapping tiles and applied only on the union of penalized tiles as

$$\mathrm{PD}_{\mathrm{post}}(i) = \mathrm{PD}_{\mathrm{pre}}(i)/\gamma(i).$$

A binary `ka_gate_map.npy` stores the penalized region. Protected pixels are enforced by setting $\gamma = 1$ on the protected set (flow mask and extreme-score pixels as in Section 3).

Implementation score conventions. Throughout the manuscript, quantitative low-FPR operating points are reported using the right-tail detector score maps intended for labeled ROC and robustness reporting (Table 1). In the exported bundles these are stored as `score_base` (baseline power Doppler score) and `score_stap_preka` (pre-KA matched-subspace STAP score), with optional baseline comparators `score_base_pdlog` and `score_base_kasai`. PD-mode scores (`score_pd_*.npy`) are legacy artifacts used for proxying/auditing and for PD-only instrumentation regimes; they are not used as the primary detector scores for labeled ROC reporting in this manuscript. When KA is enabled, the exported maps enforce a shrink-only mapping on the chosen score: in this right-tail score space, $S_{\mathrm{post}} = w\, S_{\mathrm{pre}}$ with $w \in (0, 1]$; in PD-mode, `ka_scale_map.npy` stores $\gamma = 1/w \geq 1$ and is applied as $\mathrm{PD}_{\mathrm{post}} = \mathrm{PD}_{\mathrm{pre}}/\gamma$ on penalized regions.

I Simulation details (supplementary)

I.1 k-Wave brain pilots (Brain-* profiles)

The k-Wave brain pilots used in this repository are generated by `sim/kwave/pilot_motion.py` and are orchestrated by Makefile targets in `Makefile` (`r4c-generate`, `r4c-hab-generate`, `r4c-skull-generate`, `r4c-pial-generate`). The shared acquisition/geometry settings are:

- Grid: $N_x = N_y = 240$ with $dx = dy = 90$ μm; PML size 16; $c \approx 1540$ m/s; $f_0 = 7.5$ MHz; 3 cycles.
- Angles: $(-12, -6, 0, 6, 12)^\circ$; ensembles: 5; pulses per ensemble: 64 (so $T = 320$ compounded frames); PRF = 1500 Hz.
- Tiling: 8×8 tiles, stride 3; $L_t = 8$ (conservative profile temporal aperture).

Synthetic flow/alias/clutter overlays. To make Doppler/alias structure explicit and reproducible, the pilots apply deterministic slow-time overlays on the beamformed cube (code: `sim/kwave/common.py`). The canonical Brain-OpenSkull overlays in the Makefile are:

- Flow Doppler on the flow mask: `--flow-doppler-min-hz 60`, `--flow-doppler-max-hz 180`.
- Background alias tone: `--bg-alias-hz 650` with fraction 0.3 in depth $[0.30, 0.70]$ and jitter ± 50 Hz.
- Colored temporal clutter: `--clutter-beta` 1.0 with target SNR 20 dB over depth $[0.20, 0.95]$.
- Aperture phase screen: `--aperture-phase-std 0.8` rad, correlation length 14 elements.

Brain-Pial128 uses 4 ensembles with 32 pulses each ($T = 128$ compounded frames) and concentrates shallow alias in depth $[0.15, 0.30]$ with a stronger alias fraction and a vibration tone (Makefile variables `R4C_PIAL_*`). Brain-AliasContract and Brain-SkullOR are generated by invoking `pilot_motion.py` with `--profile Brain-AliasContract` and `--profile Brain-SkullOR`, which (when `--regime` is left as `generic`) select the HAB presets `hab_contract` and `hab_v3_skull` inside `pilot_motion.py`; in the Makefile we explicitly pass the pial/deep alias and vibration parameters for these regimes (`R4C_HAB_*`) and rely on `hab_v3_skull` primarily for additional depth/aberration tagging.

I.2 Replay/analysis pipeline used for ROC and latency

Downstream processing is run on the saved pilot angle data using `scripts/replay_stap_from_run.py`, which loads angle directories (or ensembles of angle directories), forms the slow-time cube, applies the chosen baseline, runs STAP, and writes a standard output bundle (`sim/kwave/common.py:write_acceptance_bundle`). Low-FPR ROC points and feasibility telemetry are extracted using `scripts/hab_contract_check.py`.

For the conservative-profile STAP PD runs (`--stap-profile clinical`) used in the latency suite and robustness sweeps, we additionally use `scripts/run_brain_replay_sweep.py`, which:

- Reuses a fixed pilot per (regime, PRF) and replays STAP with a fixed PD configuration on 64-frame windows (`--time-window-length 64`).
- Enables conditional STAP and the PD-only fast path via environment variables: `STAP_FAST_PD_ONLY=1`, `STAP_MAX_SNAPSHOTS=64`, `STAP_SNAPSHOT_STRIDE=4`.
- Uses an MC–SVD baseline in "literature" mode (`--svd-profile literature`, Brain-* frozen energy fraction $e = 0.90$ when explicit rank/energy are not provided; sweep report `reports/brain_mcsvd_energy_sweep_seed1.csv`), and simulator-truth evaluation masks

(`--flow-mask-mode default`). Conditional STAP compute gating uses the exported label-free PD proxy mask `mask_flow_pd.npy` derived from the baseline PD brightness proxy (quantile 0.995, depth $[0.25, 0.85]$, 2 dilation iterations).

J Implementation details (supplementary)

J.1 RPCA baseline (full)

The RPCA baseline used in this repository is implemented as `sim/kwave/common.py:_baseline_pd_rpca`. It performs a windowed Principal Component Pursuit (PCP) using an inexact augmented Lagrangian method (IALM) on a downsampled Casorati matrix:

- Spatial downsampling: complex average pooling by factor 2 when $\max(H, W) > 128$.
- Temporal windows: default subwindow length $t_{\text{sub}} = 80$ (or shorter if $T < 80$); windows are processed sequentially.
- Optimization: IALM with at most 80 inner iterations per window (`max_iters_inner=min(max_iters,80)`), tolerance 10^{-4}.
- Shrinkage: $\lambda = 1/\sqrt{\max(m, n)}$ when not supplied, where $m = T_w$ and $n = H_d W_d$.
- Low-rank step: singular value thresholding using either full SVD or a randomized SVD depending on matrix size, with a per-window rank heuristic $k = \min(8, \max(2, \lfloor T_w/10 \rfloor))$.
- Output PD: sum of sparse-component energy over windows on the downsampled grid, then nearest-neighbor upsampling to the original resolution.

J.2 HOSVD baseline (full)

The tensor HOSVD baseline is implemented as `sim/kwave/common.py:_baseline_pd_hosvd`. The cube $X \in \mathbb{C}^{T \times H \times W}$ is optionally spatially downsampled (complex average pooling). Mode-0/1/2 SVDs are computed on the unfoldings and ranks are chosen either explicitly (`--hosvd-ranks`) or by energy fractions (`--hosvd-energy-fracs`, default $(0.99, 0.99, 0.99)$). The low-rank approximation is reconstructed by projecting onto the multilinear subspace, and PD is the time-averaged residual energy $\frac{1}{T}\sum_t |X(t) - L(t)|^2$. In the brain baselines used in this repository, the Makefile target `r4c-replay-hosvd` runs HOSVD with `--hosvd-spatial-downsample 2` and `--hosvd-energy-fracs 0.99,0.99,0.99`.

J.3 STAP temporal core (full)

For k-Wave pilots and replay bundles, the temporal STAP implementation used for these exported bundles lives in `sim/kwave/common.py` (`_stap_pd_tile_lcmv` and helpers) and uses the same conceptual decomposition as the main text: Hankelize slow-time tiles, estimate per-tile covariances with a robust estimator (`tyler_pca` or Huber), apply diagonal loading with condition-number targets (MVDR load in `auto` mode), whiten, and compute band-energy statistics using a fixed Doppler tone grid. The conservative-profile PD fast path and conditional STAP are enabled by environment variables (`STAP_FAST_PD_ONLY=1`, `STAP_MAX_SNAPSHOTS`, `STAP_SNAPSHOT_STRIDE`) and by skipping tiles with zero flow-mask overlap (telemetry key `stap_tiles_skipped_flow0` in `meta.json`). For reproducible run settings, the Makefile targets and scripts in `scripts/` record the exact STAP arguments passed to `scripts/replay_stap_from_run.py` (e.g. `diag_load`, `mvdr_auto_kappa`, `grid_step_rel`, `max_pts`) and export them in each bundle's `meta.json`.

J.4 Covariance blending constraints and KA modes

We permit anisotropic shrinkage, but only *within* bands. In practice we consider two KA paths:

- **Updated (commuting band–scalar).** $Q^{\mathrm{H}}Q$ is forced to be bandwise scalar with target gains (s_f, s_a, s_o) (in code: $s_f = s_o \approx 1$, $s_a \approx 1.5$), small mixing, and no off–block priors; the content of R_0 beyond (P_f, P_a) is ignored.
- **Blend.** A trace–normalized blend $R_\beta = (1-\beta)R_t^{(n)} + \beta R_0^{(n)}$ with a small diagonal lift. In practice R_0 is overridden to a band–scalar form (Pf weight 1, Pa weight $\gamma_a > 1$, Po weight 1) so the geometry is explicit; β is treated as a fixed hyperparameter per regime.

Both modes enforce bandwise structure; there are no off–block priors.

J.5 CB–1: Band–Block Blending

We construct

$$R_\beta := (1-\beta)\,\widehat{R} + \beta\big(R_{0,f} \oplus R_{0,a} \oplus R_{0,o}\big) + \delta I,$$

where $R_{0,f} = P_f R_0 P_f$, etc., for some prior R_0. There are no off–block priors; $\beta \in [0,1)$ and $\delta \geq 0$.

Interpretation: this structure makes Q nearly block–diagonal in (P_o, P_f, P_g, P_a) while still allowing bandwise control to realize OC–2.

J.6 CB–2: Bandwise Spectral Targets

We choose (β, δ) to enforce OC–2 explicitly:

$$\lambda_{\min}(Q^{\mathrm{H}}Q|_{P_f}) \in [s_f^2(1-\delta_f), s_f^2(1+\delta_f)], \quad \text{etc.}$$

with design tolerances $\delta_\bullet \leq 0.1$ and separation $\Lambda = s_a/s_f \geq 1.25$.

J.7 CB–3: Bandwise Condition Number Safeguard

We constrain bandwise conditioning:

$$\kappa(R_\beta|_{P_f}) \leq \kappa_f^{\max}, \qquad \kappa(R_\beta|_{P_a}) \leq \kappa_a^{\max},$$

avoid an incompatibility between conditioning and preserving flow Rayleigh quotients.

J.8 Localization (restricting KA in score space)

A global band–constant transform with Q cannot improve ROC if applied everywhere (it is close to a scalar reweighting of BR/GLRT). To create re–ranking, KA must be *localized* to tiles where:

- scores are dominated by alias/undesired components,
- positives are unlikely to lie.

We define localization based on a combination of alias metrics, flow coverage, depth, PD, and registration. This supplements G–1/G–2 with a generalized alias score m_{alias} from Section A.1.

J.9 G–1: Alias–dominant tile selection

Given a baseline tile z, compute per–band energies (E_o, E_f, E_g, E_a) and a generalized alias score $m_{\text{alias}}(z)$ satisfying R–2.

Data–adaptive threshold. For BG tiles (or their proxy), collect $m_{\text{alias}}(z)$ and define

$$\tau_{\text{alias}} := Q_{q_{\text{alias}}}\big(m_{\text{alias}} \mid H_0\big),$$

where Q_q is the q–quantile and $q_{\text{alias}} \in [0.7, 0.9]$ is a design choice (typical: 0.8 or 0.9).

Define an alias gate

$$g_{\text{alias}}(z) = \mathbf{1}\left\{m_{\text{alias}}(z) \geq \tau_{\text{alias}}\right\}.$$

Band–edge purity. Additionally require that guard energy is small relative to Pf+Pa:

$$\frac{E_g}{E_o + E_f + E_a + E_g} \leq \eta_{\text{edge}}, \quad \eta_{\text{edge}} \leq 0.1.$$

Interpretation: this gate selects the top tail of alias–heavy tiles among negatives, while protecting against band–edge contamination. The distribution of m_{alias} across classes is governed by R–2; empirical validation (Section A.1) must show a nontrivial difference between H_0 and H_1.

J.10 G–2: Flow coherence protection

We can optionally compute a flow coherence metric FPC (first principal component fraction) within the flow band:

$$\text{FPC} := \frac{\lambda_1(\widehat{S}_f)}{\text{tr}(\widehat{S}_f)}, \qquad \widehat{S}_f := \frac{1}{K}\sum_{k=1}^{K} u_k u_k^{\text{H}}, \quad u_k := P_f z^{(k)},$$

where $z^{(k)}$ are projected slow–time vectors across DPSS tapers or within a local ensemble.

Gate *may* require

$$\text{FPC} \leq c_{\text{pc}}, \qquad c_{\text{pc}} \in [0.4, 0.6].$$

Interpretation: flow tiles tend to exhibit higher coherence in Pf; when enabled, we suppress penalization when FPC is large, thereby reducing positive penalization probability p_+. Alias/motion tiles have lower flow coherence and pass this condition. In the current runs this check is typically left off; coverage and depth localization carry most of the class separation.

J.11 G–2': Flow coverage protection

Let c_f be the tile–wise flow coverage fraction:

$$c_f := \frac{\#\{\text{pixels labeled as flow}\}}{\#\{\text{pixels in tile}\}},$$

using a label–free proxy such as a PD–derived flow mask.

We require

$$c_f \leq c_{f,\max}, \qquad c_{f,\max} \in [0.1, 0.2],$$

and optionally $c_f \geq c_{f,\min}$ to avoid pure background noise tiles (typical $c_{f,\min} \approx 0$).

Interpretation: we prefer penalization on tiles with *low to moderate* flow coverage. This is the regime where alias/motion dominates and where down–scaling scores is most beneficial. Dense flow tiles are either left to STAP alone or are protected via the flow coherence check. (In the current implementation, coverage is treated as an *upper* bound; missing coverage defaults to "pass" in this test.)

J.12 G–3: Baseline score quantile window

Let $S_{\text{base}}(z)$ be the baseline BR/GLRT score. Within a given depth stratum, compute score quantiles $q_{0.3}, q_{0.95}$ over BG tiles. Gate requires

$$S_{\text{base}}(z) \in [q_{0.3}, q_{0.95}].$$

Interpretation: avoid penalization on the noise floor (very low scores) and on already high–confidence positives (very high scores). This focuses KA on the part of the negative tail that matters near the target FPR.

J.13 G–4: Registration quality (optional)

If motion compensation / registration is used, include a registration quality metric (e.g. PSR) and require

$$\text{PSR}_{\text{reg}} \leq \text{PSR}_{\text{max}},$$

or any similar condition ensuring that tiles with gross registration failure are handled separately.

J.14 Offline validation of gating

On a held–out labeled configuration, define

$$p_+ := \mathbb{P}\{g = 1 \mid H_1\}, \qquad p_- := \mathbb{P}\{g = 1 \mid H_0\},$$

for the combined gate g that incorporates G–1, G–2, G–2', G–3, G–4. Require:

$$p_- \geq 0.2, \quad p_- \leq 0.5, \quad p_+ \leq 0.1 p_-.$$

Interpretation: a nontrivial fraction of negatives are gated (but not too many), while positives are rarely gated. This condition is central to the SL–2 quantile shift argument.

J.15 Summary: Compact Feasibility Checklist (Updated)

KA–STAP can be expected to outperform MC–SVD (or STAP–only) in the low-FPR regime if and only if:

1. **Regime preconditions.** Band occupancy R–1, alias separation R–2, and baseline headroom R–3 are satisfied on a held–out validation set.
2. **Band design.** BG–1/BG–2/BG–3 hold for the chosen PRF/T/DPSS; Pf/Pa are physically meaningful (e.g. Pf $\approx$ 30–250 Hz, Pa $\gtrsim$ 400 Hz in brain fUS).
3. **Operator geometry.** OC–1 with $\varepsilon_{\text{mix}} \leq 0.05$ and OC–2 with $\Lambda = s_a/s_f \geq 1.25$ and $\delta_\bullet \leq 0.1$.
4. **Blending realizable.** CB–1/CB–2/CB–3 satisfied with chosen (β, δ).
5. **Gating conservative and class–differential.** G–1/G–2/G–2'/G–3/G–4 implemented; offline, $p_- \in [0.2, 0.5]$ and $p_+ \leq 0.1 p_-$.
6. **Regime sentinel.** RS–1 does not trigger (KA not out of regime).
7. **Invariance sentinel.** $\widehat{s}_a/\widehat{s}_f \notin [1-\epsilon, 1+\epsilon]$ with $\epsilon \leq 0.02$; otherwise KA is effectively scalar and can be skipped.
8. **Safety guard.** Safety–1 (monotonicity on gated flow tiles) passes.

Bundle file(s)	Meaning	Primary use / direction
`pd_base.npy`, `pd_stap.npy`	PD-mode maps (baseline / legacy "STAP").	Deterministic preprocessing (proxy masks) and legacy visualization/compatibility; not used as the primary detector score for quantitative reporting in this manuscript.
`score_pd_base.npy`, `score_pd_stap.npy`	Legacy right-tail PD score maps (baseline / PD-mode "STAP").	Optional legacy PD scores with a fixed sign convention (`score_pd`=`pd` in current bundles); used for compatibility and auditing rather than for headline ROC.
`pd_stap_pre_ka.npy`, `score_pd_stap_pre_ka.npy` (optional)	Pre-KA STAP maps (when KA is enabled).	KA auditing (pre/post) under the same sign convention.

Table 21: Acceptance-bundle PD naming/sign schema.

Bundle file(s)	Meaning	Primary use / direction
`score_base.npy`	Baseline power Doppler score (right tail).	Structural ROC; tail hygiene; proxy masks.
`score_base_pdlog.npy` (optional)	$\log(\texttt{score_base} + \epsilon)$ (right tail).	Baseline fairness / strawman avoidance.
`score_base_kasai.npy` (optional)	$\log(\lvert \sum_t y_t \overline{y_{t-1}} \rvert + \epsilon)$ (Kasai lag-1 power; right tail).	Short-ensemble Doppler baseline fairness.
`score_stap_preka.npy`, `score_stap.npy`	STAP score maps (pre/post KA; right tail).	Structural ROC; within-ensemble motion ladders; motion robustness; KA auditing.

Table 22: Acceptance-bundle score-map naming schema for labeled ROC and label-free reporting (right-tail score convention).

9. **Evaluation discipline.** GLRT controls and tail CI practices (see Section 4.1) in place for fair low–FPR comparison.

When all of the above hold, KA applies a non–mixing, alias–inflating band transform chiefly to alias–dominant tiles, pushing negative scores down while largely leaving positives unchanged. This lowers the negative α–quantile and increases TPR at fixed small FPR in a manner consistent with the SL–2 quantile shift analysis.

K Bundle artifacts and naming conventions

This appendix records the artifact naming used in the released acceptance bundles and the mapping between the mathematical score notation in the main text and the stored arrays. These details are provided for reproducibility and are not required to follow the core derivations.

Exported artifact	Property / monotonicity	Allowed claim scope (this manuscript)
`score_stap_preka.npy`	Primary STAP detector score (matched-subspace; right tail). Not a shrink-only reweighting of baseline PD.	Quantitative ROC on labeled simulations and structurally labeled phantoms; label-free motion robustness endpoints on unlabeled IQ.
`score_base.npy`	Baseline power Doppler score (right tail).	Quantitative ROC as baseline comparator (same labels/masks as STAP).
`score_base_pdlog.npy`	Monotone transform of `score_base` (right tail).	Baseline-score fairness check; does not change within-method rankings.
`score_base_kasai.npy`	Lag-1 autocorrelation magnitude score (Kasai power; right tail).	Short-ensemble baseline-score fairness check (Twinkling).
`score_stap.npy`	Post-KA score map. By construction, shrink-only relative to `score_stap_preka` on gated regions ($S_{\text{post}} \leq S_{\text{pre}}$).	KA auditing and tail-hygiene reporting; not used as the primary detector score unless stated explicitly.
`pd_base.npy`	Conventional PD brightness proxy $\text{PD}^{\uparrow}$ (nonnegative).	Deterministic preprocessing (proxy masks) and legacy visualization; quantitative reporting uses detector *scores*.
`pd_stap.npy`	PD-mode "STAP" map exported for legacy compatibility/visualization. In PD-mode it is a non-increasing reweighting of `pd_base.npy`.	Visualization / compatibility output; can suppress background tails but cannot recover flow removed by baseline clutter suppression.

Table 23: Score taxonomy and claim scope. Quantitative detection results are reported using right-tail detector scores (baseline power Doppler vs matched-subspace STAP). PD-mode maps are treated as legacy/visualization artifacts with explicit monotonicity constraints.

L Reproducibility Manifest

This appendix records a minimal reproduction recipe: exact commands, data locations, and a pinned repository state. The manifest lists the minimal set of commands/outputs needed to reproduce the main claims; additional exploratory reports referenced in the text are not required unless explicitly listed here. Run `PYTHONPATH=.pythonscripts/generate_repro_manifest.py` to refresh this manifest.

Repository state. Commit: `6db311e` (full: `6db311e65c1bea51cb6f126709d619630c4532eb`); branch: `main`; dirty: `true`.

Environment. Python: `3.12.8|packagedbyAnaconda,Inc.|(main,Dec112024,16:31:09) [GCC11.2.0]`. Platform: `Linux-6.6.87.2-microsoft-standard-WSL2-x86_64-with-glibc2.39`. Packages: `numpy`=1.26.4, `scipy`=1.14.1, `torch`=2.2.2+cpu, `matplotlib`=3.10.0.

Datasets. Paths below are the expected on-disk locations used by the scripts in this repository. Large outputs under `runs/` and `reports/` are not tracked.

- **Brain-* k-Wave pilots (simulated IQ)**: Labeled low-FPR ROC + ablations.
  Local: `runs/pilot/`.
  Source: generated locally via sim/kwave/pilot_motion.py (Makefile targets r4c-*).
- **Mace/Urban whole-brain mouse fUS (PD-only)**: PD-only telemetry + hold-out alias-gate evaluation (no Doppler IQ).
  Local: `data/whole-brain-fUS`.
  Source: `https://doi.org/10.5281/zenodo.4905862`.
- **Shin RatBrain Fig3 (LOCA-ULM) beamformed IQ**: Real-IQ ingestion, telemetry, contract + label-free robustness sweeps.
  Local archive: `data/shin_zenodo_10711806/RatBrain_Fig3.zip`.
  Extracts to: `data/shin_zenodo_10711806/ratbrain_fig3_raw`.
  Source: `https://doi.org/10.5281/zenodo.10711806`.
- **Twinkling artifact dataset (RawBCF phantoms)**: Structurally labeled phantom ROC (Gammex) + within-ensemble motion ladders + KA hygiene (calculi).
  Local: `data/twinkling_artifact/`.
  Source: `https://doi.org/10.17816/DD76511`.
- **ULM Zenodo 7883227 (rat brain kHz IQ)**: Label-free motion robustness + one-time baseline calibration sweep.
  Local archive: `data/ulm_zenodo_7883227/IQ_001_to_025.zip`.
  Extracts to: `tmp/ulm_zenodo_7883227/`.
  Source: `https://doi.org/10.5281/zenodo.7883227`.

Pre-registered selections used in this manuscript.

- **Brain seed sweep (AliasContract pilot)**: seed2-seed12 (n=11).
- **Shin all-clips telemetry**: IQData001-IQData080 (n=80), window 0:128.
- **Shin motion subset**: IQData001-005, IQData010, IQData020, IQData040, IQData060, IQData080 (n=10).
- **Mace holdout split (deduplicated)**: train: scan1/2 + scan3/6; test: scan4 + scan5.
- **Gammex flow phantom structural ROC (along-linear17)**: frames 0:85 (n=85), PRF=2500 Hz, N=17 shots.
- **Gammex flow phantom structural ROC (across-linear17)**: frames 0:200 (n=200), PRF=2500 Hz, N=17 shots.
- **Gammex flow phantom within-ensemble motion ladder**: along-linear17, frames 0:85, amps 0..2 px, kinds rw,step.
- **Twinkling calculi KA hygiene**: calcifications sequence, frames 0:50 (n=50), PRF approx 500 Hz, N=9 shots.
- **ULM 7883227 baseline sweep**: blocks 1-3, frames 0:128, MC-SVD energy-frac sweep (label-free).
- **ULM 7883227 motion sweeps**: blocks 1-3, frames 0:128, frozen baseline e=0.975, motion kinds brainlike,elastic.

Artifact commands. Each entry below lists the command(s) used to regenerate the referenced paper artifact(s). Most scripts write a JSON summary containing the full parameterization and

are safe to rerun.

(R1) **PD-mode sanity checks (PD-mode map and score S=PD)** (paper refs: Notation (PD convention)). Outputs:
`reports/pd_mode_sanity/(optionalfigs)`
Notes: Run on any generated acceptance bundle root containing pw_* directories.

```
PYTHONPATH=. python scripts/pd_mode_sanity.py \
  runs/<bundle_root>/pw_* \
  --out-dir reports/pd_mode_sanity
```

(R2) **Conditional STAP leakage ablation (full vs conditional masks)** (paper refs: Results (Brain-k-Wave): conditional STAP ablation). Outputs:
`reports/condstap_leakage.csv`
`reports/condstap_leakage.json`

```
PYTHONPATH=. python scripts/conditional_stap_leakage_ablation.py \
  --pilots runs/pilot/r4_kwave_seed1 runs/pilot/r4_kwave_seed2 \
  --profile Brain-OpenSkull \
  --out-root runs/ablation/condstap_leakage \
  --window-length 64 --window-offset 0 --disjoint-offset 64 \
  --summary-csv reports/condstap_leakage.csv \
  --summary-json reports/condstap_leakage.json
```

(R3) **Brain-* baseline fairness: MC–SVD(e) tune-once sweep** (paper refs: Baseline fairness (Brain-*)). Outputs:
`reports/brain_mcsvd_energy_sweep_seed1.csv`
`reports/brain_mcsvd_energy_sweep_seed1.json`

```
PYTHONPATH=. python scripts/brain_mcsvd_energy_sweep.py \
  --pilot runs/pilot/r4_kwave_seed1 \
  --profile Brain-OpenSkull \
  --out-root runs/sweep/mcsvd_energy_brain_seed1 \
  --window-length 64 --window-offset 0 \
  --energy-fracs 0.90,0.95,0.97,0.975,0.98,0.99 \
  --fprs 1e-4,3e-4,1e-3 \
  --summary-csv reports/brain_mcsvd_energy_sweep_seed1.csv \
  --summary-json reports/brain_mcsvd_energy_sweep_seed1.json
```

(R4) **Brain-* ROC curve figure (median+IQR over disjoint windows)** (paper refs: Figure: `brain_kwave_roc_curves.pdf`). Outputs:
`figs/paper/brain_kwave_roc_curves.pdf`
Notes: Reads precomputed per-window score/mask arrays under runs/pilot/fair_filter_matrix_full_clinical_cpu

```
PYTHONPATH=. python scripts/fig_brain_kwave_roc_curves.py \
  --runs-root runs/pilot/fair_filter_matrix_full_clinical_cpu_v2 \
  --out-pdf figs/paper/brain_kwave_roc_curves.pdf
```

(R5) **Brain-* cross-window threshold-transfer audit (STAP)** (paper refs: Appendix: cross-window threshold calibration audit (Brain-*)). Outputs:
`reports/brain_crosswindow_calibration.csv`
`reports/brain_crosswindow_calibration_summary.json`
`reports/brain_crosswindow_calibration_table.tex`
Notes: Calibrates thresholds on one 64-frame window's negatives and evaluates on disjoint windows (ordered pairs).

```
PYTHONPATH=. python scripts/brain_crosswindow_calibration.py \
  --runs-root runs/pilot/fair_filter_matrix_full_clinical_cpu_v2 \
  --alphas 1e-4,3e-4,1e-3 \
```

```
  --out-csv reports/brain_crosswindow_calibration.csv \
  --out-json reports/brain_crosswindow_calibration_summary.json
```

(R6) **Knowledge-aided prior falsifiability ablation (STAP-only vs contract vs forced)** (paper refs: KA evaluation discipline). Outputs:
`reports/ka_v2_ablation.csv`
`reports/ka_v2_ablation.json`

```
PYTHONPATH=. python scripts/ka_contract_v2_ablation.py \
  --pilot runs/pilot/r4c_kwave_hab_contract_seed2 \
  --profile Brain-AliasContract \
  --out-root runs/ablation/ka_v2_falsifiability \
  --stap-device cpu \
  --window-length 64 \
  --window-offsets 0 64 128 192 256 \
  --summary-csv reports/ka_v2_ablation.csv \
  --summary-json reports/ka_v2_ablation.json
```

(R7) **Knowledge-aided prior positive-control ablation (OpenSkull shallow-alias + high-rank ghosts)** (paper refs: KA evaluation discipline (positive control)). Outputs:
`reports/ka_v2_ablation_openskull_shallowalias_e50.csv`
`reports/ka_v2_ablation_openskull_shallowalias_e50.json`
Notes: Synthetic, replay-only KA-positive regime used as a contract-mechanics positive control; not a realism benchmark.

```
PYTHONPATH=. python scripts/ka_contract_v2_ablation.py \
  --pilot runs/pilot/r4_kwave_seed1 \
  --profile Brain-OpenSkull \
  --out-root runs/ablation/openskull_shallowalias_e50 \
  --stap-device cpu \
  --window-length 64 \
  --window-offsets 0 64 128 192 256 \
  --summary-csv reports/ka_v2_ablation_openskull_shallowalias_e50.csv \
  --summary-json reports/ka_v2_ablation_openskull_shallowalias_e50.json \
  --replay-extra \
    --svd-energy-frac 0.50 \
    --flow-doppler-min-hz 60 --flow-doppler-max-hz 180 \
    --flow-doppler-noise-amp 60 --flow-doppler-noise-mode fft_band \
    --bg-alias-hz 650 --bg-alias-fraction 0.7 \
    --bg-alias-depth-min-frac 0.12 --bg-alias-depth-max-frac 0.28 \
    --bg-alias-jitter-hz 35 \
    --flow-mask-suppress-alias-depth \
    --clutter-beta 1.0 --clutter-snr-db 20 \
    --clutter-depth-min-frac 0.20 --clutter-depth-max-frac 0.95 \
    --aperture-phase-std 0.8 --aperture-phase-corr-len 14 \
    --bg-alias-highrank-mode gw_reverb_add \
    --bg-alias-highrank-coverage 0.3 --bg-alias-highrank-amp 0.3 \
    --bg-alias-highrank-margin-px 3 \
    --bg-alias-highrank-freq-jitter-hz 25 \
    --bg-alias-highrank-drift-step-hz 12 \
    --bg-alias-highrank-pf-leak-eta 0.0 \
    --ka-score-contract-v2-mode auto
```

(R8) **Mace PD-only holdout alias-gate evaluation (label-free thresholds)** (paper refs: Table: Mace hold-out alias gate). Outputs:
`reports/mace_alias_gate_holdout.csv`
`reports/mace_alias_gate_holdout.json`

```
PYTHONPATH=. python scripts/mace_alias_gate_holdout.py \
  --out-csv reports/mace_alias_gate_holdout.csv \
  --out-json reports/mace_alias_gate_holdout.json
```

(R9) **Mace pixel-level vascular-atlas check (independent structural labels)** (paper refs: Mace section: vascular atlas pixel-level check). Outputs:

`reports/mace_vascular_pixel_eval.csv`

`reports/mace_vascular_pixel_eval.json`

Notes: Maps atlas.Vascular into each scan plane using Transformation.mat; evaluates vascular vs non-vascular pixels (structural, not clinical efficacy).

```
PYTHONPATH=. python scripts/mace_vascular_pixel_eval.py \
  --out-csv reports/mace_vascular_pixel_eval.csv \
  --out-json reports/mace_vascular_pixel_eval.json
```

(R10) **Mace atlas-alignment sanity overlay figure** (paper refs: Appendix: Mace atlas alignment sanity check). Outputs:

`figs/paper/mace_atlas_overlay.png`

Notes: Overlays atlas ROIs and atlas.Vascular contours on mean PD for representative planes (sanity check of Transformation.mat alignment).

```
PYTHONPATH=. python scripts/mace_atlas_overlay_fig.py \
  --scan-name scan1 \
  --plane-indices 5 10 15 \
  --out-png figs/paper/mace_atlas_overlay.png
```

(R11) **Telemetry regime comparison (sim vs real; contract-v2 scalars)** (paper refs: Figure: `telemetry_regime_compare.png`). Outputs:

`figs/paper/telemetry_regime_compare.png`

Notes: Overlays PfPeakFrac(flow), guard_q90, and iqr_alias_bg across Brain-* simulations, Shin real IQ, and Mace PD-only planes; intended as telemetry anchoring (not a claim that sims match real data).

```
# Sample Brain-* replay bundles (write meta.json with ka_contract_v2 telemetry):
PYTHONPATH=. python scripts/replay_stap_from_run.py \
  --src runs/pilot/r4_kwave_seed1 \
  --out runs/telemetry_regime_compare/brain_open_seed1 \
  --profile Brain-OpenSkull \
  --time-window-length 64 \
  --time-window-offset 0 --time-window-offset 64 --time-window-offset 128 \
  --time-window-offset 192 --time-window-offset 256
PYTHONPATH=. python scripts/replay_stap_from_run.py \
  --src runs/pilot/r4c_kwave_hab_contract_seed2_v2 \
  --out runs/telemetry_regime_compare/brain_aliascontract_seed2 \
  --profile Brain-AliasContract \
  --time-window-length 64 \
  --time-window-offset 0 --time-window-offset 64 --time-window-offset 128 \
  --time-window-offset 192 --time-window-offset 256
PYTHONPATH=. python scripts/replay_stap_from_run.py \
  --src runs/pilot/r4c_kwave_hab_v3_skull_seed2_v2 \
  --out runs/telemetry_regime_compare/brain_skullor_seed2 \
  --profile Brain-SkullOR \
  --time-window-length 64 \
  --time-window-offset 0 --time-window-offset 64 --time-window-offset 128 \
  --time-window-offset 192 --time-window-offset 256

# Generate the overlay histogram figure:
# (reads Shin bundles and Mace CSV from other artifacts)
```

```
PYTHONPATH=. python scripts/telemetry_regime_compare_fig.py \
  --out-png figs/paper/telemetry_regime_compare.png
```

(R12) **Mace PD-only prior dashboard (plane sweep + paper figure)** (paper refs: Figure: `mace_pdonly_contract_v2_dashboard.png`). Outputs:
`reports/mace_pdonly_contract_v2.csv`
`figs/paper/mace_pdonly_contract_v2_dashboard.png`

```
PYTHONPATH=. python scripts/mace_pdonly_contract_v2_sweep.py \
  --out-csv reports/mace_pdonly_contract_v2.csv

PYTHONPATH=. python scripts/mace_pdonly_contract_v2_dashboard_fig.py \
  --in-csv reports/mace_pdonly_contract_v2.csv \
  --out-png figs/paper/mace_pdonly_contract_v2_dashboard.png
```

(R13) **Shin all-clips prior-regime telemetry (n=80; telemetry-only)** (paper refs: Shin section: all-clips telemetry audit). Outputs:
`reports/shin_ratbrain_shinU_e970_telemetry_only_0_128_all80.csv`
`reports/shin_ratbrain_shinU_e970_telemetry_only_0_128_all80_summary.json`

```
PYTHONPATH=. python -m scripts.shin_ratbrain_allclips_contract_hygiene \
  --extract-from-zip --no-run-stap --no-score-ka-v2 \
  --profile U --svd-energy-frac 0.97 --frames-list 0:128 \
  --out-root runs/shin_ratbrain_shinU_e970_telemetry_only_0_128_all80 \
  --out-csv reports/shin_ratbrain_shinU_e970_telemetry_only_0_128_all80.csv \
  --out-json reports/shin_ratbrain_shinU_e970_telemetry_only_0_128_all80_summary.json
```

(R14) **Shin brainlike motion sweep (10 clips) + aggregate plots** Paper refs:
Table: Shin motion endpoints
Figure: `shin_motion_brainlike_batch_U_agg.png`
Figure: `shin_motion_brainlike_compare_reg1.png`
Figure: `shin_motion_brainlike_compare.png`
Outputs:
`reports/shin_motion_brainlike_batch_U/*.csv`
`reports/shin_motion_brainlike_batch_U_agg.csv`
`figs/paper/shin_motion_brainlike_batch_U_agg.png`
`figs/paper/shin_motion_brainlike_compare_reg1.png`
`figs/paper/shin_motion_brainlike_compare.png`
Notes: Window-specific batch runs (64:192, 122:250) follow the same pattern.

```
# Per-clip sweeps (canonical window 0:128; KA disabled):
for f in IQData{001,002,003,004,005,010,020,040,060,080}.dat; do
  PYTHONPATH=. python -m scripts.shin_ratbrain_motion_sweep \
    --iq-file "$f" --frames 0:128 --profile U --svd-energy-frac 0.97 \
    --motion-kind brainlike --amp-px-list 0,0.5,1,2 \
    --no-score-ka-v2 --out-root runs/shin_motion_brainlike_batch_U \
    --out-csv "reports/shin_motion_brainlike_batch_U/${f%.dat}.csv" \
    --out-png "reports/shin_motion_brainlike_batch_U/${f%.dat}.png";
done

# Aggregate curve (writes CSV + plot):
PYTHONPATH=. python -m scripts.shin_ratbrain_motion_aggregate \
  --in-dir reports/shin_motion_brainlike_batch_U \
  --out-csv reports/shin_motion_brainlike_batch_U_agg.csv \
  --out-png reports/shin_motion_brainlike_batch_U_agg.png
```

```
# Compare across windows / reg ablations (writes comparison plots):
PYTHONPATH=. python -m scripts.shin_ratbrain_motion_compare \
  --curve "0:128=reports/shin_motion_brainlike_batch_U_agg.csv" \
  --curve "64:192=reports/shin_motion_brainlike_batch_U_f64_192_reg1_agg.csv" \
  --curve "122:250=reports/shin_motion_brainlike_batch_U_f122_250_reg1_agg.csv" \
  --out-png reports/shin_motion_brainlike_compare_reg1.png
cp -f \
  reports/shin_motion_brainlike_compare_reg1.png \
  figs/paper/shin_motion_brainlike_compare_reg1.png
```

(R15) **Shin motion crop/alignment sensitivity (post-hoc)** (paper refs: Shin motion section: crop sensitivity note). Outputs:
`reports/shin_motion_brainlike_batch_U_crop_sensitivity.csv`

```
PYTHONPATH=. python -m scripts.shin_ratbrain_motion_sensitivity_posthoc \
  --run-root runs/shin_motion_brainlike_batch_U \
  --amps 0,0.5,1,2 \
  --crop-margins 0,4,8,12 \
  --align-maps-list 0,1 \
  --out-csv reports/shin_motion_brainlike_batch_U_crop_sensitivity.csv
```

(R16) **Twinkling decode sanity (RawBCF -¿ B-mode/CFM)** (paper refs: Twinkling section: RawBCF decode sanity). Outputs:
`reports/twinkling_decode_sanity/*/decode_report.json`

```
SEQ_DIR="data/twinkling_artifact/Flow in Gammex phantom"
SEQ_DIR="$SEQ_DIR/Flow in Gammex phantom (along - linear probe)"
PYTHONPATH=. python scripts/twinkling_decode_sanity.py \
  --seq-dir "$SEQ_DIR" \
  --frame-idx 0 \
  --out-dir reports/twinkling_decode_sanity
```

(R17) **Gammex flow phantom structural ROC (along-linear17, PRF=2500)** Paper refs: Twinkling section: Table (structural ROC)
Figure: `twinkling_gammex_along_mask_overlay.png`
Outputs:
`runs/real/twinkling_gammex_alonglinear17_prf2500_str6_msd_ka/`
`reports/twinkling_gammex_alonglinear17_prf2500_str6_msd_ka_with_baselines.csv`
`reports/twinkling_gammex_alonglinear17_prf2500_str6_msd_ka_with_baselines_summary.json`
`figs/paper/twinkling_gammex_along_mask_overlay.png`
Notes: Along-linear uses stride 6; this matches the fixed geometry-only stride policy (largest stride with ¿=500 tiles).

```
SEQ_DIR="data/twinkling_artifact/Flow in Gammex phantom"
SEQ_DIR="$SEQ_DIR/Flow in Gammex phantom (along - linear probe)"
OUT_ROOT="runs/real/twinkling_gammex_alonglinear17_prf2500_str6_msd_ka"
MASK_GLOB=$OUT_ROOT/*__mask_debug

PYTHONPATH=. python scripts/twinkling_make_bundles.py \
  --seq-dir "$SEQ_DIR" \
  --out-root "$OUT_ROOT" \
  --frames 0:85 \
  --prf-hz 2500 --Lt 16 \
  --tile-hw 8 8 --tile-stride 6 \
```

```
  --cov-estimator tyler_pca --diag-load 0.07 \
  --baseline-type svd_bandpass --svd-keep-min 2 --svd-keep-max 17 \
  --score-mode msd --score-ka-v2-enable --score-ka-v2-mode auto

PYTHONPATH=. python scripts/twinkling_eval_structural.py \
  --root "$OUT_ROOT" \
  --out-csv reports/twinkling_gammex_alonglinear17_prf2500_str6_msd_ka_with_baselines.csv \
  --out-summary-json \
    reports/twinkling_gammex_alonglinear17_prf2500_str6_msd_ka_with_baselines_summary.json

PYTHONPATH=. python scripts/fig_twinkling_mask_overlay.py \
  --bmode $MASK_GLOB/bmode_roi_ref_norm.png \
  --mask-flow $MASK_GLOB/mask_flow.png \
  --mask-bg $MASK_GLOB/mask_bg.png \
  --out figs/paper/twinkling_gammex_along_mask_overlay.png \
  --min-width 1200 --dpi 300
```

(R18) **Gammex flow phantom structural ROC (across-linear17, PRF=2500)** Paper refs:
Twinkling section: Table (structural ROC)
Figure: `twinkling_gammex_across_mask_overlay.png`
Outputs:
`runs/real/twinkling_gammex_across17_prf2500_str4_msd_ka/`
`reports/twinkling_gammex_across17_prf2500_str4_msd_ka_with_baselines.csv`
`reports/twinkling_gammex_across17_prf2500_str4_msd_ka_with_baselines_summary.json`
`figs/paper/twinkling_gammex_across_mask_overlay.png`
Notes: Across-linear uses stride 4; this matches the fixed geometry-only stride policy (largest stride with ¿=500 tiles).

```
SEQ_DIR="data/twinkling_artifact/Flow in Gammex phantom"
SEQ_DIR="$SEQ_DIR/Flow in Gammex phantom (across - linear probe)"
DAT_PATH="$SEQ_DIR/RawBCFCine_08062017_145434_17.dat"
PAR_PATH="$SEQ_DIR/RawBCFCine_08062017_145434_17.par"
OUT_ROOT="runs/real/twinkling_gammex_across17_prf2500_str4_msd_ka"
MASK_GLOB=$OUT_ROOT/*__mask_debug

PYTHONPATH=. python scripts/twinkling_make_bundles.py \
  --seq-dir "$SEQ_DIR" \
  --dat-path "$DAT_PATH" \
  --par-path "$PAR_PATH" \
  --out-root "$OUT_ROOT" \
  --frames 0:200 \
  --prf-hz 2500 --Lt 16 \
  --tile-hw 8 8 --tile-stride 4 \
  --cov-estimator tyler_pca --diag-load 0.07 \
  --baseline-type svd_bandpass --svd-keep-min 2 --svd-keep-max 17 \
  --score-mode msd --score-ka-v2-enable --score-ka-v2-mode auto

PYTHONPATH=. python scripts/twinkling_eval_structural.py \
  --root "$OUT_ROOT" \
  --out-csv reports/twinkling_gammex_across17_prf2500_str4_msd_ka_with_baselines.csv \
  --out-summary-json \
    reports/twinkling_gammex_across17_prf2500_str4_msd_ka_with_baselines_summary.json

PYTHONPATH=. python scripts/fig_twinkling_mask_overlay.py \
```

```
  --bmode $MASK_GLOB/bmode_roi_ref_norm.png \
  --mask-flow $MASK_GLOB/mask_flow.png \
  --mask-bg $MASK_GLOB/mask_bg.png \
  --out figs/paper/twinkling_gammex_across_mask_overlay.png \
  --min-width 780 --dpi 300
```

(R19) **Gammex within-ensemble motion ladder (along-linear17; PRF=2500)** Paper refs: Twinkling section: Figure (within-ensemble motion ladder)
Figure: `twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_ladder_rw_ci_v2.png`
Figure: `twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_ladder_step_ci_v2.png`
Outputs:
`runs/real/twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_rw_v2/`
`runs/real/twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_step_v2/`
`reports/twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_ladder_rw_ci_v2.json`
`reports/twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_ladder_step_ci_v2.json`
`figs/paper/twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_ladder_rw_ci_v2.png`
`figs/paper/twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_ladder_step_ci_v2.png`

```
SEQ_DIR="data/twinkling_artifact/Flow in Gammex phantom"
SEQ_DIR="$SEQ_DIR/Flow in Gammex phantom (along - linear probe)"
REF_ROOT="runs/real/twinkling_gammex_alonglinear17_prf2500_str6_msd_nomotion_ref_fast_f020"
OUT_RW="runs/real/twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_rw_v2"
OUT_STEP="runs/real/twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_step_v2"
REP_DIR="reports"
FIG_DIR="figs/paper"
BASE="twinkling_gammex_alonglinear17_prf2500"
SUF_RW="within_ensemble_motion_ladder_rw_ci_v2"
SUF_STEP="within_ensemble_motion_ladder_step_ci_v2"
RW_PREFIX="$REP_DIR/${BASE}_${SUF_RW}"
STEP_PREFIX="$REP_DIR/${BASE}_${SUF_STEP}"
RW_FIG="$FIG_DIR/${BASE}_${SUF_RW}.png"
STEP_FIG="$FIG_DIR/${BASE}_${SUF_STEP}.png"

# Amp=0 reference bundles for equivalence checks (20 frames):
PYTHONPATH=. python scripts/twinkling_make_bundles.py \
  --seq-dir "$SEQ_DIR" \
  --out-root "$REF_ROOT" \
  --frames 0:20 \
  --prf-hz 2500 --Lt 16 \
  --tile-hw 8 8 --tile-stride 6 \
  --cov-estimator tyler_pca --diag-load 0.07 \
  --baseline-type svd_bandpass --svd-keep-min 2 --svd-keep-max 17 \
  --score-mode msd

# Within-ensemble motion sweeps (85 frames; rw and step):
```

```
PYTHONPATH=. python scripts/twinkling_gammex_within_ensemble_motion_sweep.py \
  --seq-dir "$SEQ_DIR" \
  --out-root "$OUT_RW" \
  --frames 0:85 --mask-ref-frames 0:20 \
  --prf-hz 2500 --Lt 16 \
  --tile-hw 8 8 --tile-stride 6 \
  --cov-estimator tyler_pca --diag-load 0.07 \
  --baseline-type svd_bandpass --svd-keep-min 2 --svd-keep-max 17 \
  --score-mode msd \
  --motion-kind rw --amp-px-list 0,0.25,0.5,1,1.5,2
PYTHONPATH=. python scripts/twinkling_gammex_within_ensemble_motion_sweep.py \
  --seq-dir "$SEQ_DIR" \
  --out-root "$OUT_STEP" \
  --frames 0:85 --mask-ref-frames 0:20 \
  --prf-hz 2500 --Lt 16 \
  --tile-hw 8 8 --tile-stride 6 \
  --cov-estimator tyler_pca --diag-load 0.07 \
  --baseline-type svd_bandpass --svd-keep-min 2 --svd-keep-max 17 \
  --score-mode msd \
  --motion-kind step --amp-px-list 0,0.25,0.5,1,1.5,2

# Aggregate + bootstrap CIs + plots:
PYTHONPATH=. python scripts/twinkling_eval_motion_ladder.py \
  --root "$OUT_RW" \
  --out-csv "${RW_PREFIX}.csv" \
  --out-json "${RW_PREFIX}.json" \
  --out-png "$RW_FIG" \
  --bootstrap 1000 --bootstrap-seed 0 \
  --amp0-ref-root "$REF_ROOT" \
  --amp0-ref-frames 0:20 --amp0-ref-tol 5e-3
PYTHONPATH=. python scripts/twinkling_eval_motion_ladder.py \
  --root "$OUT_STEP" \
  --out-csv "${STEP_PREFIX}.csv" \
  --out-json "${STEP_PREFIX}.json" \
  --out-png "$STEP_FIG" \
  --bootstrap 1000 --bootstrap-seed 0 \
  --amp0-ref-root "$REF_ROOT" \
  --amp0-ref-frames 0:20 --amp0-ref-tol 5e-3
```

(R20) **Twinkling calculi KA hygiene + contract telemetry (calcifications; PRF approx 500)** Paper refs:
Twinkling section: KA hygiene on calculi
Figure: `twinkling_calculi_tail_example.png`
Outputs:
`runs/real/twinkling_calculi_calcifications_prf500_str4_Lt8_msd_ka_scm_dl015_f050/`
`reports/twinkling_calculi_calcifications_prf500_str4_Lt8_msd_ka_scm_dl015_f050_roc.csv`
`reports/twinkling_calculi_calcifications_prf500_str4_Lt8_msd_ka_scm_dl015_f050_summary.json`
Notes: Calculi runs use proxy masks (not tube structural masks). Used for contract/tail-hygiene instrumentation (not clinical detection claims).

```
BASE_DIR="data/twinkling_artifact/Twinkling artifact on calculi"
SEQ_DIR="$BASE_DIR/Twinkling and Flash artifacts on artificial calculi"
```

```
RUN_TAG="twinkling_calculi_calcifications_prf500_str4_Lt8_msd_ka_scm_dl015_f050"
OUT_ROOT="runs/real/${RUN_TAG}"
REP_PREFIX="reports/${RUN_TAG}"

PYTHONPATH=. python scripts/twinkling_make_bundles.py \
  --seq-dir "$SEQ_DIR" \
  --out-root "$OUT_ROOT" \
  --frames 0:50 \
  --prf-hz 500 --Lt 8 \
  --tile-hw 8 8 --tile-stride 4 \
  --cov-estimator scm --diag-load 0.15 \
  --baseline-type svd_bandpass --svd-keep-min 2 --svd-keep-max 9 \
  --score-mode msd --score-ka-v2-enable --score-ka-v2-mode auto \
  --mask-mode none

PYTHONPATH=. python scripts/twinkling_eval_structural.py \
  --root "$OUT_ROOT" \
  --out-csv "${REP_PREFIX}_roc.csv" \
  --out-summary-json "${REP_PREFIX}_summary.json"
```

(R21) **Twinkling contract figure pack (states/reasons + calculi tail example)** Paper refs: Twinkling section: Figure (twinkling_contract_states_reasons.png)
Twinkling section: Figure (twinkling_calculi_tail_example.png)
Outputs:
`figs/paper/twinkling_contract_states_reasons.png`
`figs/paper/twinkling_calculi_tail_example.png`
`reports/twinkling_contract_figpack.json`

```
ALONG_BASE="twinkling_gammex_alonglinear17_prf2500_str6_msd_ka_with_baselines"
ACROSS_BASE="twinkling_gammex_across17_prf2500_str4_msd_ka_with_baselines"
ALONG_SUM="reports/${ALONG_BASE}_summary.json"
ACROSS_SUM="reports/${ACROSS_BASE}_summary.json"
CALC_TAG="twinkling_calculi_calcifications_prf500_str4_Lt8_msd_ka_scm_dl015_f050"
CALC_SUM="reports/${CALC_TAG}_summary.json"
CALC_ROOT="runs/real/${CALC_TAG}"
OUT_STATE="figs/paper/twinkling_contract_states_reasons.png"
OUT_EX="figs/paper/twinkling_calculi_tail_example.png"
OUT_JSON="reports/twinkling_contract_figpack.json"

PYTHONPATH=. python scripts/twinkling_contract_figpack.py \
  --along-summary "$ALONG_SUM" \
  --across-summary "$ACROSS_SUM" \
  --calculi-summary "$CALC_SUM" \
  --calculi-root "$CALC_ROOT" \
  --out-state-reason-png "$OUT_STATE" \
  --out-calculi-example-png "$OUT_EX" \
  --out-json "$OUT_JSON"
```

(R22) **ULM 7883227 one-time baseline sweep (MC-SVD energy fraction)** (paper refs: ULM section: baseline calibration (frozen svd_energy_frac=0.975)). Outputs:
`runs/real/ulm7883227_baseline_mcsvd_energy_sweep_ULM_0000_0128/`
`reports/ulm7883227_baseline_mcsvd_energy_sweep_ULM_0000_0128.csv`
`reports/ulm7883227_baseline_mcsvd_energy_sweep_ULM_0000_0128.json`

```
PYTHONPATH=. python scripts/ulm_zenodo_7883227_baseline_sweep.py \
  --block-ids 1,2,3 --frames 0:128 \
```

```
  --svd-energy-frac-list 0.90,0.95,0.97,0.975,0.98,0.99 \
  --profile ULM --prf-hz 1000 --Lt 64 \
  --tile-h 8 --tile-w 8 --tile-stride 3 \
  --reg-enable --reg-subpixel 4 \
  --no-run-stap --bg-tail-fpr 0.001 --stability-split \
  --cache-dir tmp/ulm_zenodo_7883227 \
  --out-root runs/real/ulm7883227_baseline_mcsvd_energy_sweep_ULM_0000_0128 \
  --out-csv reports/ulm7883227_baseline_mcsvd_energy_sweep_ULM_0000_0128.csv \
  --out-json reports/ulm7883227_baseline_mcsvd_energy_sweep_ULM_0000_0128.json
```

(R23) **ULM 7883227 motion sweeps (brainlike + elastic; fairness alignment + degeneracy)** Paper refs:
ULM section: Figure (ulm7883227_motion_sweep_*_brainlike_e975.png)
ULM section: Figure (ulm7883227_motion_sweep_*_elastic_e975.png)
Outputs:
`reports/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_brainlike_e975.csv`
`reports/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_brainlike_e975.json`
`figs/paper/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_brainlike_e975.png`
`reports/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_elastic_e975.csv`
`reports/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_elastic_e975.json`
`figs/paper/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_elastic_e975.png`

```
PYTHONPATH=. python scripts/ulm_zenodo_7883227_motion_sweep.py \
  --block-ids 1,2,3 --frames 0:128 \
  --profile ULM --prf-hz 1000 --Lt 64 \
  --tile-h 8 --tile-w 8 --tile-stride 3 \
  --baseline-type mc_svd --svd-energy-frac 0.975 \
  --reg-enable --reg-subpixel 4 \
  --cov-estimator scm --diag-load 0.07 \
  --motion-kind brainlike --amp-px-list 0,0.5,1,2,3 \
  --out-root runs/real/ulm7883227_motion_sweep_ULM_brainlike_e975 \
  --out-csv reports/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_brainlike_e975.csv \
  --out-json reports/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_brainlike_e975.json \
  --out-png figs/paper/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_brainlike_e975.png
PYTHONPATH=. python scripts/ulm_zenodo_7883227_motion_sweep.py \
  --block-ids 1,2,3 --frames 0:128 \
  --profile ULM --prf-hz 1000 --Lt 64 \
  --tile-h 8 --tile-w 8 --tile-stride 3 \
  --baseline-type mc_svd --svd-energy-frac 0.975 \
  --reg-enable --reg-subpixel 4 \
  --cov-estimator scm --diag-load 0.07 \
  --motion-kind elastic --amp-px-list 0,0.5,1,2,3 \
  --out-root runs/real/ulm7883227_motion_sweep_ULM_elastic_e975 \
  --out-csv reports/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_elastic_e975.csv \
  --out-json reports/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_elastic_e975.json \
  --out-png figs/paper/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_elastic_e975.png
```

(R24) **Paper build modes (paper vs supplement vs full)** (paper refs: Build packaging (Phase 7)). Outputs:
`stap_fus_paper.pdf`
`stap_fus_supplement.pdf`
`stap_fus_methodology.pdf`

```
pdflatex stap_fus_methodology.tex   # full (default)
```

```
pdflatex stap_fus_paper.tex         # paper-only (no appendices)
pdflatex stap_fus_supplement.tex    # supplement-only (appendices only)
```

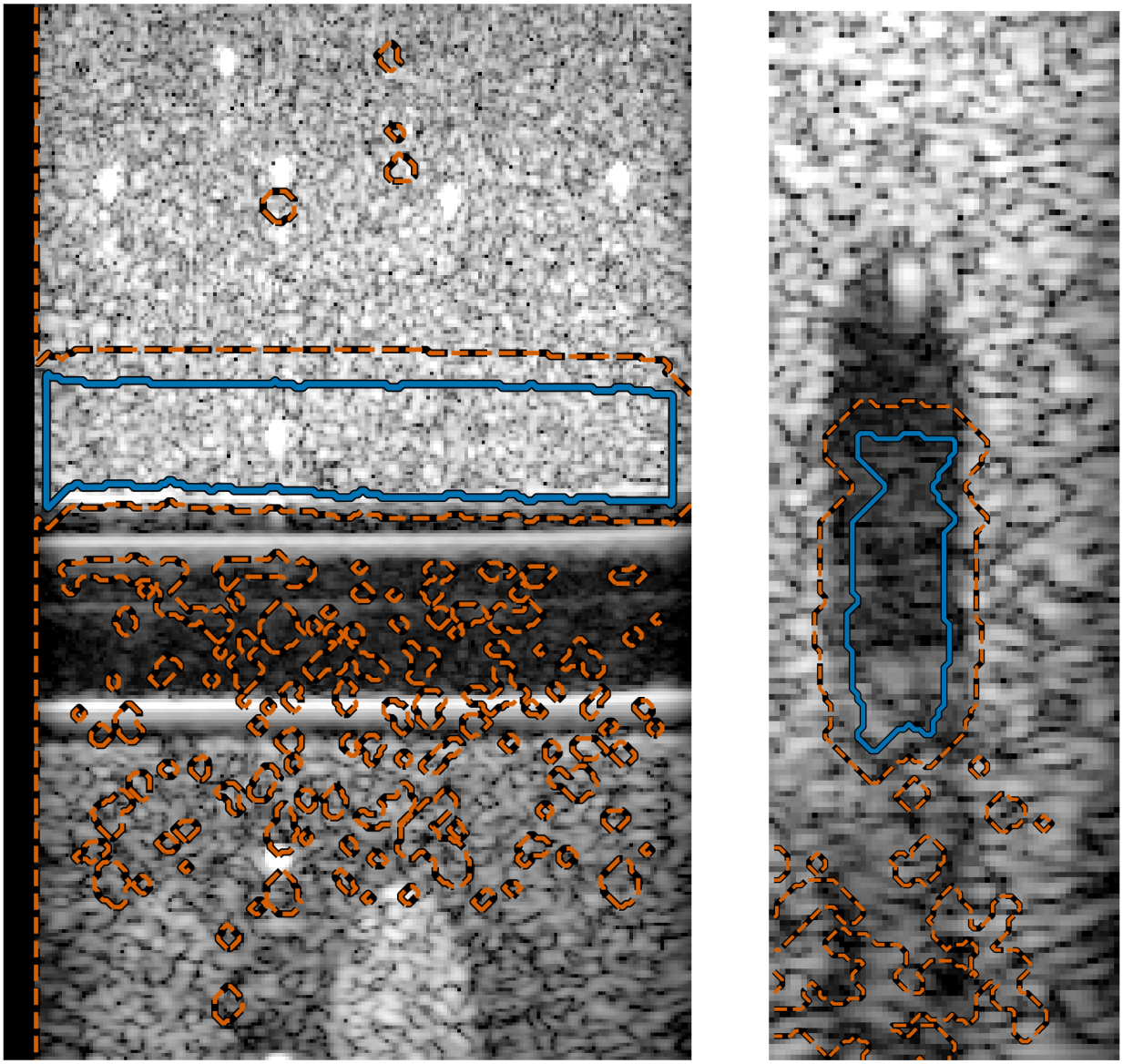

Figure 6: Gammex flow phantom structural evaluation masks derived from B-mode only. Left: along-linear view; right: across-linear view. The lumen mask is intentionally eroded to avoid wall/partial-volume pixels, and the background mask excludes a guard band around the lumen and bright reflectors. Images show the CFM-aligned B-mode ROI with contour overlays of the tube lumen (flow) mask (blue) and background mask (orange, dashed) used for structural-label ROC reporting. Reproducibility details are provided in Appendix L.

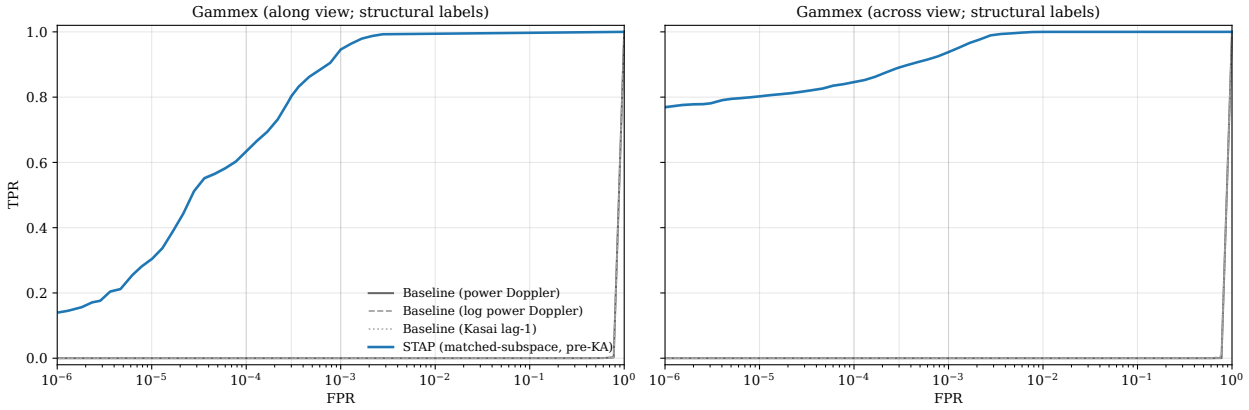

Figure 7: Gammex flow phantom structural labels: pooled ROC-style curves on along and across views, shown for baseline comparators and the pre-KA STAP matched-subspace score. The vertical guides indicate the strict low-FPR operating points reported in Table 7.

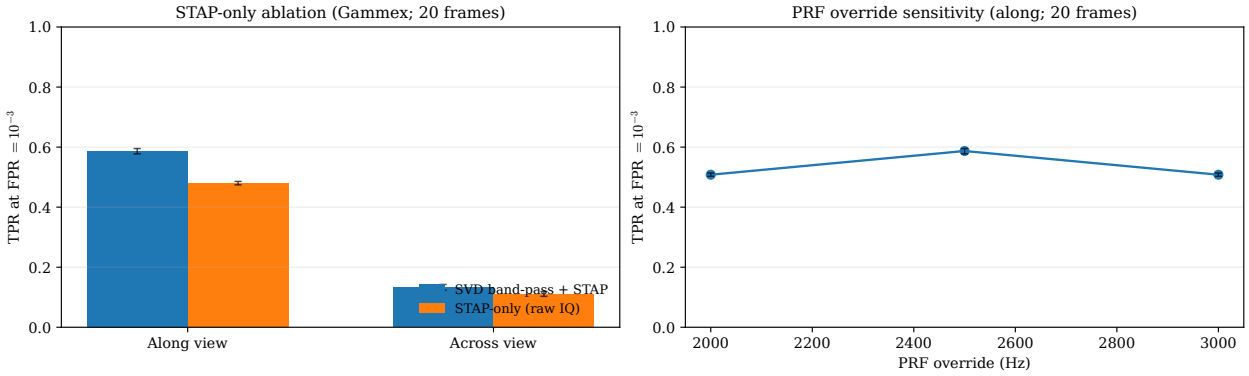

Figure 8: Gammex audits: STAP-only ablation and PRF override sensitivity (first 20 frames; along and across views). Left: STAP-only (raw IQ) vs baseline SVD band-pass + STAP at fixed FPR (10^{-3}). Right: PRF override sensitivity on the along view (PRF $\in \{2000, 2500, 3000\}$ Hz) under a frozen operating profile; this is a low-cost sanity audit of Doppler-band discretization, since PRF is constant within a sequence family and the discretization sentinels depend only on (N, PRF). Additional details are provided in Appendix B.

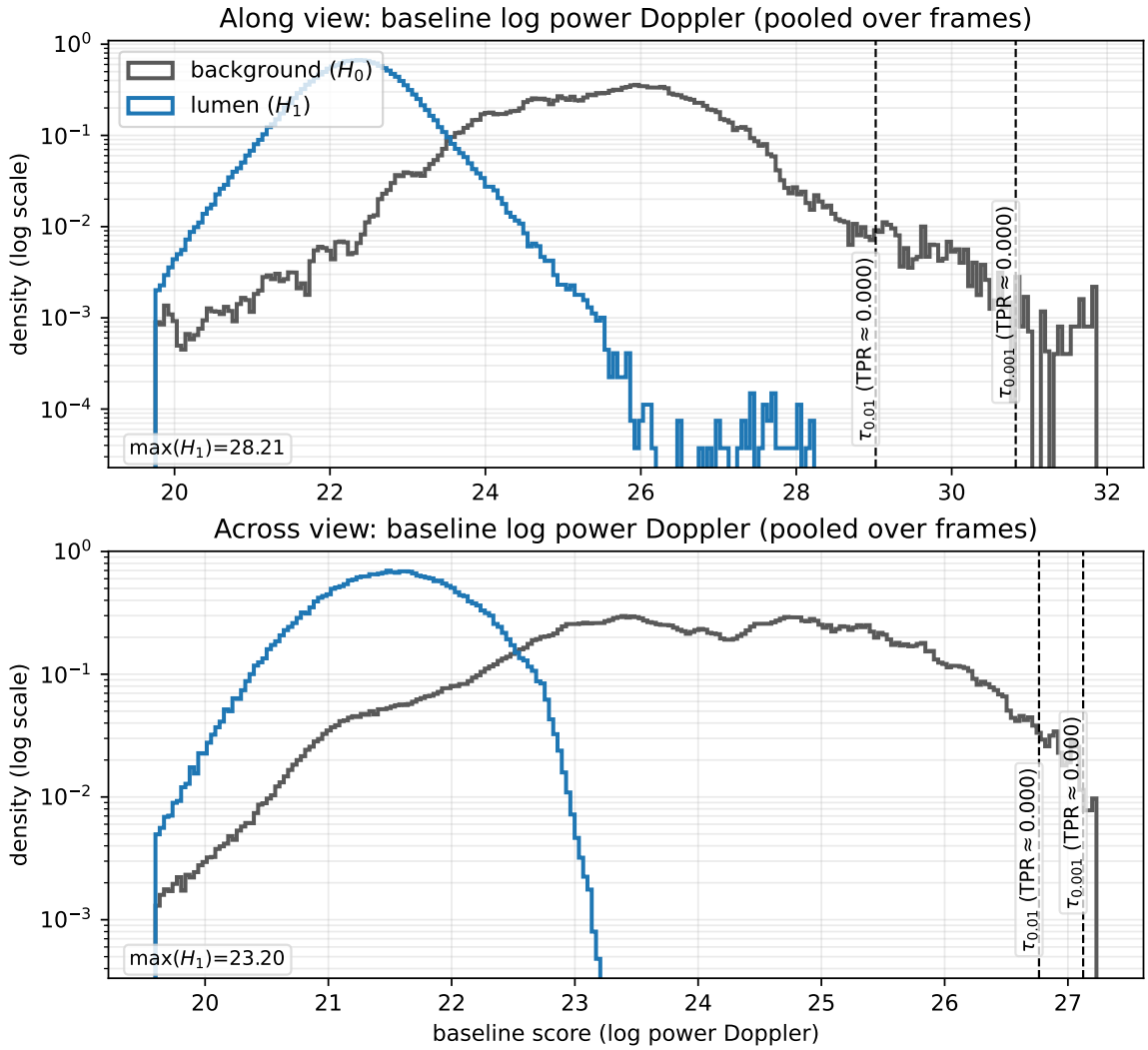

Figure 9: Why conventional baselines can have zero TPR at ultra-low FPR on Gammex short-ensemble data. The plot shows pooled score distributions for the baseline log power Doppler comparator on lumen (H_1; structural tube mask) versus background (H_0; structural background mask), for both along and across views. Vertical lines indicate background-calibrated right-tail thresholds τ_α at $\alpha \in \{10^{-2}, 10^{-3}\}$ (the main text reports $\alpha \leq 10^{-3}$; $\alpha = 10^{-2}$ is shown here for context). On both views the background tail threshold lies above $\max(H_1)$, so $\mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}$ is numerically zero at these tail operating points.

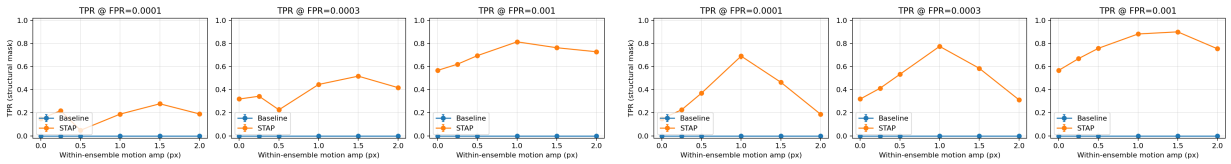

Figure 10: Gammex flow phantom within-ensemble motion ladder on the along-linear PRF=2500 sequence (85 frames). Left: random-walk per-shot translation; right: step translation. Each panel reports TPR vs motion amplitude at fixed FPR targets using B-mode structural masks. Shaded bands are 25–75% bootstrap intervals over cine frames. Reproducibility details are provided in Appendix L.

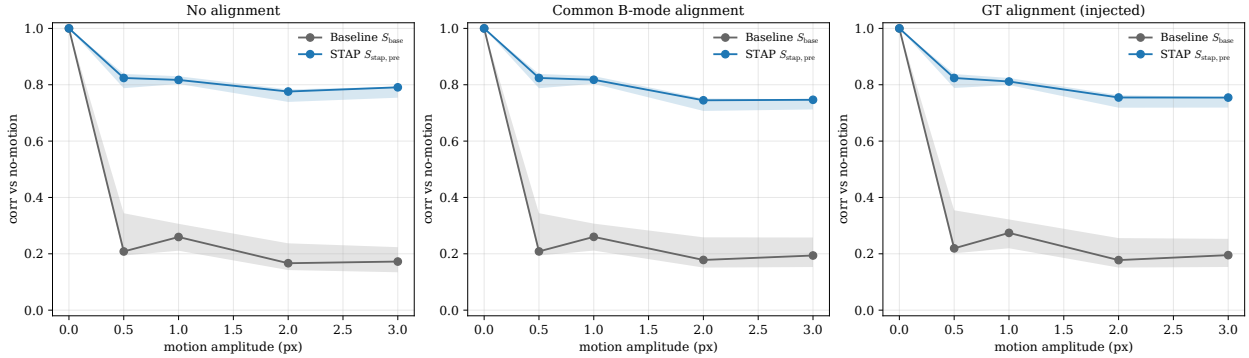

Figure 11: ULM 7883227 motion sweep (blocks 1–3; frames 0:128; frozen baseline $e = 0.975$): `brainlike` motion (rigid + elastic). Each panel reports correlation vs the no-motion reference under a different alignment mode (none / common B-mode / injected GT proxy) for the baseline score S_{base} and the STAP score $S_{\text{stap,pre}}$; shaded bands show median±IQR over blocks. Reproducibility details are provided in Appendix L.

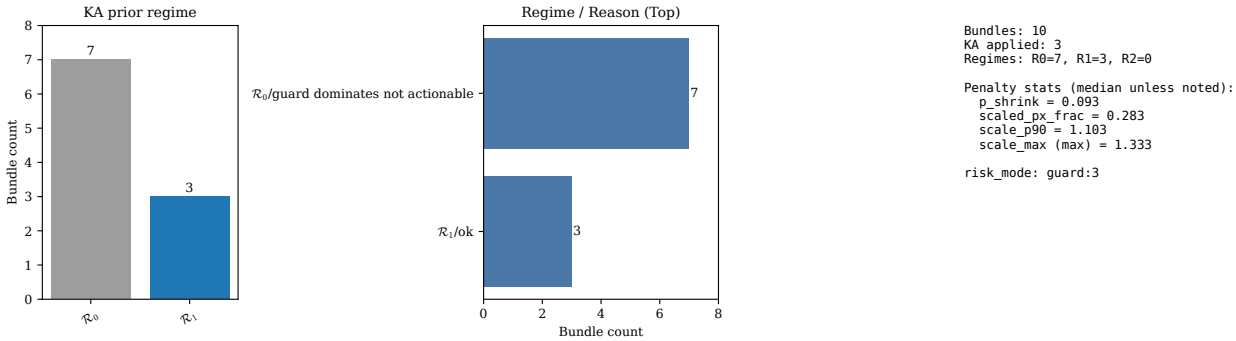

Figure 12: KA prior summary panel on a representative sweep. Left: regime counts. Middle: top (regime, reason) labels. Right: summary of shrink coverage and strength on runs where $w \not\equiv 1$.

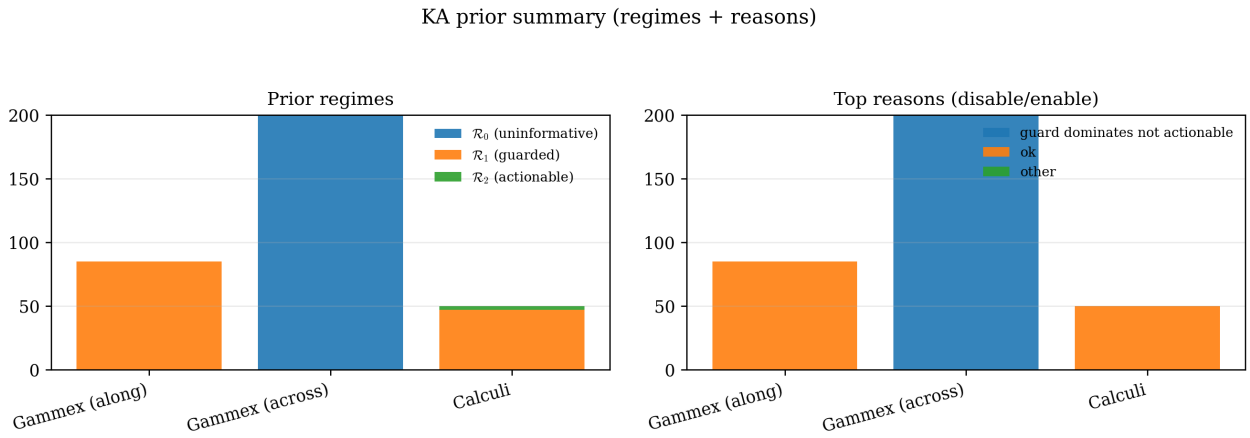

Figure 13: Twinkling KA-prior summary panel: prior (region, reason) distributions across Gammex (structural-label) and calculi (artifact-heavy) runs. Reproducibility details are provided in Appendix L.

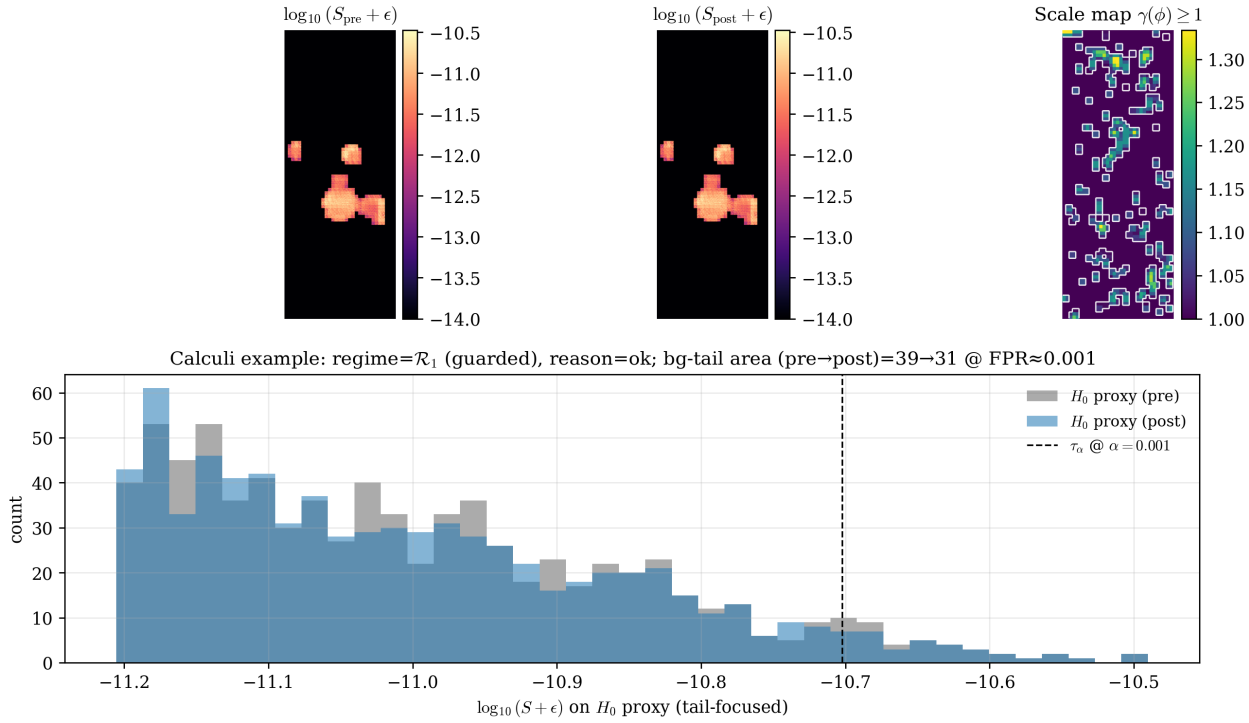

Figure 14: Representative calculi tail example (pre/post KA) showing shrink-only tail hygiene behavior when the KA prior is active ($w < 1$). Reproducibility details are provided in Appendix L.

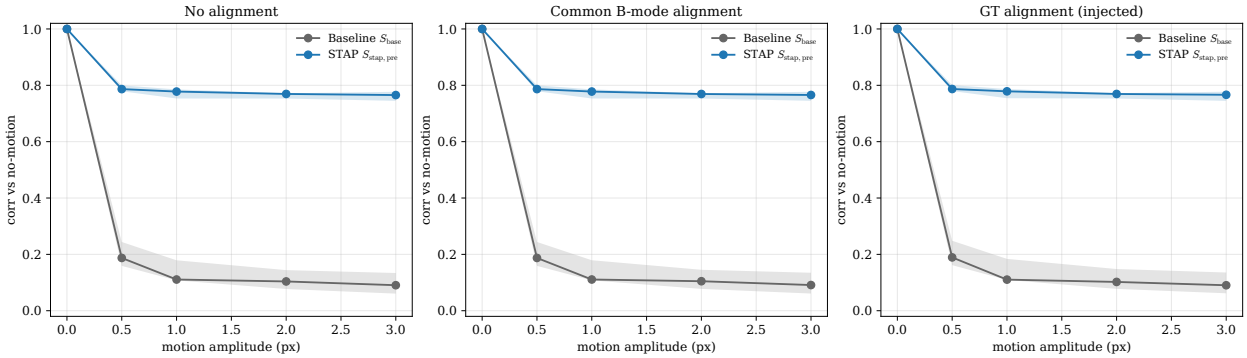

Figure 15: ULM 7883227 motion sweep under `elastic` (nonrigid-only) motion with the same frozen baseline/profile used in the main paper (`blocks 1--3`, frames 0:128, baseline MC–SVD with frozen energy fraction $e = 0.975$). Motion is injected at the beamformed IQ stage as a robustness stress test; it does not model channel-domain decorrelation physics. Reproducibility details are provided in Appendix L.

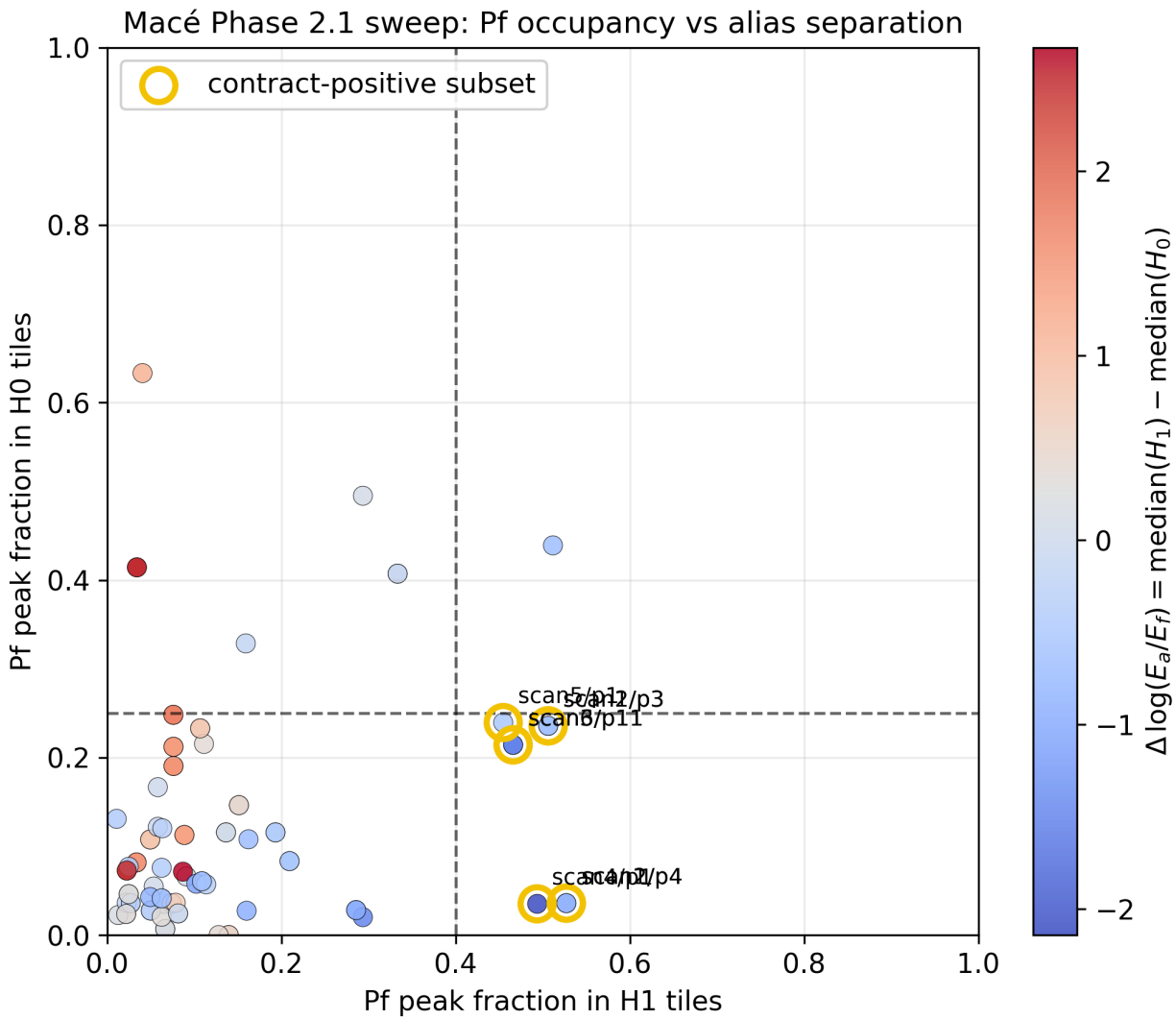

Figure 16: Macé systematic band-energy summary (six files; 84 planes; four unique acquisitions in this loader): Pf peak occupancy vs alias separation under a fixed hemodynamic STAP configuration (tiles 8×8, stride 3; $P_f = [0, 0.5]$ Hz, guard $[0.5, 1.0]$ Hz, $P_a = [1.0, 3.0]$ Hz). Each point is one (scan, plane) instance; the x-axis is the Pf peak fraction in the VIS/SC/LGd atlas stratum and the y-axis the Pf peak fraction in the CP/CA/DG stratum (offline, descriptive), and color encodes $\Delta \log(E_a/E_f) = \mathrm{median}(\mathrm{VIS/SC/LGd}) - \mathrm{median}(\mathrm{CP/CA/DG})$ (negative means alias ratios are lower in VIS/SC/LGd). Dashed lines indicate an offline descriptive threshold pair ($x > 0.4$, $y < 0.25$); they are not used for the held-out, label-free operating-point evaluation in Table 11.

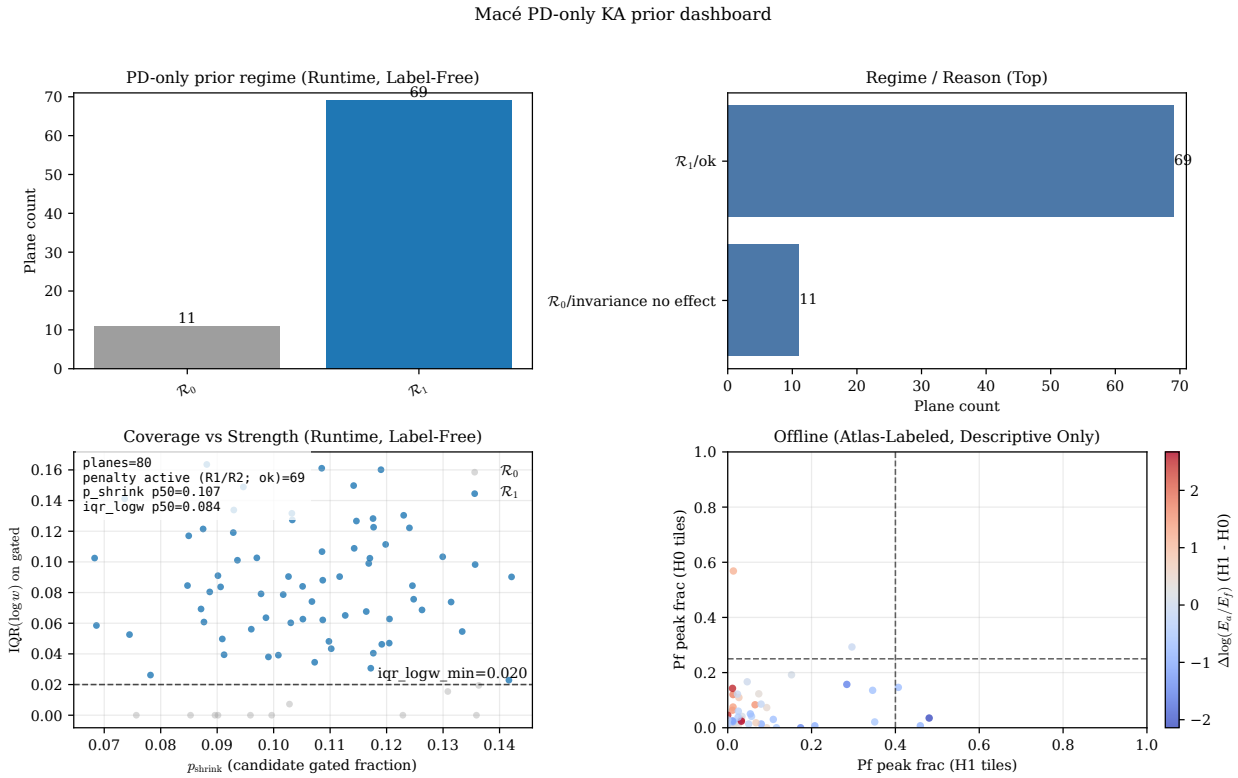

Figure 17: Macé PD-only prior summary panel over all deduplicated scan/plane instances. Top: prior regions and the most common (region, reason) pairs (runtime, label-free). Bottom-left: coverage vs strength summaries for shrink-only penalization (p_{shrink} vs IQR($\log w$) on penalized tiles), with the invariance threshold indicated. Bottom-right: offline atlas-stratified Pf-peak fractions (VIS/SC/LGd vs CP/CA/DG tiles) colored by $\Delta\log(E_a/E_f)$ (descriptive only; not a KA input). Reproducibility details are provided in Appendix L.

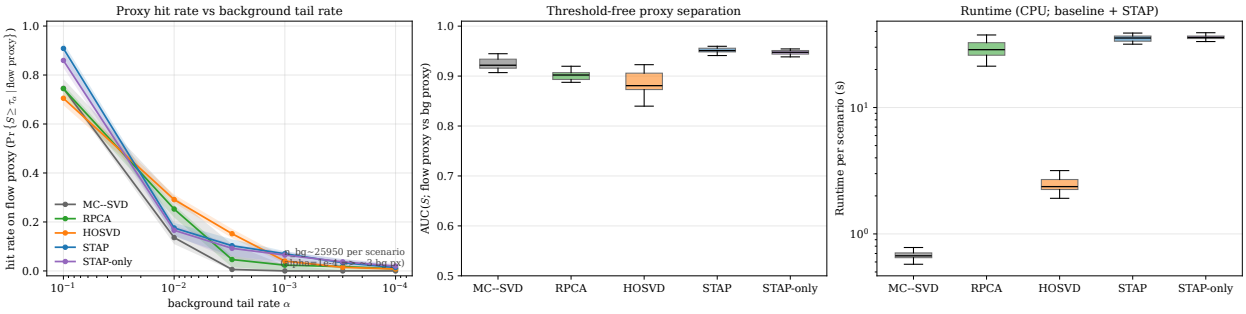

Figure 18: Shin RatBrain Fig3 baseline comparisons under label-free proxy masks (10 clips; three windows; Shin-U profile). Left: flow-proxy hit rate vs background tail rate α (median and IQR over scenarios). Middle: threshold-free AUC between flow-proxy and background-proxy score distributions. Right: runtime distribution per scenario in this CPU-only environment (baseline time for MC–SVD/RPCA/HOSVD; baseline+STAP total time for STAP variants), using compute-time telemetry captured in bundle metadata (excludes disk I/O and figure rendering). Full per-scenario outputs are in `reports/shin_ratbrain_baseline_matrix_shinU_e970_Lt64_nomaskunion_k80.csv`.

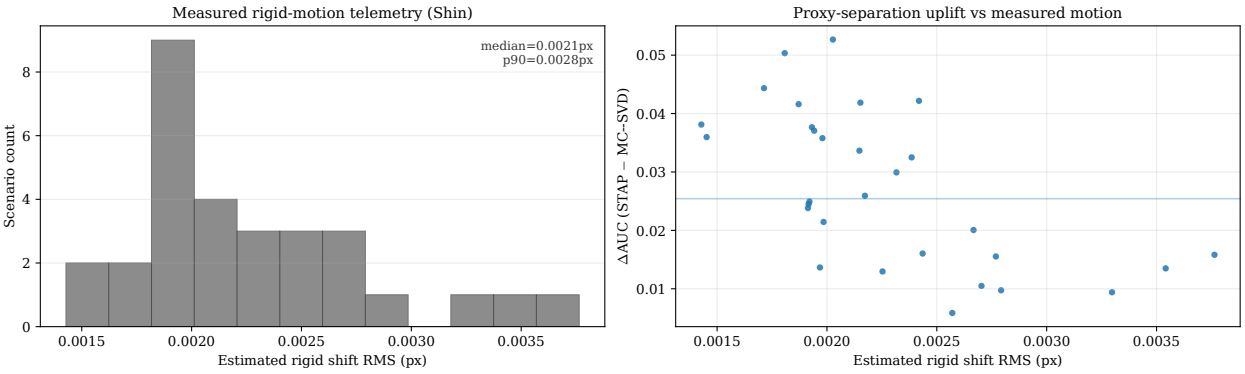

Figure 19: Shin RatBrain Fig3 measured rigid-motion telemetry across the 30 baseline-matrix scenarios (10 clips $\times$ 3 windows). Left: histogram of estimated rigid shift RMS (pixels) from phase-correlation registration used in baseline preprocessing. Right: proxy-separation uplift ΔAUC (STAP minus MC–SVD) versus the same shift statistic. Full extracted telemetry is in `reports/shin_motion_telemetry.csv`.

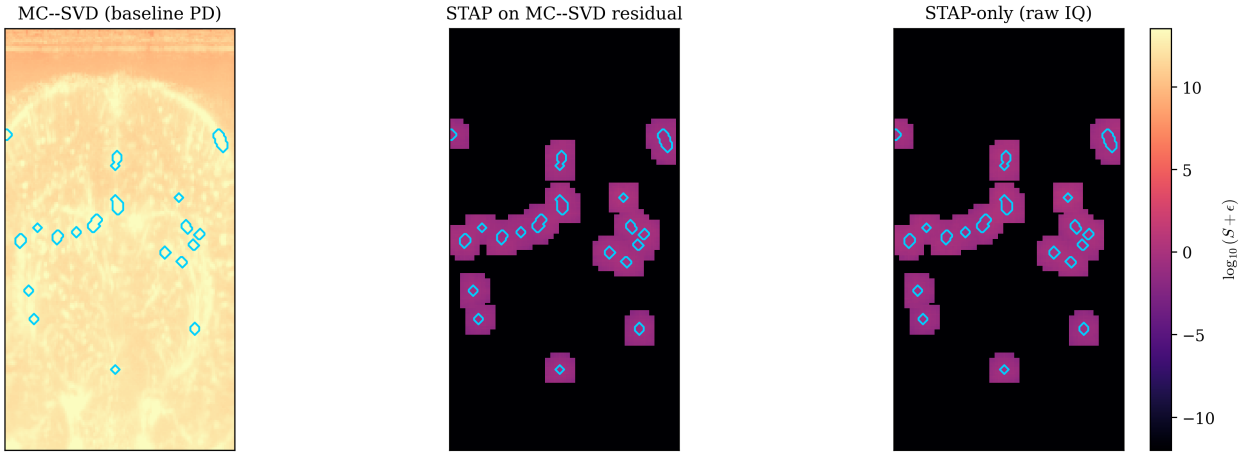

Figure 20: Representative Shin RatBrain Fig3 qualitative comparison (clip `IQData001`, window 0:128). Panels show log-scaled right-tail score maps for MC–SVD baseline PD, STAP on the MC–SVD residual, and STAP-only on raw/registered IQ, with the shared flow-proxy mask overlaid (cyan contour).

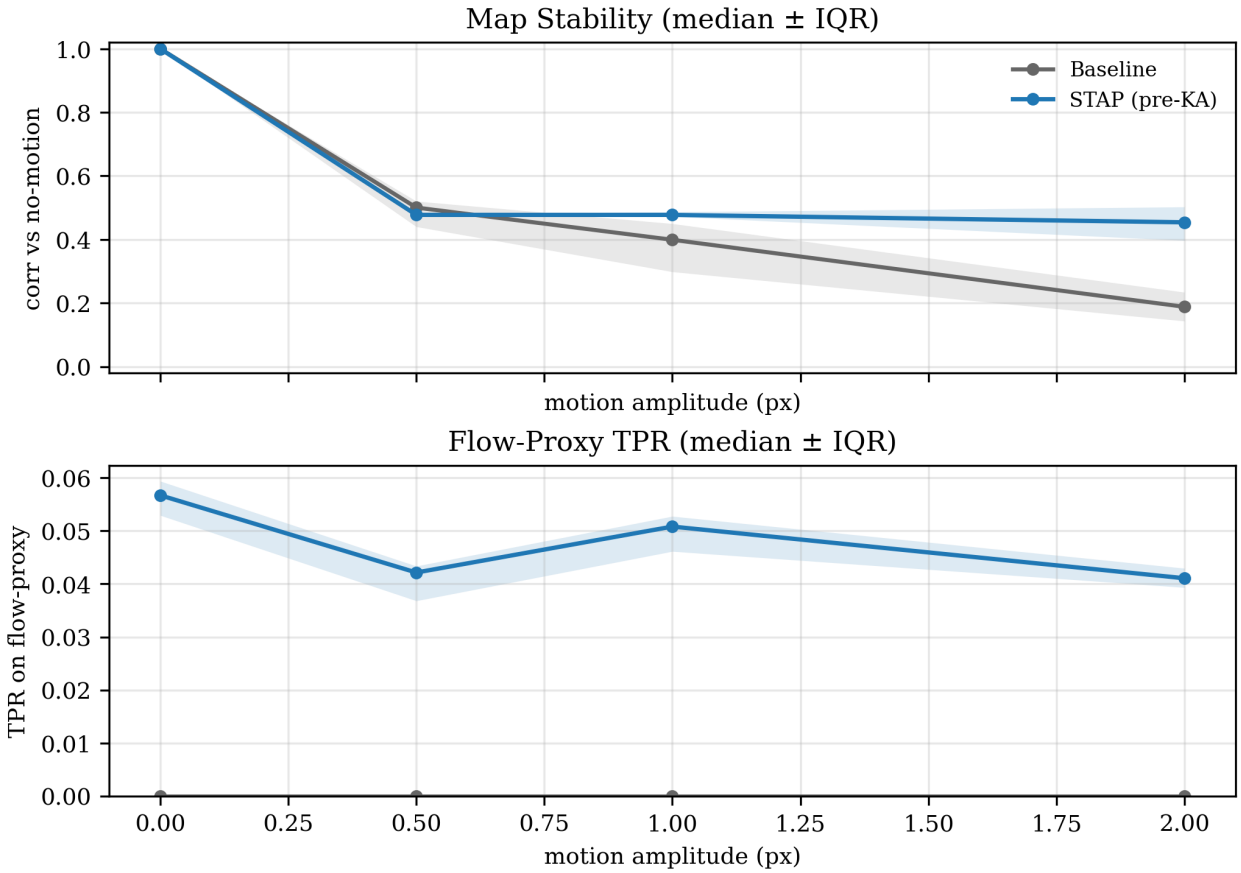

Figure 21: Shin RatBrain Fig3: brain-like residual-motion sweep (canonical window 0:128; 10 clips; median ± IQR) under the frozen Shin-U profile with rigid registration enabled and KA disabled (STAP reported pre-KA). Top: PD-mode score-map stability measured as Pearson correlation between each method's PD-mode score map at a given motion amplitude and its own no-motion reference, after phase-correlation alignment and a boundary crop. Bottom: flow-proxy hit rate (median ± IQR) at matched background-proxy tail rate = 10^{-3}, computed under the right-tail score rule $\mathbb{P}\{S_{\mathrm{PD}} \geq \tau \mid \text{bg proxy}\} = 10^{-3}$ with a fixed flow-proxy mask derived from the no-motion baseline pd_{base}.

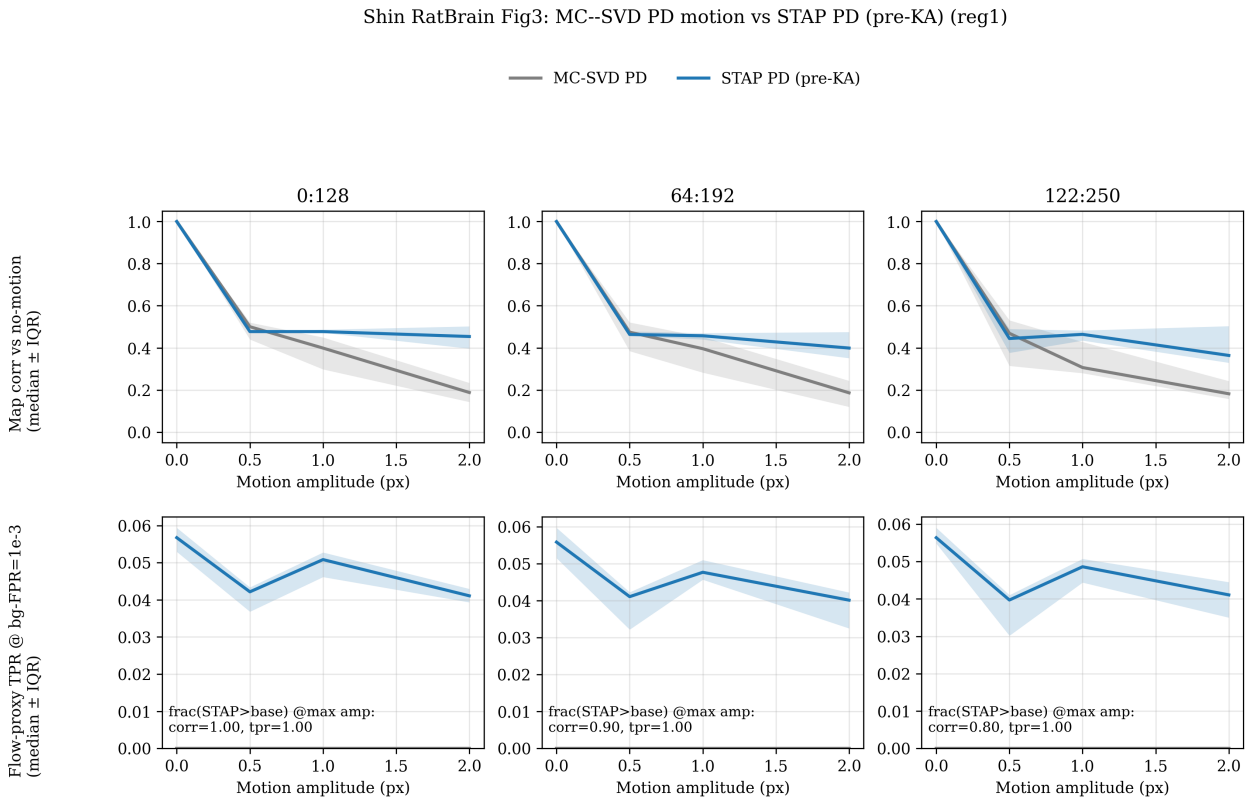

Figure 22: Shin RatBrain Fig3: brain-like residual-motion sweep across three overlapping windows (10 clips; median ± IQR) under the frozen Shin-U profile with rigid registration enabled and KA disabled (STAP reported pre-KA). Motion amplitude (x-axis) is the target displacement scale in pixels. "Brain-like" motion is applied directly to the beamformed IQ cube and combines (i) a rigid random-walk translation (70% of the amplitude), (ii) a smooth, depth-decaying elastic warp (30%; spatial smoothing $\sigma \approx 24$ px; depth weight $\propto \exp(-z/0.8)$; random-walk temporal coefficients), and (iii) per-frame micro-jitter (std $\approx 0.08\times$ amplitude). Endpoints match Figure 21. This stress test is image-domain (IQ warp) rather than a scatterer-level simulator; it is intended to mimic the "residual rigid + elastic motion after registration" mismatch in Brain-SkullOR.

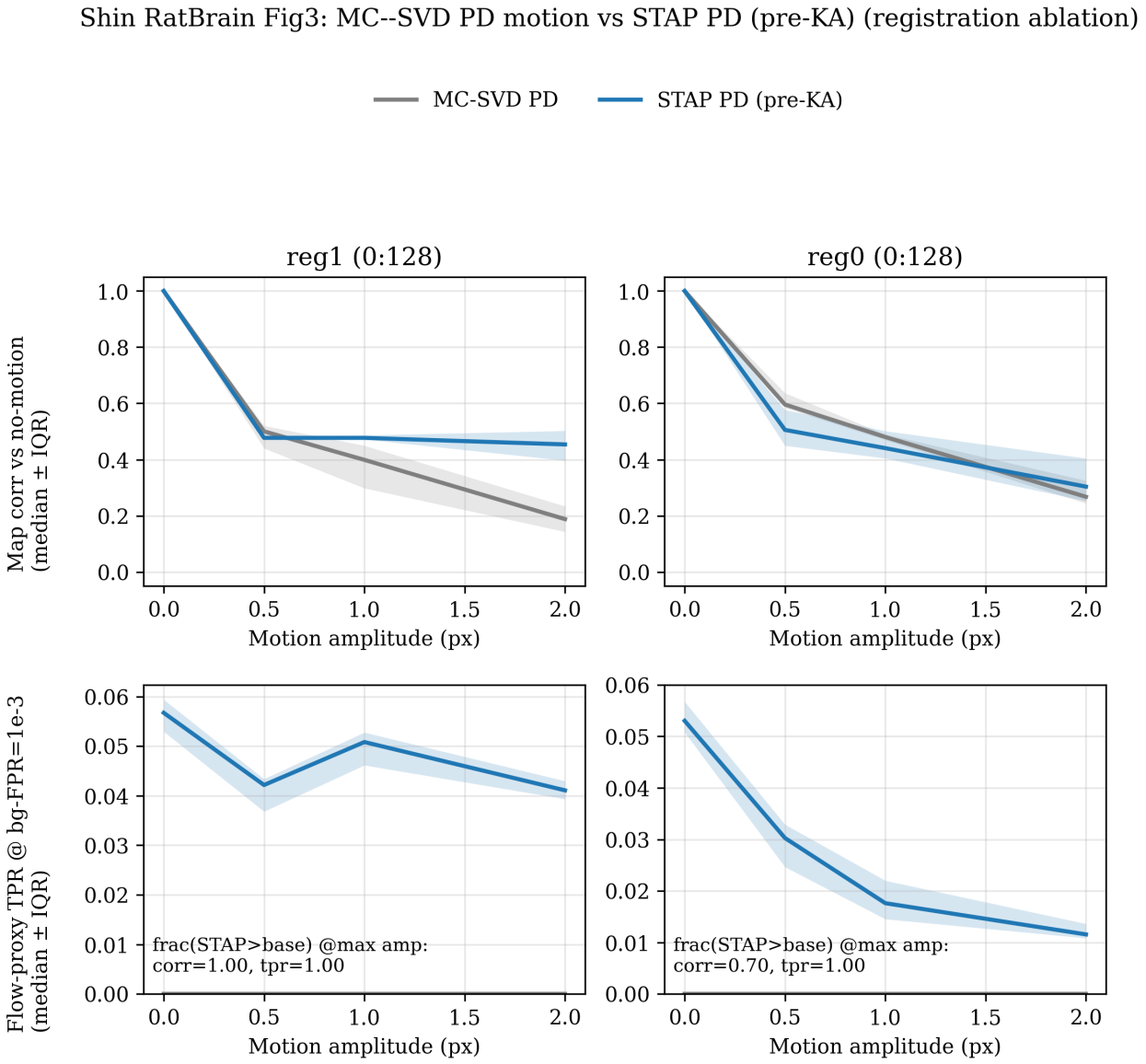

Figure 23: Shin RatBrain Fig3: registration ablation for the brain-like residual-motion sweep. The rightmost column (`reg0`) repeats the canonical window (0:128) with rigid registration disabled in the baseline and STAP pipelines, illustrating that the motion stress test contains a substantial bulk rigid component that phase-correlation registration can correct. Even with registration enabled, a smooth depth-decaying elastic warp remains (by design) and drives the robustness gap between MC–SVD and STAP.

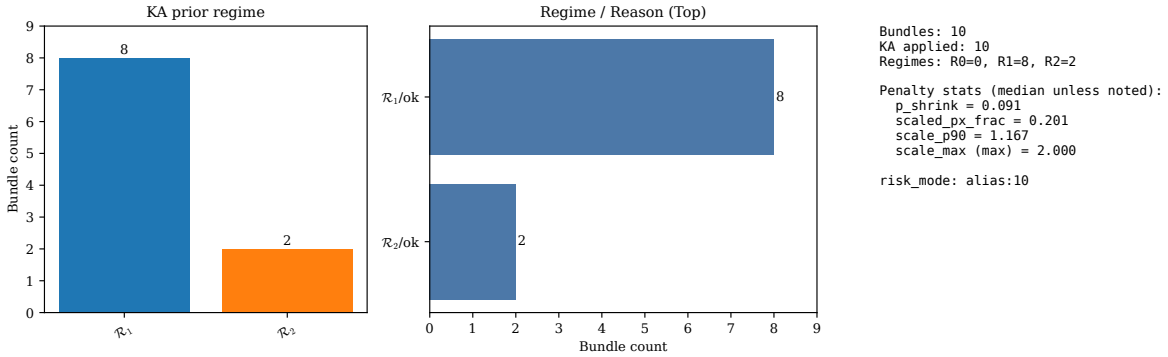

Figure 24: Shin RatBrain Fig3 (10 clips; window 0:128) KA prior summary panel under the frozen Shin-U profile (MC–SVD energy fraction ≈ 0.97, $L_t = 64$). The panel summarizes prior regions and reasons, Pf peak fractions on flow/non-background proxies, and KA shrink coverage/strength.

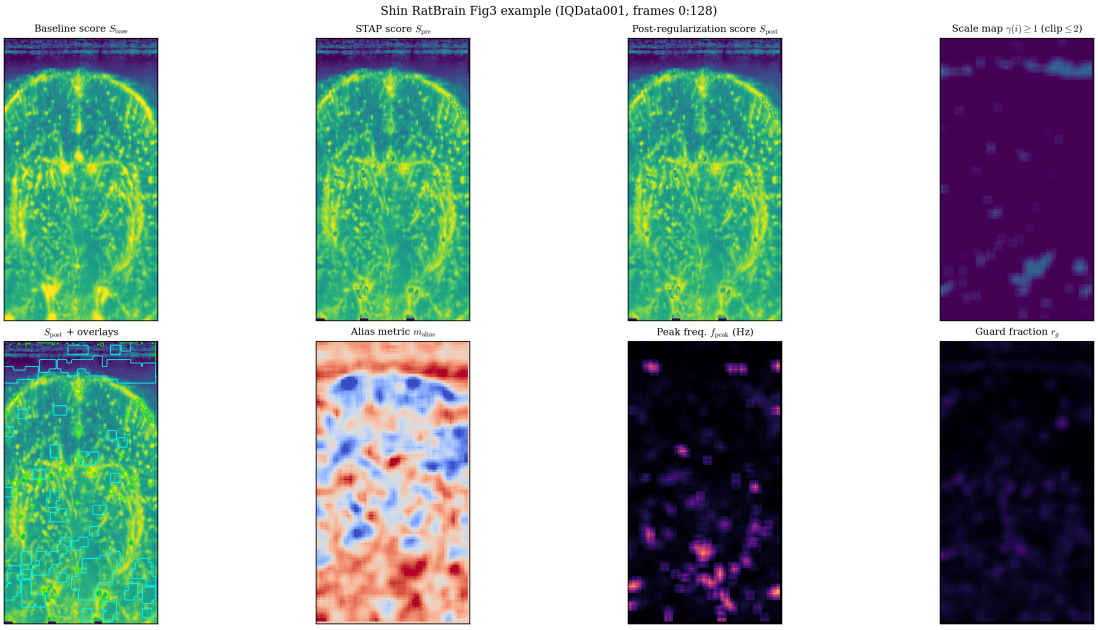

Figure 25: Representative Shin RatBrain Fig3 example (clip `IQData001`; window 0:128) illustrating the end-to-end real-IQ pipeline: baseline PD-derived map, STAP PD-mode score map (pre-KA), KA gate/scale overlay, and final PD-mode score map (post-KA), together with band-energy maps and the flow-proxy mask that defines protected tiles. The KA layer is shrink-only and preserves protected-set invariance exactly. Reproducibility details are provided in Appendix L.

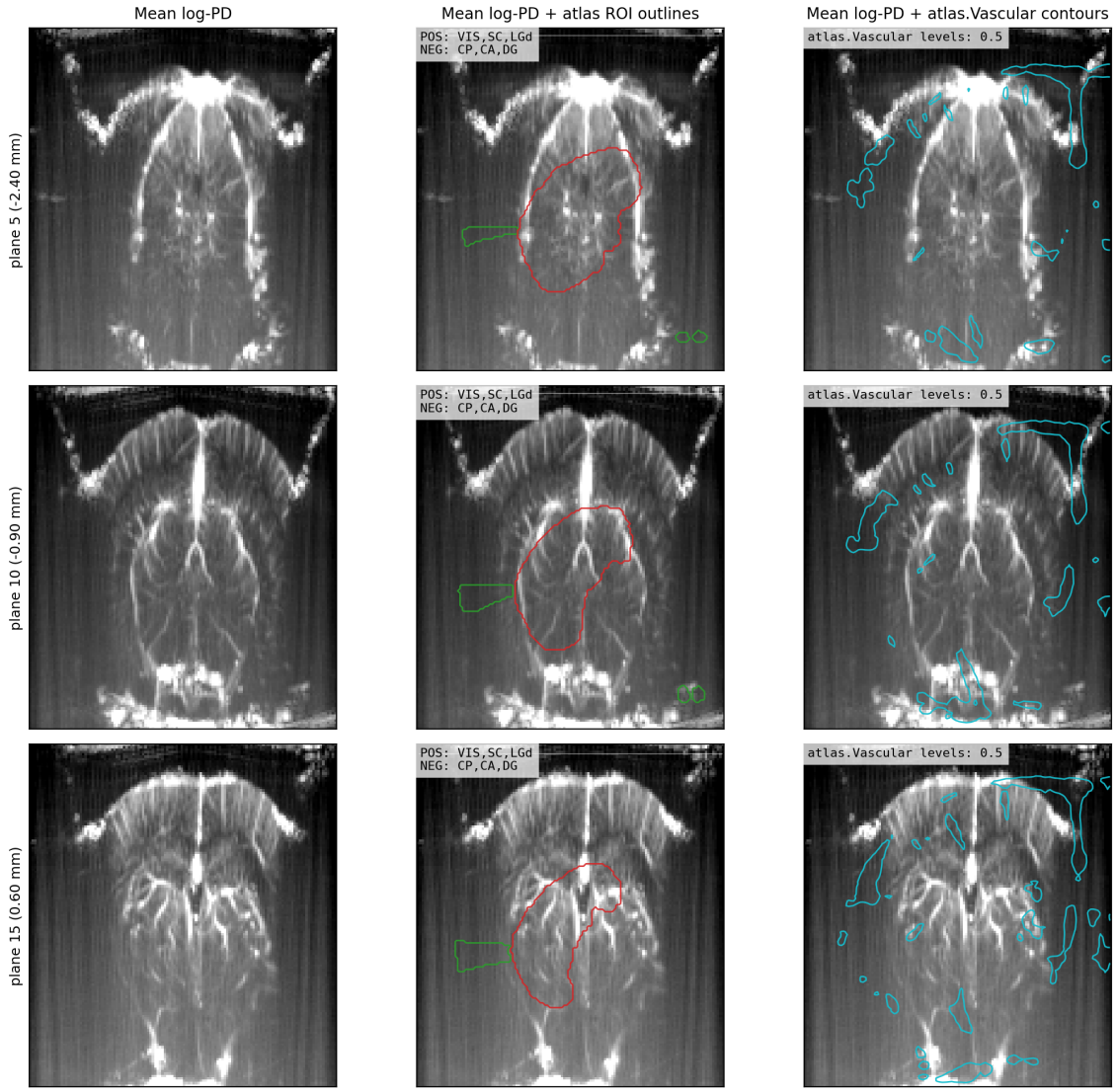

Figure 26: Macé atlas-alignment sanity figure (representative planes from `scan1`). Left: mean log-PD image. Middle: the same mean log-PD with outlines of selected atlas ROIs mapped into the scan plane using `Transformation.mat` (VIS/SC/LGd in green; CP/CA/DG in red). Right: mean log-PD with contours of `atlas.Vascular` (cyan; level 0.5) mapped into the scan plane using the same affine. This figure is generated by `scripts/mace_atlas_overlay_fig.py` and written to `figs/paper/mace_atlas_overlay.png`.